Laboratory for Industry 4.0
Smart Manufacturing Systems

UNIVERSITY OF AUCKLAND
Waipapa Taumata Rau
NEW ZEALAND

A Collaborative Industrial Robotic System with Machine-Learned Trajectory Planning and Control

Wanqing Xia

September 2024

Department of Mechanical and Mechatronics Engineering

Faculty of Engineering

The University of Auckland

A thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy in Engineering, The University of Auckland, 2024.

Abstract

The manufacturing industry is undergoing a paradigm shift from mass production to mass personalisation, driven by increasingly diverse customer demands. This transition necessitates the development of smart manufacturing systems capable of adapting to customised products and varying production levels. To support such a transition, a hybrid manufacturing mode where human collaboratively works with industrial robots is proposed, which combines the high precision, repeatability and strength of robots with the flexibility and adaptability of human workers.

This thesis presents a system implementation for collaborative industrial robots that addresses critical gaps in current industrial automation systems. By enhancing robots' ability to understand, predict, and adapt to human behaviours while maintaining high levels of precision and efficiency. This work paves the way for more flexible, safe, and productive human-robot collaboration in smart manufacturing settings.

This developed collaborative industrial robots control system utilises advanced machine learning technologies, focusing on three key components: The cyber-physical workstation provides a real-time virtual representation status and planned actions of the collaborative robt and other components, demonstrated its effectiveness in visualising complex grasping and handover tasks, enhancing human operators' understanding of robot behaviour. The object 6D pose estimation pipeline employs advanced computer vision techniques to recognise and accurately determine the position and orientation of any objects based on their CAD models, outperformed state-of-the-art methods, reaching an average recall of 75.8 on a widely used public dataset. The proactive trajectory planning algorithm utilises reinforcement learning to predict human arm movements within the shared workspace, enabling the robot to dynamically adjust its trajectory and avoid collisions. It achieved a 96.5% success rate in collision avoidance across a diverse test set of 3,675 scenarios.

To validate the practical applicability of the implemented system, comprehensive case studies were conducted which integrated all three components into a cohesive machine-learned control system. The findings and content of some part of thesis has been published or documented in international conferences and journals, underscoring the contribution of this work made to the society.

Acknowledgement

Life is a long journey full of adventures and unknowns; no one can predict what tomorrow will bring. A day in our life could be exciting, boring, or even painful, but when we reflect on the past, these feelings are compressed and encapsulated into a mere blink. Time passes, unable to be slowed or stopped.

Eleven years ago, around this time, I left Zhoucun (周村), my small hometown lies in the northeast of China, Shandong. I said goodbye to my parents, sister, relatives, and friends, venturing alone to a strange but beautiful country on the other side of the earth, New Zealand. Thanks to my friends from Mount Albert Grammar School, I never felt lonely or suffered during my high school years. My four years of undergraduate study were also fulfilling and happy, thanks to my friends and lecturers. Now, after another four years of doctoral study, I am finally concluding my student life.

At the end of my academic journey, there are so many people to whom I owe my appreciation. While I regret not being able to list every name, I must express my deepest gratitude to those who have been instrumental in my success.

First and foremost, I would like to thank my PhD supervisor, Professor Xun Xu. I have known him since my undergraduate final-year project. He is a gentleman who approaches problems calmly, treats students courteously, and handles academic matters meticulously. What I learned from him extends beyond research methodology to include interpersonal skills and the importance of being a good person—characteristics far more valuable than academic prowess alone.

My co-supervisor, Professor Weiliang Xu, is also beloved by his students, always wearing a happy expression that brings joy to those around him. I am grateful for his contributions and suggestions throughout my PhD journey. Without the help of both my supervisors, I could not have completed this PhD.

I would also like to express my appreciation to the previous and current Heads of the Mechanical and Mechatronics Engineering Department, Professor Krishnan Jayaraman and Professor Simon Bickerton, as well as the Deputy Head, Professor Kean Aw. Their support has been important for every PhD candidate in our department.

Special thanks go to the technicians in the Laboratory for Industry 4.0 Smart Manufacturing Systems, Mr. Ehsan Amiri Tehranizadeh and Mr. Trishit Ghatak. Their technical knowledge and assistance have been the cornerstone for the successful and efficient completion of all my experiments.

I am deeply appreciative of my fellow researchers. Thank you for the time we spent together, both inside and outside the Level 8 office of the engineering building. Your presence throughout my PhD journey has been a source of inspiration and relaxation. We've walked this four-year path together, supporting each other every step of the way.

I am eternally grateful to my parents for their unwavering support in all its forms. They didn't just give me life; they gave meaning to my life. As long as they are with me, I have a steadfast source of mental support and inner peace. To my younger sister, the little sunshine in my life who brings me happiness and joy: I hope you can complete your own student life happily and write your own thesis one day. To my grandparents, uncles, aunts, cousins, and friends in China: thank you all for your continued support. Though we cannot see each other often, I know you're always thinking of me and worrying about me while I study in another country.

Last, but certainly not least, I reserve my most heartfelt thanks for my wife, Xue Sun. She is a smart and strong woman who, like me, left her hometown, parents, relatives, and friends to come to New Zealand. However, her sole reason for making this sacrifice was to be with me. The support and love she provides can never be replaced by anyone else. With her presence, even a small, rented room becomes the safest harbour for me to rest and rejuvenate.

At the end, I am profoundly grateful to everyone who has been part of my academic journey. Your support, guidance, and friendship have shaped not only my research but also my character. As I move forward from student life, I carry with me the lessons learned and the memories shared. Thank you all for being an integral part of this transformative experience.

Sincerely,

夏石青

Table of Contents

Abstract i

Acknowledgement ii

Table of Contents iv

List of Figures viii

List of Tables x

Abbreviations xi

Nomenclature xiii

Statement of Contribution xv

1. Chapter 1. Introduction 1

1.1. Research Background and Motivation 2

1.1.1. Human-robot Collaboration 3

1.1.2. Collaborative Industrial Robots 5

1.1.3. Machine Learning 7

1.2. Research Assumptions and Objectives 10

1.3. Research Scope and Outcome 13

1.4. Thesis Organisation 14

2. Chapter 2. Literature Review 16

2.1. Object 6D Pose Estimation 17

2.1.1. Object Detection 17

2.1.2. Image Segmentation 21

2.1.3. Object 6D Pose Estimation 23

2.1.4. Novel Object 6D Pose Estimation 26

2.1.5. Research Gaps 28

2.2. Robot Trajectory Planning 28

2.2.1. Traditional Trajectory Planning Algorithms 29

2.2.2. Reinforcement Learning Algorithms 32
2.2.3. Deep Reinforcement Learning for Trajectory Planning.................................. 34
2.2.4. Research Gaps.. 36
2.3. Large Language Models in Robotic Operations 37
2.3.1. The Evolving Role of LLMs... 37
2.3.2. Foundational Capabilities of LLMs for Robotic Systems 39
2.3.3. Approaches to Integrating LLMs in Robotics 41
2.3.4. Research Gaps.. 45
2.4. Chapter Summary 46
3. Chapter 3. Development of the HRC Workstation 48
3.1. System Components... 49
3.1.1. UR5e Cobot 49
3.1.2. Human-robot shared workbench... 49
3.1.3. Other Key Components... 50
3.2. Implementation Details.. 52
3.2.1. Physical System Construction.. 52
3.2.2. Digital System Development 53
3.2.3. System Communication 54
3.3. System Architecture 55
3.4. System Testing Application 57
3.5. Chapter Summary 60
4. Chapter 4. Novel Object 6D Pose Estimation.. 63
4.1. Overview... 64
4.2. Proposed Method 66
4.2.1. Detection and Segmentation 66
4.2.2. Mask Selection.. 67
4.2.3. Pose Estimation.. 68

4.3. Experiment ... 69
4.3.1. Testing Dataset and Metrics ... 69
4.3.2. Experiment Result ... 70
4.3.3. Ablation Study ... 72
4.4. Chapter Summary ... 73
5. Chapter 5. Proactive Trajectory Planning ... 76
5.1. Overview ... 77
5.2. Problem Definition ... 79
5.3. Proposed Method ... 82
5.3.1. Soft-actor Critic ... 82
5.3.2. Reward Function Design ... 83
5.3.3. Sim-to-real Transfer ... 87
5.4. Experiment ... 89
5.4.1. Training Environment Setup ... 89
5.4.2. Testing Environment Setup ... 90
5.4.3. Experiment Result ... 91
5.4.4. Discussion ... 93
5.5. Chapter Summary ... 93
6. Chapter 6. Case Studies ... 96
6.1. Case Study 1: Robot Assisted Picking for Household Items ... 97
6.2. Case Study 2: System adaption to New Objects ... 100
6.3. Case Study 3: Human-Robot Collaborative Assembly ... 101
6.4. Discussion ... 103
7. Chapter 7. Conclusions and Future Work ... 105
7.1. Recap of Research ... 106
7.2. Contributions ... 106
7.3. Limitations ... 107

7.4. Recommended Future Work 108
7.5. Concluding Remarks 109
Appendix I 111
References 113

List of Figures

Figure 1–1: The shifts and drivers of manufacturing paradigm (Koren, 2010) 2
Figure 1–2: Methodologies of human co-working with robots (Matheson et al., 2019) 4
Figure 1–3: Machine learning in intelligent manufacturing (Lihui Wang, 2019) 8
Figure 1–4: Thesis structure and research roadmap 14
Figure 2–1: A summary of the CLIP algorithm (Radford et al., 2021) 20
Figure 2–2: Semantic segmentation (left) and instance segmentation (right) (Varatharasan et al., 2019) 22
Figure 2–3: An illustration of the 6D pose estimation task (Xiang et al., 2018) 24
Figure 2–4: Planning time and success rate comparison between CHOMP and RRT (Zucker et al., 2013) 30
Figure 2–5: Benchmarking DRL algorithms across six scenarios (Haarnoja et al., 2018) 34
Figure 2–6: Robot trajectory planning under different obstacle states: (a) 1D moving obstacle (Sangiovanni et al., 2018) (b) 2D moving obstacle (P. Chen et al., 2022) (c) 3D environment with 2DoF obstacle (L. Chen et al., 2022) (d) 3D free moving obstacle (Q. Liu et al., 2021) 35
Figure 3–1:Additional sensors and actuators in the collaborative robot system 51
Figure 3–2: The HRC workstation 52
Figure 3–3: Digitalised HRC workstation 54
Figure 3–4: Communication system architecture 55
Figure 3–5: The machine-learned collaborative industrial robot control system architecture 57
Figure 3–6: Process of the system testing application 58
Figure 4–1: The proposed three-staged system architecture 66
Figure 4–2: Comparison of SAM's output without (left) and with (right) bounding box 67
Figure 4–3: Processing pipeline of MegaPose (Labbé et al., 2022) 68
Figure 4–4: Objects in the YCB-Video dataset (Xiang et al., 2018) 69
Figure 5–1:Illustration of the parameters for cobot, obstacle and goal 80
Figure 5–2: The SAC algorithm's structure 83
Figure 5–3: Reward curves for different translational and rotational weight 86
Figure 5–4: Proposed framework for training and deploying the DRL model 88
Figure 5–5: Goal and obstacle sampling range in simulated environment 89
Figure 5–6: Mean reward, success rate, mean episode length and reward std obtained during validation 92

Figure 5–7: Robot end-effector's trajectory obtained from two models on the same testing case 93
Figure 6–1: The detailed task execution workflow of the collaborative robot control system 97
Figure 6–2: Robot waiting position (a) and execution sequence (b-d)....99
Figure 6–3: 6D pose estimation result of spam can (left) and bleach bottle (right) 100
Figure 6–4: The captured scene (left) and estimation result (right)....101
Figure 6–5: Replacing the intention prediction with task prediction module....102
Figure 6–6: Top view for task prediction (left) and front view pose estimation result (right) 102

List of Tables

Table 4-1: Experiment result of the proposed method benchmarking with other methods 71
Table 4-2: Experiment result for different combinations of system components 72
Table 5-1: Definition of the state variables 81
Table 4-2: List of training hyperparameters.. 90

Abbreviations

A3C	Asynchronous Advantage Actor-Critic
AGV	Automated Guided Vehicle
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
CAD	Computer-Aided Design
CNC	Computer Numerical Control
CNN	Convolutional Neural Network
CNOS	CAD-based Novel Object Segmentation
CPS	Cyber-Physical System
CV	Computer Vision
DDPG	Deep Deterministic Policy Gradient
DL	Deep Learning
DOF	Degrees of Freedom
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
DT	Digital Twin
FCN	Fully Convolutional Network
FOV	Field of View
FPS	Frames Per Second
HMI	Human-Machine Interface
HRC	Human-Robot Collaboration
ICP	Iterative Closest Point
ICT	Information and Communication Technologies
IIoT	Industrial Internet of Things

IoT	Internet of Things
LSTM	Long Short-Term Memory
MDP	Markov Decision Process
ML	Machine Learning
NLP	Natural Language Processing
PPO	Proximal Policy Optimisation
RCNN	Region-based Convolutional Neural Network
RGB	Red, Green, Blue
RGB-D	Red, Green, Blue, Depth
RL	Reinforcement Learning
RNN	Recurrent Neural Network
ROI	Region of Interest
ROS	Robot Operating System
SAM	Segment Anything Model
SARSA	State-Action-Reward-State-Action
SIFT	Scale-Invariant Feature Transform
SLAM	Simultaneous Localisation and Mapping
SOTA	State-Of-The-Art
SURF	Speeded Up Robust Features
SVM	Support Vector Machine
TCP	Tool Centre Point
TRPO	Trust Region Policy Optimisation
YOLO	You Only Look Once

Nomenclature

Symbol	Definition
$\mathcal{S}$	State
$\mathcal{A}$	Action
$\mathcal{R}$	Reward
$\mathcal{P}$	State transition function
γ	Discount factor
π	Policy
J	Expected cumulative reward
$\mathcal{P}_e$	Pose of robot end-effector
p_e	3D translation of robot end-effector
ϕ_e	3D rotation of robot end-effector
$\mathcal{P}_g$	Pose of robot goal
p_g	3D translation of robot goal
ϕ_g	3D rotation of robot goal
$\mathcal{P}_o$	Pose of obstacle
p_o	3D translation of obstacle
ϕ_o	3D rotation of obstacle
$\mathcal{V}_o$	Velocity of the obstacle
v_o	Linear velocity of obstacle
ω_o	Angular velocity of obstacle
Θ	Robot joint angles
$\mathcal{D}$	Euclidean distance between robot links and obstacle
r_c	Collision reward
r_s	Success reward

D_t	Translational distance
r_t	Translational reward
D_r	Rotational distance
r_r	Rotational reward
r_d	Distance reward
W_i	Reward weight of link i
${W_i}^p$	Physical weight of link i
N_o	Number of total objects
V	Number of viewpoints for token embedding
Cs	Vector of tokens from template images
M	Number of total viewpoints
T^K	The kth pose estimate

Chapter 1
Introduction

In recent decades, the rapid development of information and communication technologies has significantly transformed the manufacturing industry. The integration of advanced manufacturing technologies has led to a dramatic shift in the manufacturing paradigm, moving from mass production to mass customisation and mass personalisation. Today's customers demand personalised products manufactured in smaller batches while maintaining competitive costs. This shift requires manufacturing plants to quickly adapt to changing demands and implement flexible shop floors, a concept known as smart manufacturing.

Human-robot collaborative production is a key component of smart manufacturing, providing a balance between fully automated production and manual labour. Its flexibility and adaptability are key in smart manufacturing environments where product variability and customisation are common. This approach enables quick reconfiguration of production lines, supporting the agile manufacturing principles of smart factories.

This chapter begins by reviewing the background knowledge that informs this research, including an overview of smart manufacturing and human-robot collaborative production. Following this, the research assumptions, objectives, and scope are presented. The chapter concludes with an outline of the thesis structure.

1.1. Research Background and Motivation

The manufacturing industry is currently undergoing a significant transformation, driven by advancements in information and communication technologies (ICT) and other smart technologies (X. Xu, 2012). As illustrated in Figure 1–1, the era of mass production began in 1913 with the introduction of the moving assembly line, reaching its peak in 1955. This phase was characterised by high-volume production of standardised goods. In the 1980s, the advent of Computer Numerical Control (CNC) technology facilitated a shift from mass production to mass customisation. This new paradigm allowed for greater product variety while still maintaining relatively large production batches, thus offering customers more options.

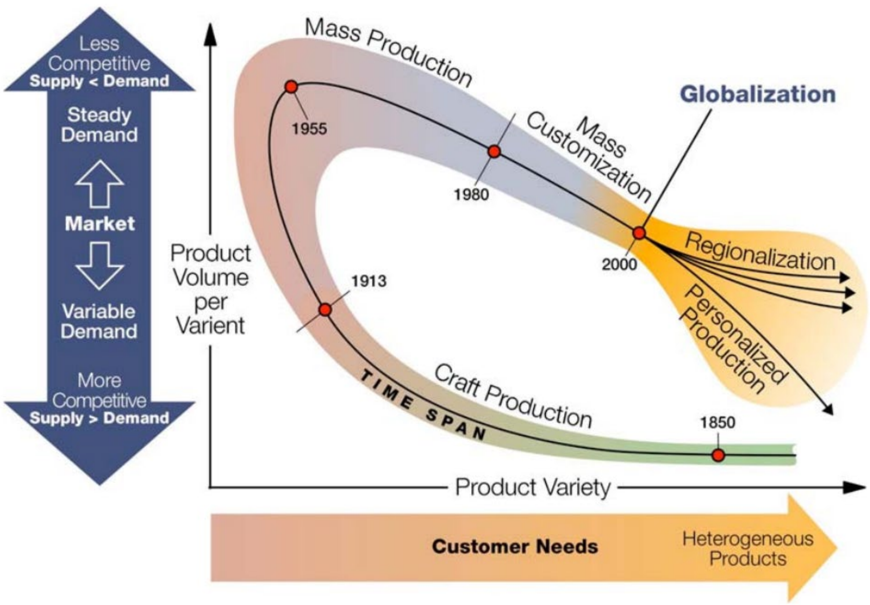

Figure 1–1: The shifts and drivers of manufacturing paradigm (Koren, 2010)

Since the turn of the millennium in 2000, globalisation has propelled the industry towards another transition: from mass customisation to mass personalisation (Koren, 2010). This latest phase is marked by an even greater increase in product variety, with production batch sizes decreasing to as small as one unit per "batch" (Kumar, 2007). This shift towards highly personalised products presents new challenges to the manufacturing industry.

The primary challenge lies in developing manufacturing systems that are more responsive, flexible, and adaptable. These systems must maintain the productivity levels achieved during the mass customisation era while accommodating small batch sizes and personalised products. Additionally, they must do so at a cost that remains competitive with traditional mass production methods (Lu et al., 2020).

To address these challenges, the concept of smart manufacturing has emerged as a comprehensive solution. Smart manufacturing leverages cutting-edge technologies to create intelligent, adaptive, and highly efficient production systems. It encompasses a wide range of smart techniques, including smart design, smart machining, smart control, smart monitoring, and smart scheduling (P. Zheng et al., 2018). By integrating these techniques, smart manufacturing aims to create a more responsive, data-driven, and interconnected manufacturing ecosystem. At the core of smart manufacturing is the idea of seamlessly combining advanced technologies with human expertise. This is where human-robot collaboration (HRC) has emerged as a prominent implementation strategy. HRC leverages the strengths of both humans and machines: it combines the high accuracy, strength, and repeatability of industrial robots with the superior flexibility and adaptability of human operators (S. Li et al., 2021; Lihui Wang et al., 2020).

In the context of HRC and smart manufacturing, machine learning (ML) has emerged as a key enabling technology (L. Wang et al., 2019). ML algorithms can significantly enhance the capabilities of HRC systems, allowing them to adapt more effectively to the varied and complex tasks inherent in personalised manufacturing (Villani et al., 2018). By analysing vast amounts of data from sensors, production processes, and human operators (Tao, Qi, et al., 2018), ML models can optimise robot behaviour, predict maintenance needs, and even improve human-robot interactions (J. Lee et al., 2015). Moreover, ML algorithms can help in real-time decision-making, allowing HRC systems to dynamically adjust to changes in product specifications or production conditions (Q. Liu et al., 2019).

1.1.1. Human-robot Collaboration

HRC in manufacturing refers to the synergistic interaction between human workers and robotic systems in industrial settings. This collaboration aims to combine the strengths of both humans (flexibility, problem-solving, and adaptability) and robots (precision, strength, and endurance) to enhance overall productivity and efficiency in manufacturing processes. In smart manufacturing environments, this collaboration often involves advanced sensors, artificial intelligence (AI), and adaptive control systems that enable robots to work safely alongside humans, sharing the same workspace and tasks (L. Wang et al., 2019).

The journey from traditional automation to collaborative systems in factories represents a significant paradigm shift in manufacturing. As shown in Figure 1–2, traditional automation focused on replacing human labour with machines, often resulting in rigid, isolated robotic cells separated from human workers by safety barriers. Human-robot co-existence represents the initial step towards integration. In this scenario, safety barriers are removed, allowing humans and robots to occupy the same general space. However, direct interaction remains limited, with each entity largely operating independently. Advancing further, synchronisation allows humans and robots to share the same workspace, but their activities are temporally separated. This approach enables more efficient use of the workspace, as robots can perform tasks when humans are not present. Cooperation represents a significant leap forward. In this scenario, humans and robots work simultaneously in the same workspace, focusing on different but complementary tasks. This arrangement allows for a more dynamic work environment, where the strengths of both humans and robots can be utilised concurrently. HRC represents the highest level of integration, emphasising a true partnership between humans and robots. In HRC, robots work independently but in close coordination with human activities, continuously adapting their actions based on human input and environmental changes (Matheson et al., 2019). This evolution has been driven by advancements in robotics, sensor technologies, and AI, enabling the development of robots that can adapt to human presence and work in harmony with human operators (S. Li et al., 2021).

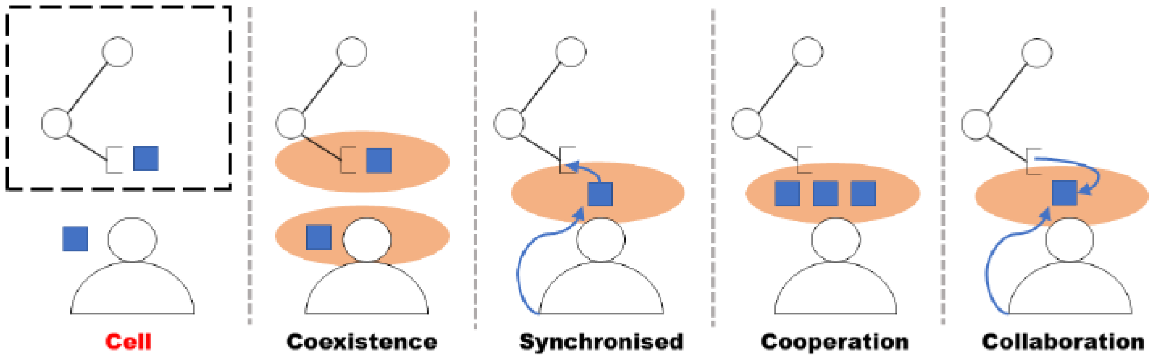

Figure 1–2: Methodologies of human co-working with robots (Matheson et al., 2019)

The benefits of HRC in manufacturing are numerous. By combining human cognitive skills with robotic precision and endurance, collaborative systems can significantly boost overall productivity (Boschetti et al., 2023). The inherent flexibility of collaborative robots allows for easy reprogramming and redeployment, enabling rapid adaptation to changing production needs. The synergy of human oversight and robotic precision often leads to fewer errors and higher product quality (Bauer et al., 2016). Moreover, by handling physically demanding or repetitive tasks, robots can reduce the risk of injury and fatigue for human workers, leading to ergonomic improvements (Lorenzini et al., 2023).

Despite these benefits, implementing collaborative systems on the factory floor comes with several challenges. Ensuring the safety of human workers in close proximity to robots remains a primary concern, necessitating advanced sensing and control systems (Robla-Gomez et al., 2017). The technical complexity of integrating collaborative robots with existing manufacturing systems and processes can be daunting, often requiring significant expertise (Malik & Bilberg, 2018). Overcoming potential resistance or fear from workers about working alongside robots is vital for successful implementation (Welfare et al., 2019). Justifying the initial investment in collaborative systems, especially for smaller manufacturers, can be challenging when considering return on investment (Bauer et al., 2016). Additionally, adhering to evolving safety standards and regulations for HRC adds another layer of complexity to implementation (Rodriguez-Guerra et al., 2021).

Looking to the future, HRC in smart manufacturing holds promising developments. The incorporation of more sophisticated AI will enable robots to better understand and predict human actions (H. Liu & Wang, 2017). More intuitive and natural interfaces, including gesture and voice control, are being developed to improve human-robot interaction (Villani et al., 2018). The scope of collaboration is expected to extend beyond manufacturing to areas such as logistics, healthcare, and construction (Ajoudani et al., 2018). Robots will likely become more capable of autonomous decision-making while maintaining safe collaboration with humans. As robots become more advanced, addressing the ethical implications of HRC will become increasingly important, shaping the future landscape of smart manufacturing (Alexander Obaigbena et al., 2024).

1.1.2. Collaborative Industrial Robots

Collaborative industrial robots, commonly known as cobots, are a new generation of robotic systems designed to work alongside humans in a shared workspace. Unlike traditional industrial robots that operate in isolation, cobots are built with the primary goal of interacting with human workers safely and efficiently. These robots are characterised by their ability to perceive and respond to their environment, including human presence and actions. Cobots are typically lighter, more flexible, and easier to program than conventional industrial robots. They are designed with rounded edges to minimise the risk of injury in case of contact with humans. The key defining feature of cobots is their ability to operate without safety fencing, allowing for direct collaboration with human workers in various tasks (Sherwani et al., 2020).

The journey from traditional industrial robots to collaborative industrial robots represents a significant paradigm shift in manufacturing automation. Industrial robots, first introduced in the 1960s, were designed to perform repetitive tasks with high speed and precision, but they required isolation from human workers due to safety concerns. As manufacturing needs evolved, demanding more flexibility and customisation, the limitations of these rigid, isolated systems became apparent. The concept of collaborative robots emerged in the late 1990s, with early prototypes designed to share workspaces with humans safely (Galin & Meshcheryakov, 2019). Over the past two decades, advancements in sensor technologies, AI, and control systems have accelerated the development of cobots. This evolution has been driven by the need for more adaptable automation solutions that can work in harmony with human skills, leading to the current generation of cobots that are becoming increasingly prevalent in modern manufacturing environments (Vicentini, 2021).

Cobots are distinguished by several key features that enable their safe and effective operation alongside humans, these features are defined in several standards and considerations to guide the development and deployment of cobots. The technical specification ISO/TS 15066:2016 (ISO, 2016), complementing the general robot safety standards ISO 10218-1 (ISO, 2011a) and ISO 10218-2 (ISO, 2011b), provides four classes of safety requirements specific to collaborative industrial robot systems.

1) Power and force limited joints inherently limit the force that cobots can exert, reducing the risk of injury if contact occurs between the robot and a human.
2) Hand-guiding functions that allow operators to physically teach the robot movements, which can be then recorded and repeated, enables intuitive programming of the robot.
3) Speed and separation monitoring continuously tracks the speed of the robot and its separation distance from the human operator. The robot's speed is adjusted based on how close the human is, potentially coming to a stop if the human gets too close
4) Safety-rated monitored stop, which allows them to pause operation when a human enters their workspace and resume automatically when the person leaves, enhancing safety without compromising productivity.

Cobots are increasingly being integrated with other smart manufacturing technologies, amplifying their capabilities and impact. Cobots can be connected to Internet of Things (IoT) networks, allowing for real-time data collection and analysis. This integration enables predictive maintenance, performance optimisation, and seamless coordination with other manufacturing systems(L. Da Xu et al., 2018). AI and ML algorithms can enhance cobot decision-making capabilities, allowing them to adapt to new situations and improve their performance over time (Jinjiang Wang et al., 2018). Digital Twin (DT) technology enables virtual replicas of cobot systems to be used for simulation and optimisation, allowing manufacturers to test new processes and configurations without disrupting live operations(Tao, Cheng, et al., 2018). Augmented Reality (AR) interfaces can be used to enhance human-cobot interaction, providing workers with real-time information and guidance for more effective collaboration (X. Liu et al., 2020). By integrating cobots with data analytics platforms, manufacturers can gain deeper insights into their production processes, identifying bottlenecks and opportunities for improvement (Zhong et al., 2017).

Despite their advantages, cobots face several limitations and challenges in industrial applications. Most cobots have lower payload capacities compared to traditional industrial robots, which can limit their applications in heavy-duty tasks (Krüger et al., 2009). To ensure safety, cobots often operate at lower speeds than traditional robots, which can impact overall productivity in some applications (Robla-Gomez et al., 2017). Designing intuitive and efficient ways for humans to interact with cobots remains a challenge, particularly in complex manufacturing environments (Aaltonen et al., 2018). While generally less expensive than traditional industrial robots, the initial investment in cobot systems can still be significant for smaller manufacturers (Bauer et al., 2016). The introduction of cobots requires workers to develop new skills in robot programming and maintenance, which can create a skill gap in some organisations (Kadir et al., 2019). Current cobots still rely heavily on human input and supervision for many tasks, limiting their ability to operate fully autonomously (L. Wang et al., 2019). While cobots are designed for safe human interaction, ensuring safety in rapidly changing or unpredictable manufacturing environments remains a challenge that requires ongoing attention and development (Lasota et al., 2017).

1.1.3. Machine Learning

ML, a subset of AI, plays a vital role in industrial AI and smart manufacturing. At its core, ML involves algorithms and statistical models that enable computer systems to improve their performance on a specific task through experience, without being explicitly programmed (Badillo et al., 2020). As shown in Figure 1–3, ML serves as a powerful tool for analysing vast amounts of data generated by manufacturing processes, equipment, and sensors. By identifying patterns, making predictions, and generating insights, ML systems can significantly enhance decision-making processes, optimise operations, and drive innovation in manufacturing environments (Lihui Wang, 2019).

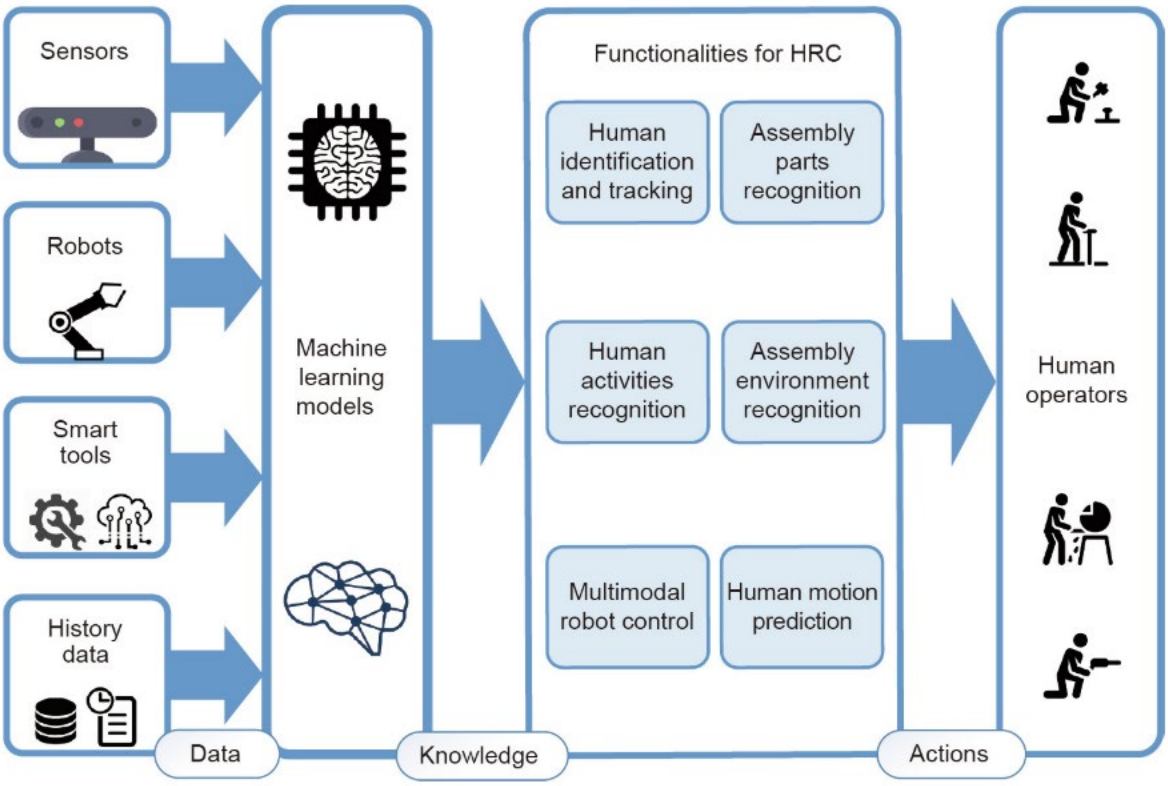

Figure 1–3: Machine learning in intelligent manufacturing (Lihui Wang, 2019)

The integration of ML with other smart manufacturing technologies creates powerful synergies that drive the evolution of Industry 4.0. When combined with robotics and automation, ML enables more adaptive and intelligent manufacturing systems capable of real-time optimisation and cognitive HRC (S. Li et al., 2021). The fusion of ML with DT technology allows for more accurate simulations and predictions of manufacturing processes (J. Lee et al., 2020). AR systems enhanced with ML can provide workers with intelligent, context-aware assistance on the factory floor (Tao, Qi, et al., 2018). As these technologies continue to converge, they pave the way for more interconnected, intelligent, and efficient smart manufacturing ecosystems.

In the realm of HRC, ML has revolutionised several critical areas, enhancing the efficiency and safety of interactions between humans and machines. Human motion prediction stands at the forefront of these advancements, employing sophisticated algorithms to analyse and anticipate human movements in real-time. By leveraging techniques such as Recurrent Neural Networks

(RNN) and Long Short-Term Memory (LSTM) models, ML systems can predict trajectories of human workers with increasing accuracy (Martinez et al., 2017).

Closely related to motion prediction is robot trajectory planning, where ML algorithms dynamically generate and adjust robot trajectories. Unlike traditional trajectory planning methods, ML-based approaches can rapidly adapt to changing environments and human presence. Reinforcement Learning (RL) techniques, in particular, have shown promise in this area, allowing robots to learn optimal trajectories through repeated interactions with their environment and human co-workers (Ibarz et al., 2021). These adaptive trajectory planning systems significantly reduce the risk of collisions while maximising operational efficiency.

The interface between humans and robots has been greatly enhanced through ML-powered systems. Deep learning models, particularly Convolutional Neural Networks (CNN) for visual data (Cao et al., 2017) and transformers for audio processing (Y. Wang et al., 2020), have dramatically improved the accuracy of these recognition systems. This enables more natural and intuitive communication between humans and robots, allowing workers to control robotic systems through gestures or voice commands without the need for complex programming or extensive training.

Adaptive Control represents another fundamental application of ML in HRC. Here, ML algorithms enable robots to dynamically adjust their behaviour based on the specific task at hand and the characteristics of their human collaborators. Through techniques like online learning and adaptive neural networks, robots can fine-tune their force application, speed, and precision in real-time (Zhijun Li et al., 2017). This adaptability is particularly valuable in tasks requiring delicate manipulation or when working with materials of varying properties.

These applications of ML in HRC are enormous and cannot be all listed here but they are not isolated, often work in concert to create more intelligent, responsive, and safe collaborative environments. As these technologies continue to evolve, we can expect even more seamless and productive interactions between humans and robots in smart manufacturing settings and beyond.

Implementing ML systems in manufacturing environments comes with its own set of challenges. Data quality and consistency are often major hurdles, as manufacturing data can be noisy, incomplete, or inconsistent across different machines or production lines (Yan et al., 2017). Integrating ML systems with existing manufacturing execution systems (MES) and enterprise resource planning (ERP) systems can be complex, requiring careful planning and execution (Tao, Qi, et al., 2018). The interpretability of ML models is another challenge, particularly in regulated industries where decision-making processes need to be transparent and explainable (Wuest et al., 2016). Additionally, the need for specialised skills in data science and ML can be a barrier for many manufacturers, necessitating investment in training or external expertise (G. Li et al., 2021). Ensuring the security and privacy of sensitive manufacturing data in ML systems is also a critical concern that needs to be addressed (Tuptuk & Hailes, 2018).

1.2. Research Assumptions and Objectives

This research is mainly based on three assumptions, as follows:

Assumption 1: **The trend towards mass personalisation in the manufacturing industry will continue.**

This assumption sets the stage for this research. Mass personalisation is the manufacturing that the world is currently experiencing. This assumption posits that the trend towards greater personalisation in manufacturing processes will persist and likely accelerate in the coming years.

This trend is driven by several factors:

- Consumer demand: Modern consumers increasingly expect products tailored to their individual preferences and needs. This shift in consumer behaviour is pushing manufacturers to offer more customisable options (Kumar, 2007).
- Data analytics and IoT: The proliferation of data collection through the IoT and advanced big data analytics capabilities allow manufacturers to better understand and respond to individual customer preferences (Zhong et al., 2017).
- Supply chain innovations: Improvements in supply chain management and just-in-time manufacturing techniques enable more flexible production processes that can accommodate personalisation (R. Novais et al., 2019).
- Competition: As some manufacturers successfully implement mass personalisation strategies, others are likely to follow suit to remain competitive in the market.

The continuation of this trend implies that manufacturers will increasingly need to adapt their processes, technologies, and business models to accommodate greater levels of product customisation while maintaining efficiency and cost-effectiveness. This shift may have far-reaching implications for various aspects of manufacturing, including product design, production planning, inventory management, and customer relationship management.

Assumption 2: The proliferation of cobots in industrial environments will continue to accelerate.

Collaborative robots, or cobots, are designed to work alongside human workers, combining the strengths of human flexibility and decision-making with robotic precision and endurance. This assumption is built on top of Assumption 1, posits that the adoption of cobots in industrial settings will continue to grow rapidly in the coming years due the demand of mass personalised production.

Key factors driving this trend include:

- Enhanced Safety Features: Cobots are equipped with advanced sensors and safety systems, allowing them to operate in close proximity to humans without the need for safety cages (Villani et al., 2018).
- Flexibility and Ease of Programming: Cobots are typically easier to program and repurpose than traditional industrial robots, making them suitable for a wider range of tasks and more adaptable to changing production needs (Vicentini, 2021).
- Cost-Effectiveness: As technology advances and production scales up, cobots are becoming more affordable, making them accessible to small and medium-sized enterprises (Zsifkovits et al., 2020).
- Labour Shortages and Ergonomics: Cobots can address labour shortages in manufacturing and improve ergonomics by taking on repetitive or physically demanding tasks (Chiacchio et al., 2018).
- Quality and Productivity Improvements: The precision of cobots, combined with human oversight, can lead to improvements in product quality and overall productivity (Lihui Wang et al., 2020).

Assumption 3: Enhancement of cobot control algorithms is essential for advancing HRC in smart manufacturing environments.

This assumption posits that current control algorithms for cobots need significant improvement to fully realise the potential of HRC in smart manufacturing settings. While cobots have made substantial progress, their control systems still face challenges in achieving seamless, efficient, and safe collaboration with human workers in complex, dynamic manufacturing environments.

Key aspects supporting this assumption include:

- Adaptability and Learning: Current control algorithms often struggle to adapt quickly to changing tasks or unpredictable human behaviours. Improved algorithms should incorporate advanced ML techniques for real-time adaptation (Ogenyi et al., 2021).
- Safety and Risk Assessment: Enhanced control algorithms are needed to better predict and respond to potential safety risks in dynamic HRC scenarios, going beyond simple collision avoidance (Robla-Gomez et al., 2017).
- Natural Interaction: Improved algorithms should facilitate more natural and intuitive interaction between humans and cobots, including better interpretation of human gestures, voice commands, and intentions (L. Wang et al., 2019).
- Task Optimisation: Advanced control algorithms are required to optimise task allocation and execution between humans and cobots, maximising the strengths of both (Chand & Lu, 2023).
- Multi-Cobot Coordination: As smart manufacturing environments become more complex, control algorithms need to manage coordination among multiple cobots and humans simultaneously (Iqbal & Riek, 2017).
- Integration with IoT and AI: Control algorithms should better integrate with broader smart manufacturing systems, including IoT devices and AI-driven decision-making processes (Humayun, 2021).

With the assumptions that have been made, the objectives for this research are:

Objective 1: Develop and validate the HRC workstation.

a) Build the physical HRC workstation with selected enabling components.
b) Design a high-fidelity 3D virtual digital representation of the physical HRC workstation, including cobots, working table, and relevant equipment.
c) Implement real-time data synchronisation between the physical workstation and its digital counterpart, focusing on cobot movements.
d) Test the basic functions of the workstation with industrial applicable test cases.

Objective 2: Develop a robust and versatile robot vision pipeline for accurate 6D pose estimation of novel objects in the collaborative workspace.

a) Implement a 6D pose estimation algorithm that can accurately determine the position and orientation of identified objects, even the object was never seen by the trained algorithm.
b) Evaluate the performance of the vision pipeline in terms of accuracy, processing speed, and generalisation to novel objects with public datasets.

Objective 3: Develop and implement an adaptive robot trajectory planning algorithm for proactive collision avoidance with dynamic human arm movements in 3D space.

a) Develop a series of cases to train the robot control algorithm with RL.
b) Implement a proactive trajectory planning algorithm that can dynamically adjust the robot's trajectory based on predicted human movements while optimising task efficiency.
c) Conduct testing and validation of the algorithm in simulated collaborative working scenarios, measuring metrics such as collision avoidance success rate, trajectory optimality, and task completion time.

Objective 4: Validate and demonstrate the integrated collaborative industrial robot system through a series of industry-relevant case studies.

a) Implement a multi-modal sensing system combining RGB-D cameras and microphone for intuitive interaction between human and robot.
b) Design and implement case studies representing different industrial applications (e.g., assembly, quality inspection, material handling) that incorporate the systems developed in Objectives 1, 2, and 3.
c) Compile and synthesise the results from all case studies to draw broader conclusions about the effectiveness, limitations, and potential improvements of the developed collaborative industrial robot system in smart manufacturing environments.

1.3. Research Scope and Outcome

This research focuses on developing a collaborative industrial robot system for mass personalised production. To enhance the system's generalisability, adaptability, and safety, machine learning algorithms are developed and integrated. Bounded by the pre-defined assumptions, this work aims to achieve the four objectives mentioned in previous section, which are: development of the physical and digital HRC workstation, design a novel object 6D pose estimation pipeline, implementing a proactive robot trajectory planning algorithm and develop a case study that applicates and validates all systems. All hardware related to this project is already available in the Laboratory for Industry 4.0 Smart Manufacturing Systems (LISMS).

1.4. Thesis Organisation

Figure 1–4 presents the overall research roadmap of this work, which is structured into seven chapters. The content of each chapter is briefly introduced in the following paragraphs.

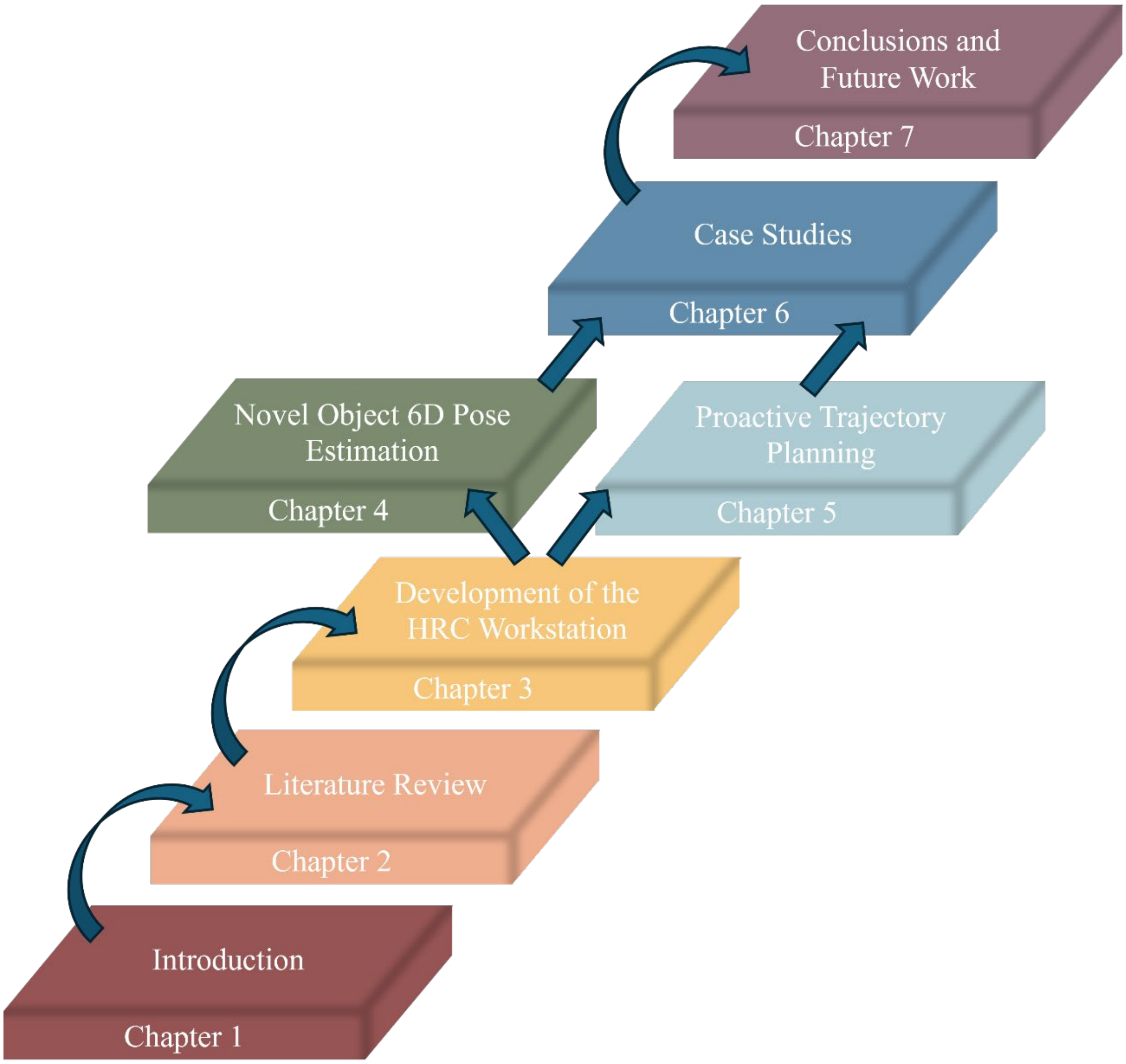

Figure 1–4: Thesis structure and research roadmap

Chapter 1 sets the stage for the entire thesis, introducing the concept of mass personalisation and its relationship to HRC, cobots and ML technologies. It outlines the motivation behind this research, highlighting the growing importance of HRC in industrial settings. The chapter also presents the research assumptions, objectives, and the overall structure of the thesis, providing readers with a roadmap for the work that follows.

Chapter 2 provides a comprehensive overview of existing research in the field of collaborative robotics, focusing on machine vision and trajectory planning algorithms. It critically analyses current approaches, identifying gaps in knowledge and areas where further research is needed. This chapter establishes the theoretical foundation for the thesis and justifies the research direction taken in subsequent chapters.

Chapter 3 details the development of the HRC workstation, including the physical building and digitalising it to form a digital replica of the physical workstation. The chapter discusses the key components used to create the system, the digitalisation process, communication method and how it facilitates real-time simulation and optimisation of collaborative tasks between humans and robots. This chapter also provide the overall architecture of the collaborative industrial robot system that is aimed to achieve in this project.

Chapter 4 addresses the challenge of identifying and locating novel objects in the dynamic workspace. It presents a new method for estimating the 6D pose (position and orientation) of novel objects, which is essential for robots’ adaptability and flexibility. The chapter details the computer vision and ML techniques employed, as well as the system's performance in various scenarios.

Chapter 5 focus on the critical aspect of robot movement in collaborative spaces, this chapter presents a novel approach to proactive trajectory planning. It describes the algorithms developed to anticipate human movements and dynamically adjust robot trajectory to ensure safe and efficient collaboration. The chapter also discusses the implementation of these algorithms and their integration with the digital model from Chapter 3.

Chapter 6 presents a comprehensive case study of HRC in an industrial setting building on the technologies developed in previous chapters. It demonstrates how the object pose estimation and proactive trajectory planning work together in real-world scenarios. The chapter analyses the effectiveness of the integrated system, discussing both successes and challenges encountered during the case study.

Chapter 7 summarises the key findings and contributions of the thesis. It reflects on how the research objectives were met and discusses the implications of the work for the field of collaborative robotics. Additionally, this chapter outlines potential directions for future research, identifying areas where further advancements could be made to enhance HRC in industrial environments.

Chapter 2 Literature Review

This chapter presents a comprehensive literature review focusing on two key areas integral to this research: Robot Trajectory Planning and Object Pose Estimation. Building upon the research objectives and scope defined in Chapter 1, this review delves deeper into the current state of the art in these fields.

The chapter explores each area in detail, examining recent developments, current trends, and state-of-the-art techniques. Particular attention is given to identifying research gaps and open problems within each field, as well as at their intersections.

By critically analysing the existing literature, this chapter aims to position our research within the broader context of these rapidly evolving domains. The insights gathered here will serve as a foundation for the subsequent work, highlighting opportunities for novel contributions and guiding the direction of the research efforts.

2.1. Object 6D Pose Estimation

Object 6D pose estimation is an advanced Computer Vision (CV) task that aims to determine an object's position and orientation in three-dimensional space from a RGB or RGBD image. Unlike object detection, which focuses on locating objects within an image using bounding boxes, or image segmentation, which classifies each pixel into different entities, 6D pose estimation provides a more complete understanding of an object's spatial configuration. It builds upon these foundational tasks by not only identifying and localising objects but also determining their precise 3D position (x, y, z coordinates) and 3D orientation (roll, pitch, yaw angles) relative to the camera (Y. Li et al., 2018). This capability is vital for applications such as robotic manipulation, augmented reality, and autonomous driving, where accurate spatial understanding of objects is essential for interaction or navigation in complex environments (Zhu et al., 2022).

This section begins with an introduction to object detection and instance segmentation algorithms in CV, as they form the foundation for and are closely related to object 6D pose estimation. Following this, we review ML methods for 6D pose estimation. Special attention is given to novel object 6D pose estimation, because of its key role in smart manufacturing settings. The section concludes by identifying current research gaps in this field.

2.1.1. Object Detection

Object detection has been a fundamental task in CV for decades, evolving significantly with the advent of deep learning techniques. Initially, traditional methods relied on handcrafted features and sliding window approaches, which were computationally expensive and often lacked robustness (Georgiou et al., 2020). The introduction of Convolutional Neural Networks (CNNs) (LeCun et al., 1998) marked a pivotal moment in the field, dramatically improving both accuracy and efficiency.

The evolution of CNN-based object detection methods has been marked by a series of innovations, each addressing limitations of its predecessors and pushing the boundaries of accuracy and efficiency. These methods can be broadly categorised into two main approaches: two-stage detectors and single-stage detectors, with further distinctions made between anchor-based and anchor-free techniques.

Two-stage detectors, pioneered by the R-CNN family, operate on the principle of first generating region proposals and then classifying these proposals. The original R-CNN used selective search to generate potential object regions, applying a CNN to each region independently (Girshick et al., 2014). This approach, while groundbreaking, was computationally expensive. Fast R-CNN (Girshick, 2015) improved upon this by introducing Region of Interest (RoI) pooling, allowing the network to process the entire image once and then extract features for each region proposal. This innovation significantly reduced computation time and enabled end-to-end training. Faster R-CNN (Ren et al., 2015) further refined this approach by introducing the Region Proposal Network (RPN), which replaced the selective search algorithm with a fully convolutional network for generating region proposals. This made the entire detection pipeline trainable end-to-end and markedly improved both speed and accuracy.

In contrast, single-stage detectors aim to predict bounding boxes and class probabilities in a single forward pass of the network. YOLO (You Only Look Once) (Redmon et al., 2016) was a pioneer in this category, dividing the image into a grid and predicting bounding boxes and class probabilities for each grid cell simultaneously. This approach achieved real-time performance, albeit initially with some sacrifice in accuracy. SSD (Single Shot Detector) (W. Liu et al., 2016) built upon this concept by utilising multiple feature maps from different network layers and a set of default bounding boxes with various scales and aspect ratios. This multi-scale approach allowed SSD to handle objects of different sizes more effectively, bridging the gap in accuracy with two-stage detectors while maintaining speed advantages.

The distinction between anchor-based and anchor-free methods represents another important dimension in object detection. Anchor-based methods, which include most traditional approaches like Faster R-CNN and SSD, rely on predefined anchor boxes as references for object detection. These anchors serve as prior knowledge about object shapes and sizes, helping the network make more accurate predictions. However, the design of these anchors often requires careful tuning and can be dataset specific.

Anchor-free methods, on the other hand, attempt to eliminate the need for predefined anchors. Models like CornerNet (Law & Deng, 2018) and CenterNet (Duan et al., 2019) fall into this category, directly predicting keypoints or center points of objects along with other properties like size and offset. This approach simplifies the detection pipeline and reduces the number of hyperparameters related to anchor box design. Anchor-free methods have shown promising results, especially in scenarios where object shapes are highly variable or unpredictable.

As the field progressed, researchers introduced various innovations that cut across these categories. Feature Pyramid Networks (FPN) (T.-Y. Lin, Dollar, et al., 2017) became a popular addition to both two-stage and single-stage detectors, enabling the efficient construction of multi-scale feature pyramids. This technique significantly improved the detection of objects at different scales, particularly benefiting small object detection. Another notable advancement was the introduction of focal loss by RetinaNet (T.-Y. Lin, Goyal, et al., 2017), which addressed the class imbalance problem inherent in object detection tasks. By down-weighting the loss contribution from easy examples, focal loss allowed models to focus on hard, misclassified examples during training, significantly boosting the accuracy of single-stage detectors.

As the field continues to advance, the lines between these categories have begun to blur. Modern object detection systems often incorporate elements from multiple approaches, leveraging the strengths of different techniques to achieve optimal performance. This trend towards hybrid approaches, combined with ongoing innovations in network architectures and training strategies, continues to drive progress in the field of object detection, paving the way for more accurate, efficient, and versatile systems.

In recent years, researchers began to explore alternatives to CNNs that could capture long-range dependencies more effectively. This exploration led to the introduction of transformer architectures in CV. Originally designed for natural language processing (NLP) tasks (Vaswani et al., 2017), transformers proved to be remarkably effective in image analysis as well. The self-attention mechanism in transformers allows for global context modelling, which is particularly beneficial for understanding complex scenes and detecting objects with varying scales and occlusions.

Vision Transformers (ViT) (Dosovitskiy et al., 2021) demonstrated that pure transformer architectures could achieve state-of-the-art performance on image classification tasks. This success quickly translated to object detection, with models like DETR (DEtection TRansformer) (Carion et al., 2020) showcasing the potential of end-to-end object detection using transformers. DETR eliminated the need for many hand-designed components present in CNN-based detectors, such as non-maximum suppression or anchor generation, by framing object detection as a direct set prediction problem. The integration of transformers in object detection continued to evolve, with hybrid approaches combining the strengths of CNNs and transformers. Models like Swin Transformer (Z. Liu et al., 2021) introduced hierarchical feature representations, addressing some of the limitations of vanilla ViTs in capturing fine-grained local features crucial for accurate object localisation.

As the fields of computer vision and natural language processing continued to converge, researchers sought ways to leverage the semantic understanding capabilities of large language models (LLMs) for visual tasks. This pursuit led to the development of CLIP (Contrastive Language-Image Pre-training) (Radford et al., 2021), a groundbreaking approach that bridges the gap between visual and textual modalities. CLIP was developed by OpenAI, represents a significant leap forward in object detection and image understanding. As illustrated in Figure 2–1, by training on a vast dataset of image-text pairs from the internet, CLIP learns to associate visual features with natural language descriptions. This approach enables zero-shot learning, where the model can detect and classify objects it has never explicitly been trained on, simply by leveraging its understanding of language and visual concepts.

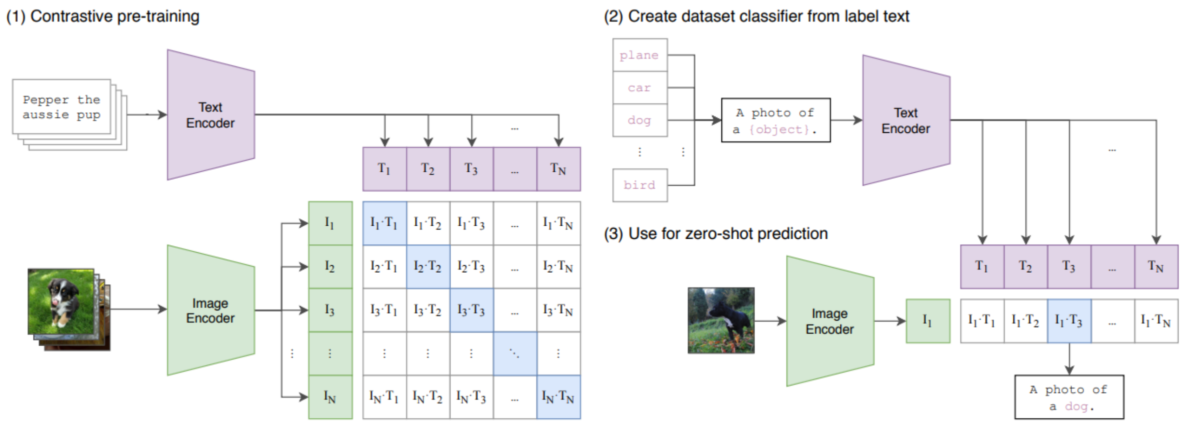

Figure 2–1: A summary of the CLIP algorithm (Radford et al., 2021)

The power of CLIP lies in its versatility and generalisation capabilities. Unlike traditional object detectors that are limited to a fixed set of predefined categories, CLIP can be prompted with arbitrary text descriptions to detect a wide range of objects, attributes, and even abstract concepts. This flexibility opens new possibilities for object detection in diverse and dynamic environments. Moreover, CLIP's ability to ground language in visual representations has paved the way for more advanced multi-modal models. These models can perform complex reasoning tasks that require both visual and linguistic understanding, such as visual question answering, image captioning, and open-vocabulary object detection.

The integration of LLM-like capabilities with object detection, as exemplified by CLIP, represents a significant shift in how we approach CV tasks. It moves beyond simple localisation and classification towards a more holistic understanding of visual scenes, context, and semantics. This paradigm shift promises to enable more intelligent and adaptable vision systems capable of interpreting the visual world in ways that more closely align with human perception and reasoning.

2.1.2. Image Segmentation

Image segmentation is a fundamental task in CV which has undergone significant advancements over the years, evolving from traditional methods to sophisticated deep learning approaches (Garcia-Garcia et al., 2017). This field aims to assign a semantic label to each pixel in an image, effectively understanding the scene at a granular level. The journey of image segmentation mirrors the broader progress in CV, marked by increasing accuracy, efficiency, and generalisation capabilities.

In the early days of semantic segmentation, researchers relied on handcrafted features and traditional ML algorithms. These methods often involved extracting low-level features such as colour histograms, texture descriptors, and edge information, followed by classification using algorithms like Support Vector Machines (SVMs) or random forests (Shotton et al., 2008). While these approaches laid the groundwork for pixel-wise classification, they struggled with complex scenes and lacked the ability to capture high-level semantic information effectively.

The advent of deep learning, particularly CNNs, marked a turning point in semantic segmentation. The Fully Convolutional Network (FCN) (Long et al., 2015) was a landmark paper that adapted classification networks for dense prediction tasks. FCNs replaced fully connected layers with convolutional layers, allowing the network to output spatial maps instead of classification scores. This architecture could process images of arbitrary size and produce correspondingly-sized output maps, making it well-suited for semantic segmentation tasks.

Building upon FCNs, researchers developed more sophisticated architectures to address challenges such as preserving spatial information and capturing multi-scale context. The U-Net (Ronneberger et al., 2015) architecture, originally designed for biomedical image segmentation, introduced skip connections between encoder and decoder paths, allowing the network to combine low-level features with high-level semantic information. This approach significantly improved the segmentation of fine details and boundaries.

DeepLab series (L.-C. Chen et al., 2017; L. C. Chen, Papandreou, et al., 2018; L. C. Chen, Zhu, et al., 2018; C. Liu et al., 2019), starting with DeepLabv1 and evolving through multiple versions, brought several innovations to the field. These include atrous (dilated) convolutions to enlarge the receptive field without increasing computational cost, atrous spatial pyramid pooling (ASPP) to capture multi-scale context, and the use of fully connected Conditional Random Fields (CRFs) as a post-processing step to refine segmentation boundaries. The pursuit of real-time semantic segmentation led to the development of efficient architectures like ENet (Paszke et al., 2016) and ICNet (H. Zhao et al., 2018), which aimed to balance accuracy and speed for applications such as autonomous driving. These models employed techniques like early down-sampling, factorised convolutions, and multi-branch designs to reduce computational complexity while maintaining reasonable accuracy.

Attention mechanisms (Vaswani et al., 2017), which had shown great success in NLP, were also incorporated into semantic segmentation models. The PSPNet (Pyramid Scene Parsing Network) (H. Zhao et al., 2017) used a pyramid pooling module to aggregate context information at different scales, effectively capturing global context. Similarly, the Attention U-Net (Oktay et al., 2018) introduced attention gates to highlight salient features and suppress irrelevant regions, improving the network's ability to focus on important parts of the image.

Recent advancements have focused on developing models that can generalise across diverse domains and tasks, while also progressing from semantic-level to instance-level image segmentation, as shown in Figure 2–2. Mask R-CNN (K. He et al., 2017) extended the Faster R-CNN object detection framework to perform instance segmentation, simultaneously detecting objects and generating a pixel-wise mask for each instance. This approach bridged the gap between object detection and image segmentation, paving the way for more unified vision models.

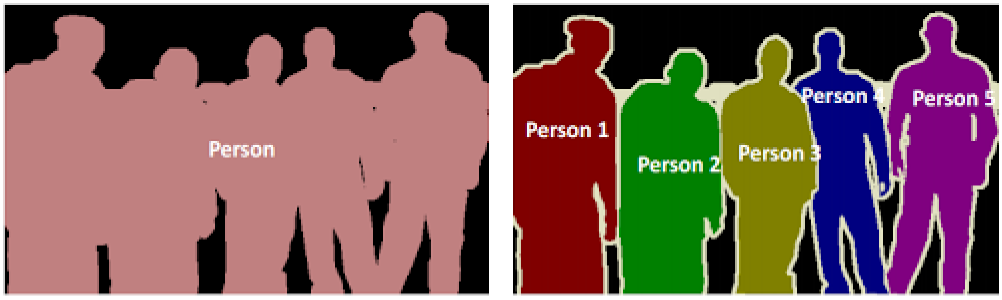

Figure 2–2: Semantic segmentation (left) and instance segmentation (right) (Varatharasan et al., 2019)

The trend towards more versatile and generalisable models culminated in the development of foundation models for CV tasks, including instance segmentation. A prime example of this is the Segment Anything Model (SAM) (Kirillov et al., 2023) introduced by Meta AI in 2023. SAM represents a significant leap forward in the field, offering a prompt-able segmentation model that can generate masks for any object in an image based on various types of prompts, including points, boxes, or free-form text.

SAM's architecture combines a powerful image encoder based on the ViT with a prompt encoder and a lightweight mask decoder. This design allows the model to understand and segment objects it has never seen during training, effectively performing zero-shot segmentation. The model's versatility extends to its ability to handle different types of segmentation tasks, from semantic segmentation to instance semantic segmentation and even open-vocabulary segmentation.

The introduction of SAM and similar foundation models marks a paradigm shift in image segmentation. Instead of training specialised models for specific datasets or tasks, these models offer a more general-purpose approach to image understanding. They can be easily adapted to new domains or specific applications through fine-tuning or prompt engineering, significantly reducing the need for large, task-specific datasets and extensive training times. For example, CNOS (CAD-based Novel Object Segmentation) (Nguyen et al., 2023) utilises SAM and built a baseline for novel object segmentation and has been used in many novel object 6D pose estimation algorithms.

This evolution towards more flexible and generalisable models aligns with the trends in object detection algorithms, where the focus is shifting from narrow, task-specific models to more general-purpose AI systems. In the context of image segmentation, this means moving beyond pixel-wise classification to a more holistic understanding of image content, where the model can interpret and segment scenes based on high-level concepts and natural language descriptions.

2.1.3. Object 6D Pose Estimation

Object 6D pose estimation is a vital task in CV, robotics, and AR, aiming to determine both the 3D position and 3D orientation of objects in a scene (B. Wang et al., 2020), illustrated in Figure 2–3. This field has seen significant advancements over the past decade, driven by the increasing demand for accurate object manipulation in robotics and precise object placement in AR applications.

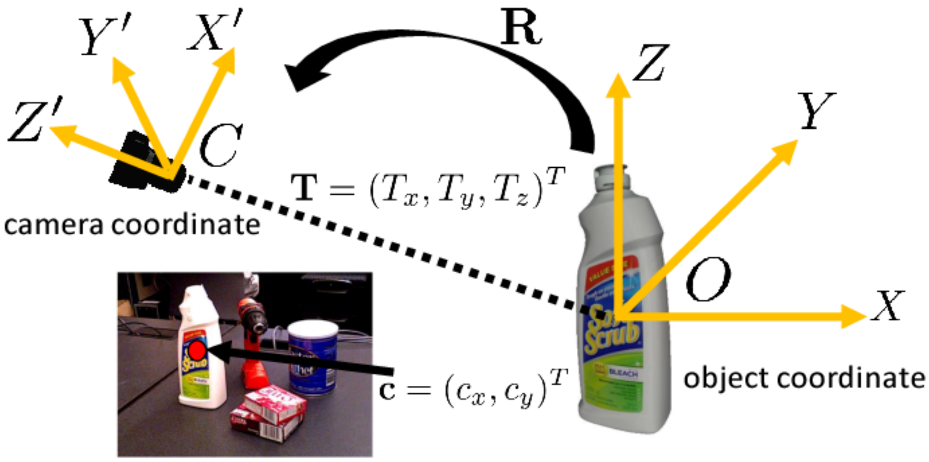

Figure 2–3: An illustration of the 6D pose estimation task (Xiang et al., 2018)

Traditional approaches to 6D pose estimation often relied on feature matching techniques. These methods typically involved extracting local features from both the input image and 3D object models, followed by establishing correspondences between these features. Algorithms like SIFT (Scale-Invariant Feature Transform) (Lowe, 2004) and SURF (Speeded Up Robust Features) (Bay et al., 2006) were commonly used for feature extraction. Once correspondences were established, techniques such as RANSAC (Random Sample Consensus) (Fischler & Bolles, 1981) were employed to estimate the 6D pose. While these methods could be effective in certain scenarios, they often struggled with texture-less objects, occlusions, and varying lighting conditions.

The advent of deep learning brought about a paradigm shift in 6D pose estimation. One of the early influential works in this direction was PoseCNN (Xiang et al., 2018), which introduced a novel network architecture for estimating 6D object pose from a single RGB image. PoseCNN decoupled the pose estimation problem into separate tasks of translation estimation and rotation estimation, using a novel loss function that directly regressed 3D translation and a quaternion representation for rotation.

Following PoseCNN, numerous deep learning-based approaches emerged, each addressing different aspects of the 6D pose estimation problem. DPOD (Dense Pose Object Detector) (Zakharov et al., 2019) introduced a dense correspondence learning approach, where the network learned to predict UV maps and object masks, which were then used to establish 2D-

3D correspondences for pose estimation. This method showed improved performance, especially for symmetric objects.

Another significant development was the introduction of PVNet (Peng et al., 2019), which proposed a pixel-wise voting network for 6D pose estimation. PVNet predicts pixel-wise unit vectors pointing to predefined keypoints and uses these votes to localise the keypoints, even under occlusion and truncation. This approach demonstrated robust performance, particularly for occluded and truncated objects.

The integration of depth information alongside RGB data has also been a direction in 6D pose estimation. DenseFusion (C. Wang et al., 2019) takes this approach, proposing an end-to-end network that processes dense RGB-D data. The method extracts pixel-wise dense feature embeddings from the RGB image and fuses these features with geometric information from the depth image. This fusion-based approach showed superior performance compared to methods using either modality alone.

More recent works have focused on improving the accuracy and efficiency of 6D pose estimation. CDPN (Coordinates-based Disentangled Pose Network) (Zhigang Li et al., 2019) introduced a novel representation learning approach, disentangling an object's coordinates from its attributes. This method showed state-of-the-art performance on several benchmarks while maintaining real-time inference speeds.

The need for more efficient and lightweight models led to the development of EfficientPose (Bukschat & Vetter, 2020), which leverages the EfficientNet (Tan & Le, 2019) architecture as a backbone for 6D pose estimation. EfficientPose demonstrated competitive accuracy while significantly reducing the model size and inference time, making it suitable for deployment on edge devices.

In the realm of real-time applications, CosyPose (Labbé et al., 2020) introduced a multi-view approach for 6D object pose estimation. By leveraging temporal information from video sequences, CosyPose achieves high accuracy while maintaining real-time performance, making it suitable for robotic manipulation tasks.

Most recently, transformer-based architectures have made their way into 6D pose estimation. TransPose (X. Lin et al., 2024), inspired by the success of transformers in various computer vision tasks, applies a transformer-based architecture to capture global context for more accurate pose estimation. This approach has shown promising results, particularly in handling complex scenes with multiple objects.

As the field continues to evolve, there is an increasing focus on developing methods that can generalise well across different object categories and handle more challenging scenarios such as cluttered environments and partially occluded objects. Researchers are also exploring ways to reduce the reliance on large, annotated datasets, which can be time-consuming and expensive to create for 6D pose estimation tasks.

2.1.4. Novel Object 6D Pose Estimation

Novel object 6D pose estimation is a subset of object pose estimation, which has emerged as a crucial research direction in CV and robotics. This field aims to estimate the 6D pose of objects that were never seen during training, addressing the challenges proposed by increasingly time-consuming and expensive for creating dataset for and train 6D pose estimation algorithms. The approaches in this domain can be broadly categorised into category-level and instance-level methods, each with its own set of challenges and solutions.

Category-level methods aim to estimate the pose of objects belonging to known categories but with unseen instances. These approaches leverage the shared characteristics within object categories to generalise to novel objects. One of the pioneering works in this area is NOCS (Normalized Object Coordinate Space) (H. Wang et al., 2019), it introduced a category-level canonical representation for objects, allowing the network to learn a shared object-centric coordinate system for each category. This approach enabled the estimation of 6D pose and size for unseen object instances within known categories.

Building upon NOCS, SPD (Shape Prior Deformation) introduced a shape prior deformation network (Tian et al., 2020). SPD learns to deform a category-level shape prior to match the observed depth map, effectively handling intra-class shape variations. This method showed improved performance, especially for objects with significant shape differences within the same category. CASS (Category-level Articulated Object Pose and Shape Estimation) extended the category-level approach to articulated objects (X. Li et al., 2020). CASS learns a canonical part-based representation for articulated object categories, enabling the estimation of pose and joint parameters for unseen articulated objects. DualPoseNet (J. Lin et al., 2021) introduced a two-stream network architecture for category-level pose estimation. One stream estimates the object's pose in a category-specific canonical space, while the other predicts the transformation from the canonical space to the camera space. This dual-stream approach improved the handling of intra-class variations. SAR-Net (Self-supervised Autoencoding Rotation) (H. Lin et al., 2022) proposed a self-supervised learning approach for category-level pose estimation. By leveraging unlabelled data and a novel autoencoding rotation framework, it demonstrates improved generalisation to unseen objects.

Instance-level methods focus on estimating the pose of completely unseen objects, often without prior knowledge of their category or shape. This allows the instance-level method can generalise to novel objects in unseen categories. These approaches typically rely on object reconstruction or CAD models.

DeepIM is one of the earlier significant contributions to the field (Y. Li et al., 2018). It introduced a method of using a neural network to iteratively refine initial pose estimates based on rendered image. Although not specifically designed for novel objects, it is demonstrated that can be applied to novel objects. This iterative approach laid important groundwork for future developments in pose estimation, providing a framework for progressively improving pose predictions.

Apart from methods that require CAD models, reconstruct object from reference images and then estimate pose with query image is an alternative approach. Thus, algorithms like Gen6D are proposed. However, reconstruct from refences images has a significant drawback in many applications, which is the data collection for reference images. In some cases, reference images are more time-consuming or harder to acquire than CAD models. Thus, in recent research, CAD model based novel object are more adopted.

Multiple works have explored template matching as a core technique. The general idea is to render many synthetic views of the object's CAD model and match the input image against these templates to estimate the pose. However, the way of how the rendered templates is used differs in difference methods. Nguyen et al. proposed a straightforward solution for matching templates. For each object, they generate 2,536 viewpoints and 36 in-plane rotations for each rendered image, yielding 92,232 templates (Nguyen et al., 2022). A query image is matched to the template with the highest similarity using a trained CNN. While straightforward, this method consumes significant memory for each object, limiting its application when dealing with a large collection of objects. OSOP (Shugurov et al., 2022) proposes dense matching of 2D-2D correspondences between query images and templates. It then projects 2D keypoints to 3D to refine the pose using PnP and RANSAC. This approach allows for more precise pose estimation. GigaPose (Nguyen et al., 2024) uses templates to identify 4 DoF of the object, then calculates the remaining in-plane rotation and translation using 2D-2D correspondences. This approach saves the effort of finding 2D-3D correspondences and improves efficiency. MegaPose (Labbé et al., 2022) proposed a fully machine-learned approach where rendered templates are iteratively generated to create increasingly closer representations of the query image. While accurate and memory-efficient, this method consumes significant time for online rendering. ZeroPose (J. Chen et al., 2023) proposed using templates only to identify and segment the objects, then calculates the 6D pose by matching point cloud information with the CAD model. This approach takes advantage of the robustness of point cloud data and offers a good balance between accuracy and efficiency.

2.1.5. Research Gaps

While object detection, segmentation, and pose estimation have developed concurrently, novel object 6D pose estimation has emerged as a comprehensive research field that encompasses object detection and segmentation as essential preliminary steps. These steps are used to isolate objects from the background, thereby improving estimation speed and accuracy. However, current novel 6D pose estimation algorithms face significant challenges in terms of generalisation and efficiency.

A major limitation of existing approaches is their reliance on methods like CNOS to identify novel objects. Which generates multiple proposals for each query image and compare them with all templates in a database. This approach, while functional, introduces substantial computational overhead and can limit both efficiency and accuracy, especially when dealing with large numbers of objects or in real-time applications.

Furthermore, the generalisation capability of current algorithms remains insufficient. Many methods struggle when confronted with objects that differ significantly from their training data in terms of shape, texture, or category. This lack of robust generalisation hinders the practical application of these algorithms in diverse, real-world scenarios where encountering truly novel objects is common.

2.2. Robot Trajectory Planning

Trajectory planning is a fundamental challenge in robotics, involving the computation of a collision-free route for a robot from a specified start point to a goal location within a given environment (B. Wang et al., 2020). This field has evolved significantly over time and researchers have developed a variety of algorithmic approaches to address this problem.

With the advent of AI, ML techniques have increasingly been applied to trajectory planning, opening new avenues for research and development. Within the ML domain, RL has emerged as a particularly powerful and popular approach for addressing trajectory planning challenges (Kober et al., 2013).

Given this evolution, this section will review the field of trajectory planning in three main parts. We will begin by examining traditional trajectory planning algorithms, which form the foundation of the field. Next, we will explore RL algorithms that can be used for robotic arm trajectory planning. Then, we will review the SOTA robot trajectory planning method using RL techniques, reflecting the current trend in AI-driven solutions. Finally, the research gap is identified to guide the research effort.

2.2.1. Traditional Trajectory Planning Algorithms

Traditional trajectory planning methods can be broadly categorised into two main approaches: offline planning and online planning. Offline planning assumes a static environment with complete information, while online planning deals with dynamic obstacles and partially known environments (Smierzchalski & Michalewicz, 2005).

Offline planning approach includes graph-based, sampling-based and optimisation-based methods. Graph-based methods discretise the configuration space and search for a path using graph search algorithms. A* (Hart et al., 1968) is widely used in robotic arm trajectory planning due to its efficiency and optimality guarantees. It uses a heuristic function to guide the search towards the goal, significantly reducing computational time compared to exhaustive searches. However, its performance degrades in high-dimensional spaces typical of multi-DOF robotic arms (Ferguson et al., 2005). While less efficient than A* for single-query problems, Dijkstra's algorithm (Dijkstra, 2022) is useful in multi-query scenarios where the same environment is queried repeatedly with different start and goal configurations. While theoretically sound, graph-based methods often struggle with the "curse of dimensionality" in high-DOF robotic arms, leading to prohibitive computational costs (Karaman & Frazzoli, 2011).

Sampling-based methods address the limitations of graph-based approaches in high-dimensional spaces by randomly sampling the configuration space. Probabilistic Roadmap (PRM) samples configurations and connects them to form a roadmap (Kavraki et al., 1996). It is particularly effective for multi-query problems in complex, static environments. Variations like Lazy PRM (Bohlin & Kavraki, 2000) have improved its efficiency for robotic arm planning. Rapidly-exploring Random Trees (RRT) grows a tree from the start configuration by randomly sampling and extending towards unexplored areas (LaValle, 1998). RRT-Connect (Kuffner & La Valle, 2000) grows trees from both start and goal, significantly improving performance for robotic arm planning. RRT* (Karaman & Frazzoli, 2011) is an asymptotically optimal version of RRT. It has shown impressive results in planning for high-DOF robotic arms, providing near-optimal paths given sufficient computational time. Sampling-based methods have become the de facto standard for offline planning in robotic arms due to their ability to handle high-dimensional spaces efficiently (Elbanhawi & Simic, 2014).

Optimisation-based methods formulate trajectory planning as an optimisation problem, seeking to minimise a cost function that typically includes factors such as path length, smoothness, obstacle avoidance, and sometimes task-specific criteria. As shown in Figure 2–4, optimisation-based methods usually require less computation time but compromise on planning success rate. Covariant Hamiltonian Optimisation for Motion Planning (CHOMP) (Zucker et al., 2013) uses functional gradient techniques to iteratively improve a trajectory, which performs simultaneous optimisation of smoothness and obstacle avoidance. Its effectiveness has been demonstrated on a 7-DOF arm in cluttered environments. Stochastic Trajectory Optimisation for Motion Planning (STOMP) (Kalakrishnan et al., 2011) uses stochastic optimisation to generate and refine trajectories that can handle non-differentiable cost functions and able to escape local optima due to its stochastic nature. TrajOpt (Schulman et al., 2016) formulates trajectory optimisation as a sequential quadratic programming problem. It uses convex decompositions of obstacles for efficient collision checking and demonstrated to have faster convergence compared to CHOMP in many scenarios.

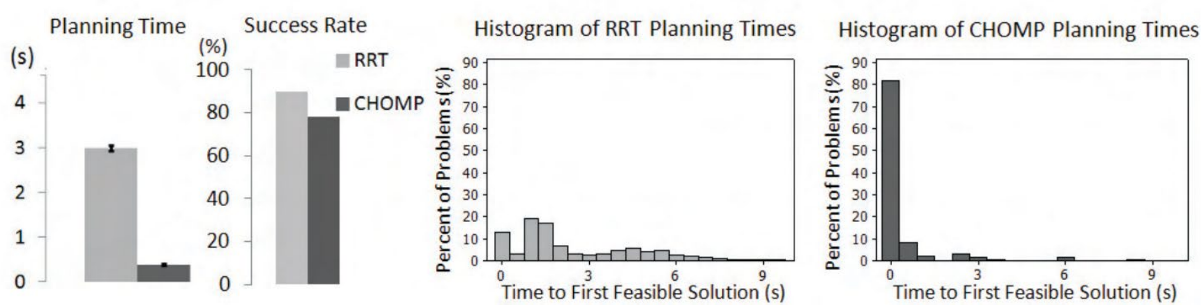

Figure 2–4: Planning time and success rate comparison between CHOMP and RRT (Zucker et al., 2013)

Offline methods for trajectory planning in robotics offer a global perspective, allowing for comprehensive analysis of the entire solution space. These methods excel in finding optimal paths in well-defined, static environments where all parameters are known in advance. By leveraging powerful computational resources without real-time constraints, offline planning can generate highly efficient and smooth trajectories for robotic arms. However, offline methods have significant drawbacks. Their reliance on complete environmental knowledge makes them inflexible in the face of unexpected changes. The time-intensive nature of their computations means they cannot quickly adapt to new situations. Furthermore, the solutions they provide, while optimal for the modelled environment, may become suboptimal or even invalid when real-world conditions deviate from the model. Online methods are critical for robotic arms operating in dynamic or uncertain environments, they often sacrifice optimality for real-time performance, making them suitable for reactive behaviours in dynamic environments.

Artificial Potential Fields (APF) treats the robot as a particle under the influence of artificial forces. The goal exerts an attractive force, while obstacles exert repulsive forces (Khatib, 1985). While computationally efficient, it suffers from local minima issues, particularly problematic for complex robotic arm configurations (Ge & Cui, 2002). Dynamic Window Approach (DWA) searches the velocity space of the robot (Fox et al., 1997). Although primarily developed for mobile robots, adaptations for robotic arms have been proposed, especially for human-robot collaboration scenarios (Flacco et al., 2012). Elastic Bands deform an initial path in response to moving obstacles (Quinlan & Khatib, 1993). This method has been successfully applied to robotic arm manipulation in dynamic environments (Brock, 2002). Incremental Trajectory Optimisation for Real-Time Replanning (ITOMP) interleaves planning and execution for real-time performance, which allows for real-time replanning in dynamic environments (Park et al., 2012).

Modern robotic systems often employ a combination of offline and online optimisation where initial trajectories may be optimised offline, then adjusted online using faster, simplified methods. This hybrid approach aims to combine the benefits of thorough offline planning with the adaptability of online methods. Anytime Repairing A* (ARA*) can quickly finds a suboptimal solution and iteratively improves it (Likhachev et al., 2003). This approach is particularly useful for robotic arms that need to start moving quickly but can refine their path during execution. PRM with local replanning allows for efficient path adaptation in partially dynamic environments. This approach has shown promise in robotic arm applications where parts of the environment remain static while others change (Jaillet & Siméon, 2004).

Traditional trajectory planning methods have laid the foundation for robotic arm motion planning. Offline methods, particularly sampling-based approaches like RRT and its variants, have proven highly effective for complex, high-DOF robotic arms in static environments.

Online methods, while less optimal, provide necessary reactivity for dynamic scenarios. However, these traditional methods face challenges in highly dynamic or uncertain environments, especially when real-time performance is critical (B. Wang et al., 2020). They also often struggle to incorporate complex constraints or optimise for multiple objectives simultaneously, which are common requirements in advanced robotic arm applications. These limitations have motivated the development of learning-based approaches, including RL methods, which promise to address some of these challenges by learning from experience and adapting to new situations more flexibly. Nonetheless, traditional methods remain relevant, often serving as building blocks or benchmark comparisons for newer approaches.

2.2.2. Reinforcement Learning Algorithms

RL has emerged as a powerful paradigm in ML, enabling agents to learn optimal behaviours through interaction with their environment. To review the current state-of-the-art of trajectory planning algorithms enabled by RL, a variety of RL algorithms need to be reviewed first. The RL algorithms can be divided into two categories: on-policy and off-policy methods.

On-policy methods learn and improve the policy that is currently being used for action selection. They are well-suited for online learning scenarios where the agent needs to learn and improve in real-time as it interacts with the environment. This makes them valuable in robotics and other real-world applications where continuous adaptation is necessary. The journey of on-policy methods began with simple algorithms like REINFORCE (Willia, 1992). While groundbreaking, REINFORCE suffered from high variance in gradient estimates, leading to unstable learning. This limitation sparked research into more sophisticated approaches.

To address the high variance issue, researchers introduced baseline methods. The Actor-Critic architecture (Konda & Tsitsiklis, 2003) emerged as a significant improvement, using a value function as a baseline to reduce variance in policy gradient estimates. However, these methods still struggled with sample efficiency and stability in complex environments.

A major leap forward came with the introduction of trust region methods. Trust Region Policy Optimisation (TRPO) was a landmark algorithm that enforced a constraint on the size of policy updates (Schulman et al., 2015). This constraint helped maintain stable learning by preventing drastic changes to the policy. TRPO showed impressive performance on a range of continuous control tasks but was computationally expensive and complex to implement.

Building on the insights from TRPO, Proximal Policy Optimisation (PPO) simplified the approach while maintaining performance (Schulman et al., 2017). PPO uses a clipped objective function to discourage large policy changes, achieving similar stability to TRPO but with greater simplicity and better sample efficiency. Due to its ease of implementation and robust performance, PPO has become one of the most popular on-policy algorithms.

A significant breakthrough in on-policy methods came with the introduction of Asynchronous Advantage Actor-Critic (A3C) (Mnih et al., 2016). A3C addressed two major challenges in deep reinforcement learning: the need for expensive GPU hardware and the lack of diversity in experience data. A3C utilises multiple agent instances running in parallel on separate threads of a CPU. Each agent interacts with its own copy of the environment, accumulating gradients over several steps before applying them asynchronously to a global network. This approach offers three advantages: improved stability, increased efficiency, and better exploration.

Off-policy methods can reuse past experiences, even if they were generated by a different policy. The learning is more efficient particularly when data collection is expensive or time-consuming. The off-policy journey began with Q-learning (Watkins & Dayan, 1992), which learned value functions from arbitrary data. However, Q-learning struggled with function approximation in continuous state and action spaces.

A significant breakthrough came with the Deep Q-Network (DQN) (Mnih et al., 2013), which first successfully combined Q-learning with deep neural networks and started the era of Deep Reinforcement Learning (DRL). DQN introduced experience replay and target networks to stabilise learning, achieving human-level performance on Atari games. However, DQN was limited to discrete action spaces.

To handle continuous action spaces, algorithms like Deep Deterministic Policy Gradient (DDPG) (Lillicrap et al., 2016). extended the ideas from DQN. DDPG used an actor-critic architecture with deterministic policy gradients, enabling off-policy learning in continuous domains. However, DDPG could be unstable and sensitive to hyperparameters. Addressing the instabilities in DDPG, Twin Delayed DDPG (TD3) (Fujimoto et al., 2018) introduced three key improvements: clipped double Q-learning to reduce overestimation bias, delayed policy updates, and target policy smoothing. These changes significantly improved stability and performance.

Normalized Advantage Functions (NAF) (Gu et al., 2016) offered an alternative approach by representing Q-functions in a special form that allowed for easy maximisation over actions. While innovative, NAF's performance was often surpassed by other methods.

A major advancement came with Soft Actor-Critic (SAC) (Haarnoja et al., 2018). SAC introduced the concept of entropy regularisation, encouraging exploration by maximising both expected return and entropy. This approach led to improved exploration and robustness. As illustrated in Figure 2–5, SAC has shown state-of-the-art performance on many continuous control benchmarks, often surpassing both on-policy methods like PPO and other off-policy methods.

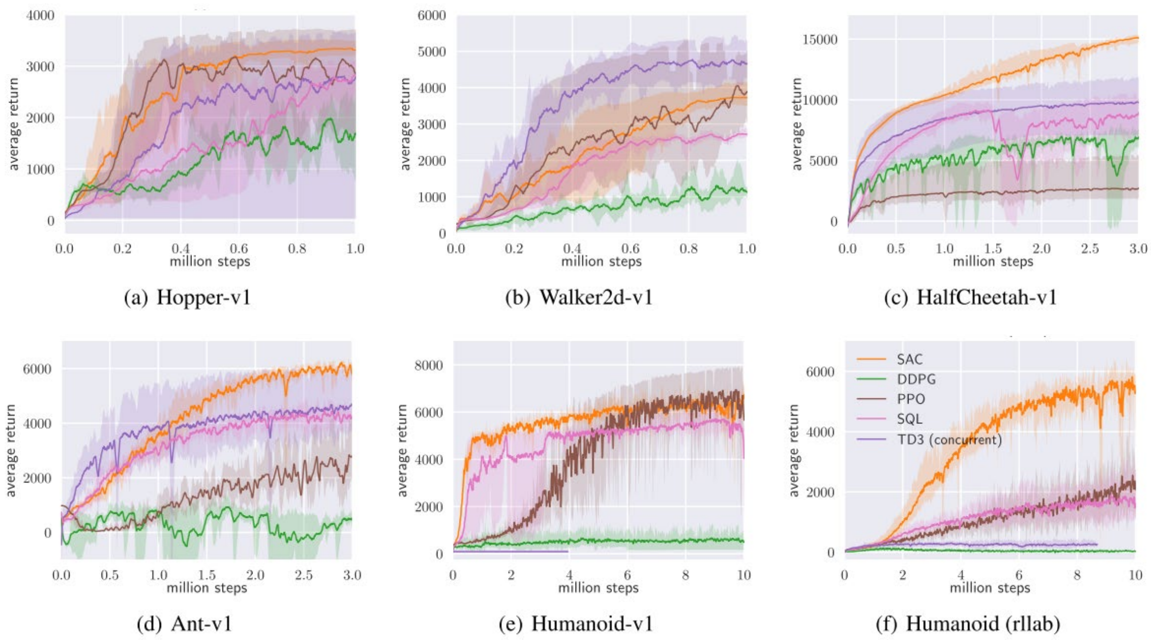

Figure 2–5: Benchmarking DRL algorithms across six scenarios (Haarnoja et al., 2018)

2.2.3. Deep Reinforcement Learning for Trajectory Planning

In the field of DRL aided manipulator trajectory planning and control, researchers have explored various environmental setups to develop collision avoidance strategies. These setups typically involve a fixed robotic manipulator learning to reach a target position or pose while avoiding moving obstacles. The complexity of these scenarios can be categorised based on the degrees of freedom (DoF) of the obstacle movement, ranging from simple linear motion to more complex trajectories in 3D space, shown in Figure 2–6.

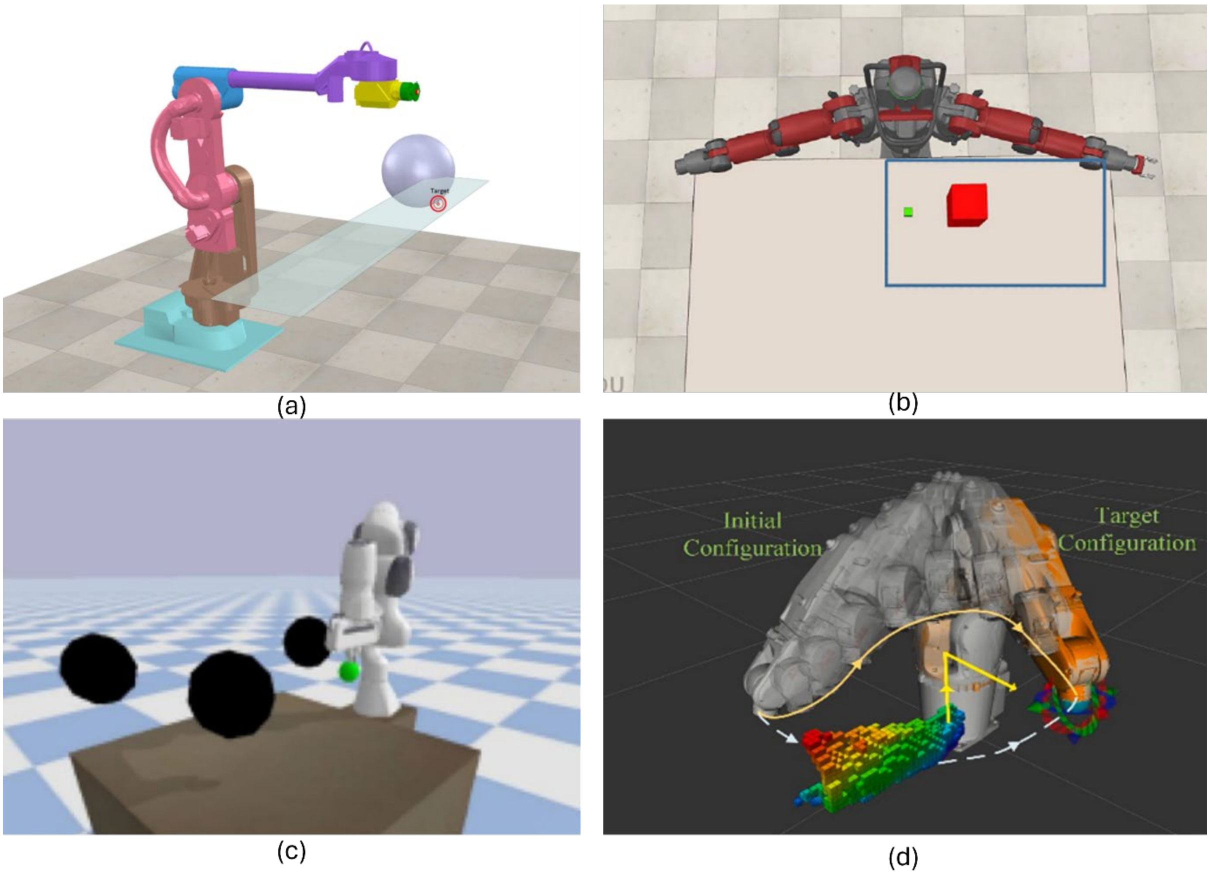

Figure 2–6: Robot trajectory planning under different obstacle states: (a) 1D moving obstacle (Sangiovanni et al., 2018) *(b) 2D moving obstacle* (P. Chen et al., 2022) *(c) 3D environment with 2DoF obstacle* (L. Chen et al., 2022) *(d) 3D free moving obstacle* (Q. Liu et al., 2021)

At the simplest level, the scenario where an obstacle move periodically along a linear track was investigated (L. Chen et al., 2022; Sangiovanni et al., 2018). This one-dimensional motion constraint simplifies the problem, making it an ideal starting point for developing and testing trajectory planning algorithms enabled by DRL. The predictable nature of the obstacle's movement allows for more straightforward modelling and training of the robotic system.

Increasing in complexity, the two-dimensional setting where obstacles can move freely across a tabletop surface was explored (P. Chen et al., 2022). This setup more closely mimics real-world scenarios, such as collaborative workspaces where objects or human hands might move unpredictably on a shared work surface. The additional degree of freedom introduces more challenges in obstacle prediction and avoidance strategies.

Further elevating the difficulty, some studies have positioned multiple obstacles in three-dimensional space while still restricting their movement to a planar domain (L. Chen et al., 2022; El-Shamouty et al., 2020). While the obstacles move in a 2D plane, their elevated position requires the manipulator to consider vertical clearance in its trajectory planning. The performance of DDPG and SAC under the same scenario was tested and compared on a 7DoF robotic arm (Q. Liu et al., 2021). The result shows SAC can significantly outperform DDPG in the high DoF and continuous space setting.

As research advances to 3D free-moving obstacles, the scenario becomes more challenging. This setting best fits the HRC context, where a human arm sharing the same workspace with a collaborative robot can be viewed as a free-moving 3D obstacle. The simplest approach is to monitor and model the human hand as a point obstacle moving in space.

Zindler et al. developed a hybrid approach where the manipulator's global trajectory is pre-planned and dynamically adjusted using a DRL-learned policy to maintain a safe distance from the human hand (Zindler et al., 2022). Their method, trained exclusively in simulation, transferred well to real robots and was tested with both precise motion capture systems and less stable skeleton tracking. Li et al. combined the APF method with DRL, introducing concepts of Distance and Force reinforcement factors (H. Li et al., 2021). This hybrid approach demonstrated superior performance compared to traditional APF, especially for dynamic obstacles and those near targets. Liu et al. adopted a fully learned approach, where the robot learns every action towards the goal through trial-and-error (Q. Liu et al., 2021). Their proposed Intrinsic Reward-DDPG method introduced a novel reward structure comprising both extrinsic and intrinsic components, aiming to maximise the expected accumulated fitness over time. This approach not only achieved real-time free-moving object avoidance but also tackled a more complex obstacle representation. Unlike simpler point-based models, they represented the human arm as a cylindrical object moving in space. This more realistic modelling presents additional challenges, as it requires the system to account for the rotational movement of the obstacle, significantly increasing the complexity of the avoidance task.

These advancements collectively contribute to creating more robust, flexible, and efficient obstacle avoidance systems capable of handling the complexities of real-world dynamic environments in HRC scenarios.

2.2.4. Research Gaps

While current research in robot trajectory planning has made significant advance using DRL, a critical gap remains in the realm of proactive obstacle avoidance. Existing methods predominantly focus on reactive real-time avoidance, where the robot responds to obstacles as they are detected. However, the vision of smart manufacturing and truly symbiotic cobot control demands a higher level of cognition from robotic systems. To achieve this, robots need to transition from merely reacting to actively avoid human movements and potential obstacles. This proactive approach would not only increase working efficiency by minimising sudden trajectory changes but also significantly reduce collision risks by planning ahead. Current DRL based trajectory planning methods, while powerful and successful, lack the predictive capabilities necessary for this level of cognitive interaction. Developing robot trajectory planning algorithms that can learn to forecast human actions and environmental changes, rather than simply respond to them, represents an essential next step in realising the full potential of smart manufacturing and creating genuinely collaborative human-robot workspaces. This shift from reactive to proactive planning could revolutionise cobot control, enabling smoother, safer, and more efficient interactions between humans and robots in shared workspaces.

2.3. Large Language Models in Robotic Operations

The integration of Large Language Models (LLMs) into robotic operations marks a significant paradigm shift, extending their capabilities far beyond their origins in natural language processing (Jeong et al., 2024). This review synthesises recent advancements, methodologies, and persistent challenges in this field. LLMs are demonstrably transforming robotics by enabling more intuitive human-robot interaction, sophisticated task planning, and enhanced perceptual understanding through multimodal integration (Jiaqi Wang et al., 2024). Methodologically, the field is characterised by the rise of Vision-Language Models (VLMs), diverse strategies for grounding language in physical actions, including symbolic methods, embedding-based techniques, and hybrid approaches, as well as innovative learning paradigms such as Learning from Demonstration (LfD) and RL guided by rewards generated LLMs (Han et al., 2025). Numerous specialised frameworks have emerged, targeting applications in robotic manipulation, navigation, multi-robot systems, and complex task execution (X. Xiao et al., 2025). Ultimately, LLMs are paving the way for more intelligent, adaptable, and collaborative robotic systems, with profound implications for various industries and the pursuit of artificial general intelligence.

2.3.1. The Evolving Role of LLMs

As LLMs have progressed, they have moved from handling purely textual data to engaging with multimodal inputs, such as audiovisual signals, and crucially, to formulating robot actions. This shift has enabled LLMs to act as a bridge between abstract human language and concrete robotic behaviours, fostering a new era of "embodied intelligence." (Y. Ma et al., 2024) In this context, artificial agents can perceive, learn, reason, and act within dynamic real-world or simulated environments, a development fundamental to the emergence of more autonomous and adaptable robotic systems.

Recent advances highlight the ability of LLMs not only to process and generate natural language but also to interpret complex instructions and generate appropriate robotic responses. This multimodal processing capability is essential for effective robotic operation, allowing robots to interact with their environments in a more holistic and human-like manner (Kim et al., 2024). By connecting human intent with actionable commands, LLMs are transforming the design, control, and utilisation of robots across a diverse array of applications.

A significant advantage of LLMs lies in their generalisation ability, which allows robotic systems to adapt to novel scenarios, environmental changes, or entirely new tasks with minimal reprogramming (Huang, Abbeel, et al., 2022). This flexibility is invaluable for deploying robots in dynamic and unpredictable real-world settings. The extensive training of LLMs on vast datasets enables them to understand context, reason about the world, and translate high-level human instructions into concrete robotic actions. As a result, advanced robotic systems are becoming more accessible and versatile.

One of the most profound impacts of LLMs is the potential "democratisation" of robot instruction (H. Xiao & Wang, 2023). Traditionally, programming and operating robots required specialised technical knowledge and complex coding skills. LLMs, however, excel at interpreting natural language, making it possible for operators without robotics expertise to communicate with and direct robots using everyday language. This shift marks a move from expert-driven control paradigms to more accessible and user-friendly interaction models. The broader implication is that robotics technology is becoming available to a much wider range of users, which could significantly accelerate its adoption across new domains and industries (Driess et al., 2023).

Moreover, the role of LLMs within robotic systems is evolving beyond isolated task execution. There is a growing trend towards LLMs functioning as a foundational, almost "operating system-like" layer within robotic architectures (Jeong et al., 2024). While early applications focused on specific tasks, such as language understanding, current research demonstrates that LLMs can now contribute across the entire spectrum of core robotics functions, including communication, perception, planning, and control. For example, frameworks such as LLMBot are being developed to enable robots to interpret language, plan future actions, and execute these plans with a level of cognitive sophistication previously unseen (Ibrahim et al., 2025).

The concept of an "LLM as a Robotic Brain" reinforces the idea that LLMs are maturing into a central reasoning and interface layer, responsible for coordinating the interplay between various robotic sub-systems and translating abstract goals into executable actions. This architectural shift carries profound implications for the future of intelligent robot design and deployment.

2.3.2. Foundational Capabilities of LLMs for Robotic Systems

The successful integration of LLMs into robotics relies on several foundational capabilities that these models provide. Chief among these is their proficiency in natural language understanding, commonsense reasoning, and their adaptability through in-context learning, all of which are proving instrumental in advancing both robotic autonomy and human-robot interaction.

A primary contribution of LLMs to robotics is their ability to serve as highly intuitive natural language interfaces. This capability is revolutionising how humans interact with robots, and how robots communicate with each other (Ahn et al., 2022). LLMs enable far more fluid and efficient exchanges, moving beyond the predefined, rigid command structures and protocols that have historically constrained robotic systems. As a result, robots can now share high-level information in a manner that is far more natural and accessible.

Crucially, this natural language interface empowers operators, regardless of whether they have specialist robotics expertise to guide robots through complex missions using everyday language. This accessibility greatly expands the potential user base for advanced robotic technologies (Kim et al., 2024). In the field of human-robot interaction (HRI), LLMs facilitate more natural and engaging exchanges, enabling robots to understand nuanced instructions, respond appropriately to queries, and generate clear explanations for their actions and decisions. For instance, the TalkWithMachines (Abbas & Beleznai, 2024) framework has been developed to leverage both LLMs and VLMs to enhance interpretability and transparency in industrial robotic systems, allowing operators to better understand robot states and intentions.

Beyond their linguistic abilities, LLMs serve as powerful reasoning engines, a capability that is increasingly harnessed for complex robotic tasks (Huang, Xia, et al., 2022). They exhibit advanced reasoning and language comprehension skills, allowing them to formulate precise and efficient action plans based on natural language instructions. This includes parsing intricate contexts, resolving ambiguities, and understanding implicit information often omitted from explicit commands.

The capacity of LLMs to process and internalise vast amounts of textual data during pre-training endows them with a broad repository of commonsense knowledge and inferential abilities. Such reasoning is critical for robots operating in unstructured human environments, as it enables them to make plausible inferences about the world, object properties, and the likely outcomes of their actions. As a result, LLMs are increasingly effective in high-level task planning, where they can decompose complex, overarching goals into sequences of manageable sub-tasks (G. Wang et al., 2023).

A hallmark of modern LLMs is their remarkable capability for in-context learning, which supports strong zero-shot and few-shot learning abilities (Huang, Abbeel, et al., 2022). This means LLMs can often adapt to new tasks, environments, or contexts with minimal or even no task-specific training data, a valuable trait in robotics, where unpredictability and novel situations are the norm.

This inherent flexibility allows LLM-equipped robots to handle unforeseen events or new tasks by dynamically breaking them down into performable sub-steps, or by making informed inferences about novel elements (Liang et al., 2023). Methods such as instruction tuning and alignment tuning are increasingly used to adapt LLMs for specific robotic domains or objectives. Furthermore, providing few-shot prompts, short sets of examples that demonstrate desired behaviours, can significantly enhance the decision-making capabilities of LLMs in targeted robotic applications.

The commonsense reasoning and few-shot adaptation capabilities displayed by LLMs point to a deeper underlying mechanism. These models are trained on massive, internet-scale datasets comprising both text and code, encompassing not just linguistic patterns, but also extensive descriptions of the world, human activities, physical processes, and causal relationships (Bisk et al., 2020). Their ability to demonstrate commonsense reasoning and infer appropriate actions in novel situations suggests that LLMs acquire more than surface-level language skills; they seem to form an “implicit world model”. This internal representation encodes knowledge about objects, their properties, event sequences, and even basic physics (Plaat et al., 2024). In robotics, this means LLMs do not simply “parrot” learned text, but instead leverage this implicit model to inform reasoning, planning, and interaction strategies. This represents a significant step beyond simple pattern-matching and is a key driver of their growing utility in embodied AI.

2.3.3. Approaches to Integrating LLMs in Robotics

The integration of LLMs into robotic systems encompasses a wide range of methodological approaches. These span from leveraging multimodal inputs to address the symbol grounding problem, employing diverse machine learning paradigms, to refining interaction through advanced prompt engineering. This section examines these key methodologies and highlights frameworks that typify current research trends.

A critical advancement in applying LLMs to robotics is the development and adoption of multimodal capabilities. While text-only LLMs are highly effective for language understanding and generation, they inherently lack direct experiential exposure to the physical world. This limitation can restrict their effectiveness for tasks that demand real-world grounding and context-sensitive decision-making (Alayrac et al., 2022). To address this, researchers are increasingly integrating real-world sensory modalities, such as vision, audio, and haptic feedback into language models, thereby establishing more robust connections between abstract linguistic concepts and concrete physical perceptions.

Multimodal Large Language Models (MLLMs), often termed VLMs when vision is a primary modality, are designed to process and reason over information from multiple sources simultaneously. These models typically fuse sophisticated language understanding with visual processing capabilities, and sometimes incorporate action models to guide robotic behaviour . As a result, robots can understand the content of images or sensor streams and integrate this perceptual data with linguistic instructions or contextual knowledge. For instance, models such as GPT-4o are employed to break down high-level textual instructions into detailed task plans, while concurrently analysing visual information to interpret environmental changes depicted in images (Hurst et al., 2024). Similarly, frameworks like TalkWithMachines leverage VLMs to create more interpretable industrial robotic systems, in which visual data plays a central role in both task execution and system transparency (Abbas & Beleznai, 2024).

A fundamental challenge for LLM-driven robotics is the "symbol grounding problem." This problem involves connecting abstract symbols, such as words or linguistic constructs to a robot’s sensorimotor experiences and concrete actions within the physical world (Harnad, 1990). Without proper grounding, language commands risk remaining uninterpretable or ambiguously related to physical reality. Research in this area encompasses a broad spectrum of approaches, from highly structured symbolic methods to more flexible embedding-based techniques (Ibarz et al., 2021).

One avenue for symbol grounding is to map natural language commands to manually defined formal representations of meaning. These can include logical formalisms such as First-Order Logic (FOL) or Linear Temporal Logic (LTL), as well as structured planning languages like the Planning Domain Definition Language (PDDL). Such methods offer the advantage of precise meaning representation, reduce the complexity of the learning problem by providing a constrained symbolic space, and often support interpretability and formal safety guarantees. Increasingly, LLMs are used as translators, converting natural language instructions into these formal languages. For instance, frameworks like LLM+P use LLMs to generate PDDL representations from task descriptions, which can then be solved by classical symbolic planners(B. Liu et al., 2023). Another related approach involves LLMs generating executable code, such as Python or JavaScript that interacts with predefined robot APIs to effect actions, exemplified by the “Code as Policies” paradigm (Liang et al., 2023).

At the opposite end of the spectrum are embedding-based and end-to-end learning approaches. These methods aim to map natural language instructions and perceptual data (such as images or sensor readings) directly into high-dimensional vector spaces (embeddings). These embeddings can then be used to directly inform or generate low-level robot policies, such as target end-effector poses or joint-state commands. The main advantage of such approaches is their potential for greater generality and flexibility, as they bypass the need for manually specified symbolic structures. By learning direct mappings from input to action, these systems can, in principle, adapt to a wider range of tasks and environments—provided sufficient training data are available. However, these methods usually require significantly larger datasets and more computational resources for training compared to symbolic approaches. Notable systems employing such strategies include PaLM-E (Driess et al., 2023) and Perceiver-Actor (Shridhar et al., 2023).

Recognising the complementary strengths and limitations of purely symbolic and purely embedding-based methods, much recent work has focused on hybrid strategies that combine aspects of both. These approaches seek to capitalise on the precision and structure of symbolic representations while retaining the flexibility and learning capacity of embedding-based methods. For example, the SayCan framework maps natural language to a predefined vocabulary of robot skills, where each skill represents a learned policy but is treated as a discrete symbolic action at a higher level (Ahn et al., 2022). Another common hybrid approach involves LLMs generating intermediate representations that are neither fully symbolic plans nor direct low-level actions. These may include subgoals, high-level action descriptions, or reward functions, which are then used to guide a separate learning or planning module (Z. Wang et al., 2023).

Beyond symbol grounding, the integration of LLMs into robotics is characterised by the adaptation and application of various machine learning paradigms to facilitate skill acquisition and performance improvement. Learning from Demonstration (LfD), or imitation learning, offers an intuitive way to teach robots by "showing" them what to do rather than explicitly programming every step (Ravichandar et al., 2020). This approach significantly simplifies the robot training process. Language-conditioned imitation learning further refines this method, allowing human demonstrators to provide natural language instructions alongside physical demonstrations, thereby creating a richer learning signal that combines linguistic guidance with observed behaviour.

LLMs are increasingly important in automating the generation of demonstrations. For example, systems like GenSim (Lirui Wang et al., 2023) leverage LLMs to automatically generate diverse robotic simulation environments and produce expert demonstrations within these settings. However, such approaches may face limitations when applied to long-horizon tasks or those requiring precise awareness of the robot's state and environment. The Moto-GPT (Y. Chen et al., 2024) framework marks another significant advancement, with models pre-trained on vast quantities of unlabelled video data to learn generalisable motion priors, which can then be fine-tuned and transferred to robotic manipulation tasks.

RL represents another powerful paradigm for robot training, where agents learn optimal behaviours through trial and error, receiving feedback in the form of rewards or penalties. LLMs are increasingly integrated into RL workflows in innovative ways. A notable application is the use of LLMs to generate reward functions directly from natural language task descriptions. The EUREKA framework (Y. J. Ma et al., 2023), for example, employs GPT-4 to create reward functions for RL agents based on textual goals. LLMs can also generate subgoal descriptions that serve as prior knowledge or intermediate targets for RL policies, effectively decomposing complex tasks into more manageable learning problems. Furthermore, RL from Human Feedback (RLHF), a technique instrumental in aligning LLMs with human preferences in NLP is being adapted to further refine robot behaviours based on human evaluations (Bai et al., 2022).

The core capabilities of LLMs are largely derived from their initial pre-training phase, during which they learn from extensive and diverse corpora of text and code. This pre-training endows them with broad world knowledge, linguistic fluency, and general reasoning abilities (Jiaqi Wang et al., 2024). However, for specific robotic applications, these general-purpose models often require further adaptation via fine-tuning. Fine-tuning enables an LLM to be specialised for particular objectives, domains, or robotic embodiments using smaller, task-specific datasets. The Moto-GPT framework exemplifies this two-stage process: it first uses autoregressive pre-training on latent motion tokens extracted from video data to learn general motion understanding, and then applies a co-fine-tuning strategy to adapt these learned priors to specific robot manipulation tasks. Similarly, the ERRA system (C. Zhao et al., 2023) employs Soft Prompt Tuning (SPT), a parameter-efficient fine-tuning technique, to adapt LLMs for robot task planning, while preserving their general knowledge and reasoning capabilities acquired during pre-training.

How instructions and context are presented to LLMs, a practice known as prompt engineering is crucial for eliciting desired behaviours and achieving reliable performance in robotic applications. Effective prompting can guide the LLM's reasoning, incorporate relevant context, and constrain its outputs for more suitable robotic execution. Various prompt engineering techniques are utilised. Few-shot prompting, where a prompt includes a small number of input-output examples, can substantially improve an LLM's generalisation to similar new tasks. Chain-of-Thought (CoT) prompting encourages the LLM to generate intermediate reasoning steps before arriving at a final answer and has been shown to enhance performance on complex reasoning tasks relevant to planning and decision-making in robotics (Kojima et al., 2022). For specific robotic functions, structured prompts are often designed to elicit particular types of outputs, such as interactive grounding of language to objects or actions, scene graph generation from perceptual data, few-shot task planning, or code generation for reward functions. Achieving reliable and accurate outputs from advanced MLLMs like GPT-4o often requires carefully crafted and sometimes lengthy prompts, frequently demanding significant domain expertise (Kim et al., 2024).

The ongoing research into LLMs for robotics has given rise to a variety of specialised frameworks and architectures, each designed to tackle distinct challenges or enable particular capabilities. These systems frequently combine LLMs with other AI techniques and robotics principles to develop more effective solutions. One prominent trend is the adoption of hybrid and modular architectures. Many state-of-the-art frameworks are moving away from monolithic, end-to-end LLM approaches, instead integrating LLMs with other specialised components, such as symbolic planners (e.g., LLM+P), dedicated safety modules (e.g., ROBOGUARD) (Ravichandran et al., 2025), or multiple collaborating LLM agents with distinct roles within the system (P. Li et al., 2025). This modularity enables systems to harness the core strengths of LLMs, such as natural language understanding and commonsense reasoning, while mitigating known weaknesses (including hallucination, safety concerns, and limitations in long-horizon reasoning) by incorporating components that are more deterministic, verifiable, or specialised for critical functions like precise planning logic or safety enforcement. This shift reflects a maturation in system design towards more robust, practical, and reliable robotic solutions.

2.3.4. Research Gaps

Despite the transformative potential and rapid advances in applying LLMs to robotics, a range of significant challenges continue to impede their widespread, reliable, and safe deployment. These obstacles span fundamental technical limitations of the models themselves, persistent issues in evaluation and standardisation, and broader ethical and security concerns.

A pervasive challenge for both LLMs and especially MLLMs is their tendency to "hallucinate" or "confabulate", generating responses that may be inaccurate, factually incorrect, inconsistent with visual or sensory input, or entirely fabricated. Such hallucinations seriously undermine the reliability and trustworthiness of LLM-powered robotic systems, particularly in safety-critical contexts, where actions based on false information can result in severe consequences. The causes of hallucination are multifaceted. In MLLMs, they may stem from insufficient or noisy training data, biases inherent in the data, limitations of the vision model's perceptual ability, the language model's prior knowledge overruling current sensory evidence, misalignments in the integration of visual and textual features, or problems occurring during training or inference. For LLMs deployed with Retrieval-Augmented Generation (RAG), hallucinations can also arise from the retrieval process itself, such as retrieving irrelevant, outdated, or misleading documents or from errors in the generation stage, where the LLM might be misled by noisy or conflicting context, struggle with long-context reasoning (the so-called "middle curse"), misalign generation with the query, or reach the limits of its own capabilities.

Ensuring the safety, reliability, and robustness of LLM-driven robots is paramount, given their direct interaction with the physical world, where mistakes can lead to property damage or even harm to people. LLMs, by design, may generate plans or actions that are contextually plausible but physically unsafe, at times prioritising task completion over risk mitigation. Traditional robot safety verification and validation processes may fall short in addressing the new vulnerabilities introduced by the probabilistic and opaque nature of LLMs. In response, dedicated safety frameworks are emerging. Nevertheless, achieving consistent and fault-tolerant decision-making remains an unresolved challenge, as emergent abilities such as in-context learning are not yet fully predictable or reliably robust.

A further obstacle lies in the computational demands of large LLMs, which due to their extensive parameter counts can introduce significant inference latency. For robotic applications requiring real-time control, perception, and interaction, such delays can compromise both effectiveness and safety. Reliance on cloud-hosted, closed-source LLM APIs exacerbates this issue by adding network latency and API call limitations, which can hinder both the development of embedded robotic systems and their deployment in real-time commercial settings.

Addressing these performance constraints requires advances in model optimisation techniques such as quantisation (reducing the precision of model weights), pruning (removing less critical model parameters), and knowledge distillation (training smaller, more efficient models to replicate larger ones). The development of specialised hardware accelerators, such as GPUs and TPUs optimised for LLM inference, and the adoption of LLMOps (Large Language Model Operations) practices for efficient deployment and serving are also crucial steps towards achieving real-time performance in robotics.

2.4. Chapter Summary

This chapter provided a comprehensive review of the current state-of-the-art in traditional trajectory planning algorithms, RL algorithms, DRL enabled robot trajectory planning methods, object detection techniques, image segmentation algorithms, and 6D pose estimation methods. Identified research gaps with a particular focus on their applications in HRC and smart manufacturing environments.

In the field of robot trajectory planning, significant advancements have been made using DRL. Current methods excel at reactive, real-time obstacle avoidance, allowing robots to respond effectively to detected obstacles. However, a critical gap exists in proactive obstacle avoidance capabilities. The vision of smart manufacturing and truly symbiotic cobot control system demands a higher level of cognition from robotic systems, necessitating a transition from merely reacting to actively avoiding human movements and potential obstacles. This proactive approach would not only increase working efficiency by minimising sudden trajectory changes but also significantly reduce collision risks by planning ahead. Developing robot trajectory planning algorithms that can learn to forecast human actions and environmental changes, rather than simply respond to them, represents a crucial next step in realising the full potential of smart manufacturing and creating genuinely collaborative human-robot workspaces.

In the realm of object perception, this review revealed that while object detection, segmentation, and pose estimation have developed concurrently, novel object 6D pose estimation has emerged as a comprehensive research field that encompasses object detection and segmentation as essential preliminary steps. These steps are used to isolate objects from the background, thereby improving estimation speed and accuracy. However, current novel 6D pose estimation algorithms face significant challenges in terms of generalisation and efficiency. A major limitation of existing approaches is their reliance on computationally intensive methods like CNOS to identify novel objects, which can limit both efficiency and accuracy, especially when dealing with large numbers of objects or in real-time applications.

These findings highlight several key research opportunities. First, there is a need to develop proactive trajectory planning algorithms that can learn to predict and avoid potential obstacles and human movements. Second, improving the efficiency and generalisation capabilities of novel object 6D pose estimation algorithms is essential. Finally, integrating advanced trajectory planning with improved object detection and pose estimation could significantly enhance cobot control in smart manufacturing environments.

In conclusion, this literature review underscores the significant progress made in robot trajectory planning and object perception, while also identifying key areas for future research. Addressing these gaps could lead to more intelligent, efficient, and collaborative robotic systems, ultimately advancing the field of cobot control system and smart manufacturing. The transition from reactive to proactive planning, coupled with more generalised and efficient object perception, has the potential to revolutionise cobot control, enabling smoother, safer, and more efficient interactions between humans and robots in shared workspaces.

Chapter 3 Development of the HRC Workstation

The HRC workstation is the shared workspace where human and robot can work collaboratively together. Development of the physical and digital HRC workstation is essential before enabling it with machine-learned algorithms. This first research objective establishes the foundation for the entire study, with the physical and digital systems working as a foundation to achieve subsequent goals.

This chapter provides a detailed account of the process of developing the HRC workstation, which comprises a shared working table for human and robot interaction, a UR5e robotic arm, and various other components. The chapter starts with introduces all the physical components used to construct the HRC workstation, describing their functionality and roles within the system. The discussion then delves into the implementation details of the physical and digital workstation, as well as the communication between two systems. Following the system's construction, the overall system architecture is presented, illustrating the machine learning enabled system that is aimed to achieve in this project. Then the chapter describes the development of a simple, yet effective testing application designed to verify the system's basic functionality. This practical demonstration serves to validate the digital model's fidelity to its physical counterpart. At the end, the chapter concludes with a comprehensive summary that discusses the significance of this work.

3.1. System Components

In this section, the components that used to build the physical collaborative industrial robot system are introduced. These components were selected carefully to make the system safe, efficient and intelligent.

3.1.1. UR5e Cobot

The UR5e cobot stands as the cornerstone of this collaborative robotics project in the LISMS lab. The UR5e offers configurable safety settings, providing flexibility in its operational parameters. Users can adjust power and speed limits to suit specific tasks or work environments, further enhancing the robot's adaptability and safety profile.

Building upon this safety foundation, the UR5e has improved precision and sensitivity, due to its higher resolution position sensors. These refinements allow for more accurate operations, particularly when handling small objects, thus expanding the robot's potential applications. Complementing these technical advancements is the UR5e's intuitive programming interface. This user-friendly system supports hand-guided programming, allowing operators to physically manipulate the robot to teach new movements and tasks, thereby streamlining the programming process. These combined features – improved safety, precision, programmability, and flexibility, make the UR5e an ideal choice for the collaborative robot system being developed in this project.

3.1.2. Human-robot shared workbench

The human-robot shared workbench is a key element in the human-robot interaction setup, custom-designed using Vention's platform. This approach allows for a tailored solution that precisely meets the project's requirements while benefiting from Vention's comprehensive parts supply and installation guidance.

The workbench design features a spacious layout measuring approximately two meters in length and 90 centimeters in width. This configuration is strategically planned to optimise the collaboration between human operators and the UR5e cobot. Two pillars, positioned on the left and right sides of the workbench, provide mounting points for monitors, cameras, and other necessary equipment, enhancing the system's versatility and monitoring capabilities. Central to the design is the placement of the UR5e cobot on the rear side of the workbench, mounted on a linear track. This positioning is carefully considered to complement the cobot's 85-centimeter reach. When a human operator stands at the front of the workbench facing the robot, their head naturally falls outside the cobot's working envelope. This thoughtful arrangement limits potential collision points to the operator's arms, significantly enhancing safety in the shared workspace.

An additional advantage of utilising Vention's design platform is the availability of exact CAD file for the workbench. This file ensures precise fabrication of the workbench components and can be integrated into the digital system. By incorporating the accurate digital representations, the system can be configured to avoid collisions between the cobot and the workbench structure. This integration of physical design and digital control further enhances the safety and efficiency of the collaborative workspace, demonstrating a holistic approach to HRC design.

3.1.3. Other Key Components

The collaborative robot system incorporates several key components beyond the UR5e cobot and the customised workbench, each playing a critical role in realising effective human-robot collaborative manufacturing.

Mounted on the UR5e cobot, the Azure Kinect camera, as shown in Figure 3–1(a), serves as a versatile sensory input device. This advanced camera system can capture visual data in three distinct formats: RGB, depth, and infrared (IR). The depth sensing capability is particularly valuable, as it enables precise measurement of distances between the camera and objects in its field of view. This feature proves instrumental in two critical applications: object 6D pose estimation and human pose recognition. Such capabilities are fundamental to ensuring safe and efficient human-robot interaction, allowing the system to accurately perceive and respond to its environment and the human operator's movements. Additionally, the Azure Kinect's built-in microphone array opens up possibilities for voice-based human-robot interaction, allowing operators to issue voice commands to the cobot.

The Vention linear track depicted in Figure 3–1(b) is mounted on the rear side of the workbench, significantly expands the cobot's operational range by providing an additional 855 mm of linear movement. While the two-meter-long workbench offers enough collaborative space, it also creates areas that would otherwise be beyond the cobot's reach. In a human-robot collaborative manufacturing scenario, the cobot often needs to pick and place parts for the human operator, including from locations outside the immediate workbench area.

Figure 3–1:Additional sensors and actuators in the collaborative robot system

The linear track solves this problem by extending the cobot's working envelope. This extension allows the cobot to interact with areas beyond the confines of the workbench, a capability that becomes particularly important when integrating additional equipment into the manufacturing process. For instance, the extended reach facilitated by the linear track enables the cobot to interact with a conveyor belt system, which can be used to transport parts from other machines to the HRC workstation.

The Robotiq FT300-S force and torque sensor, illustrated in Figure 3–1(c), is a vital addition to the system. While the UR5e cobot comes equipped with built-in torque sensors in each joint, these sensors have a limitation: their data cannot be accessed by external software. To overcome this constraint, the FT300-S sensor is installed on the robot's end-effector mount. This placement allows for precise measurement of forces and torques exerted on the end-effector in all three spatial dimensions.

The significance of the FT300-S sensor lies in its potential to enhance human-robot interaction. When properly programmed, this sensor enables the cobot to interpret human intentions by associating specific force directions or magnitudes with particular commands. This capability opens up new possibilities for intuitive and natural interaction between the human operator and the cobot. For example, an operator can signal task success or failure to the cobot by simply applying an upward or downward force.

The Robotiq 2F85 parallel gripper shown in Figure 3–1(d) is a two-finger gripper with a maximum opening width of 85mm. While it doesn't match the dexterity of a human hand, the gripper is well-suited for a wide range of pick-and-place tasks commonly encountered in human-robot collaborative scenarios. The gripper's design strikes a balance between versatility and simplicity. Its parallel grip motion allows it to securely handle a variety of object shapes and sizes, making it adaptable to different manufacturing tasks.

3.2. Implementation Details

In the previous section, the components that used to make the collaborative robot system are introduced. In this section, how they were integrated together are explained in detail. The integration can be conceptualised as a three-step process: constructing the physical system, developing the digital system, and establishing bidirectional communication between the physical and digital realms.

3.2.1. Physical System Construction

In Figure 3–2, the HRC workstation built in LISMS lab is presented. It is assembled with the workbench serving as its foundation. Constructed from blue steel frames, the workbench supports a fixed linear track on its rear side. This track enables the cobot to move horizontally, significantly expanding its operational range.

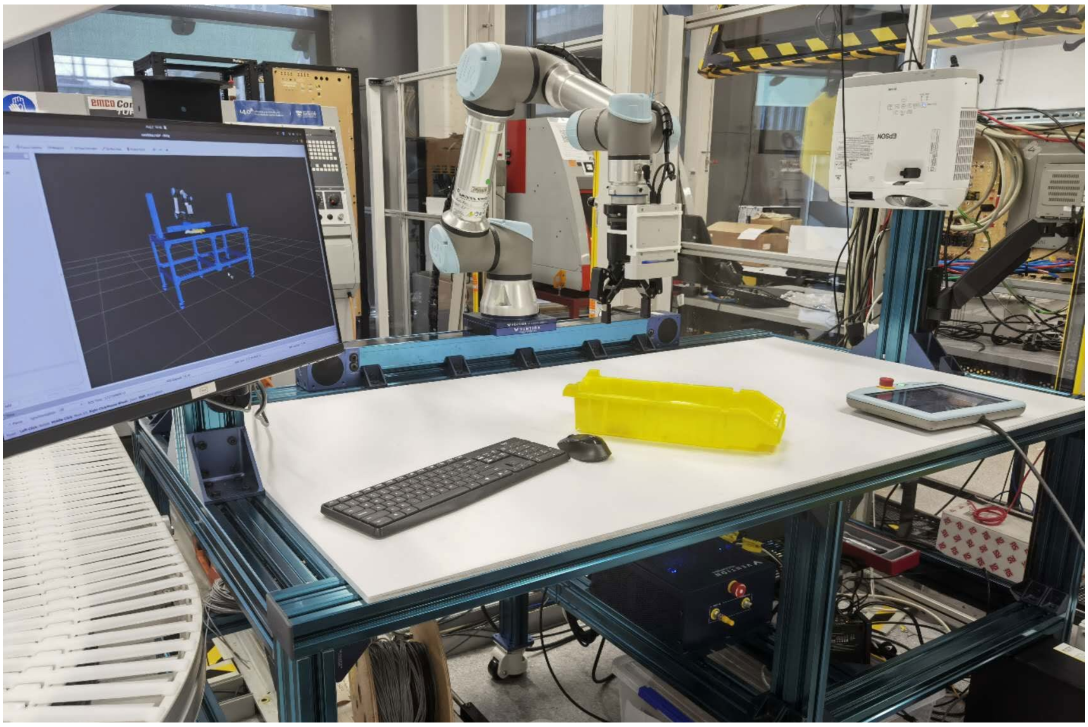

Figure 3–2: The HRC workstation

Mounted atop the linear track, the cobot is equipped with a force-torque sensor at its end-effector attachment point. This sensor ensures precise detection of all forces applied to the camera or gripper. A custom 3D-printed holder secures the camera above the sensor and gripper, positioned to prevent the gripper's fingers from obstructing the camera's field of view (FOV).

At the heart of the workstation is a desktop PC, functioning as the central hub for control and communication. This computer controls the operations of the cobot, linear track, and gripper while processing feedback data from these devices, as well as input from the camera and sensor. This integrated approach allows for real-time decision-making and system management.

To provide visual feedback to the human operator, a monitor is installed on the pillar. This display offers a real-time view of both the physical system's status and its digital counterpart, ensuring seamless interaction between the human operator and the robotic components.

3.2.2. Digital System Development

The digitalised HRC workstation must precisely mirror its physical counterpart. This digital replica demands accuracy in size, length, and orientation of all components to prevent system failures. The creation of this digitalised system relies on precise modelling of each system component. The workbench has been manufactured from its original CAD file, can be directly imported into the digital system without modification. For other off-the-shelf components, their respective CAD files are readily available from the manufacturers' official websites.

In the virtual environment, these individual components are integrated using the Unified Robot Description Format (URDF) file. The URDF file defines the spatial relationships between each part and its "parent" component. The implemented URDF file is provided in Appendix I for reference.

With this file and the established communication between the physical and digital systems, the entire digitalised workstation can be visualised and manipulated in software like Rviz, as illustrated in Figure 3–3. Furthermore, the digital system incorporates camera data visualisation capabilities. For instance, a yellow box placed on the physical workbench can be used to demonstrate how colour and depth information captured by the camera is accurately represented in the digital system.

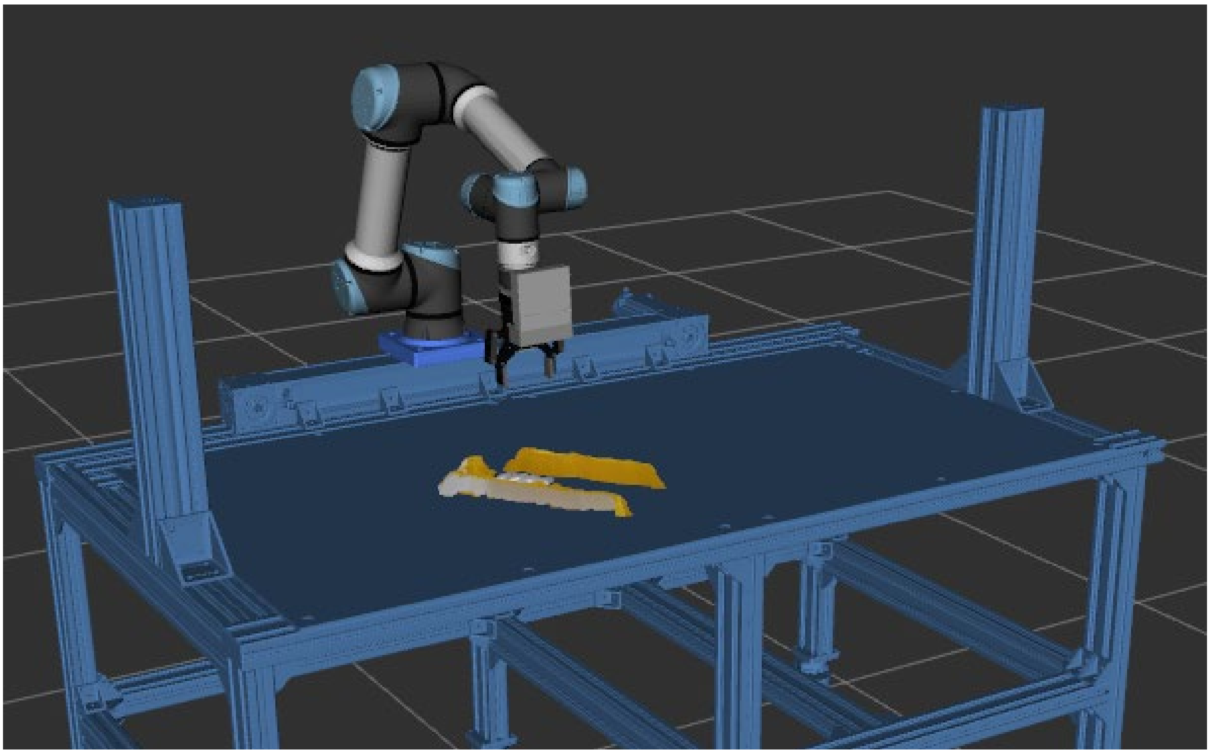

Figure 3–3: Digitalised HRC workstation

3.2.3. System Communication

The communication system of this project is built upon the Robot Operating System (ROS), a framework that runs on top of existing operating systems. Despite its name, ROS is not a traditional operating system but rather a comprehensive toolkit designed to simplify the creation of complex and robust robot behaviour across various robotic platforms. It offers a range of features, including a message-passing platform, package management, hardware abstraction, and libraries for common functionality.

In this project, we leverage ROS's message-passing platform to facilitate communication between the physical and digital systems. This platform operates on the concept of publishers and listeners interacting through topics. Any process within ROS can publish to or listen to a topic, subject to the topic's specified message type. This design ensures the independence and robustness of each process.

As shown in Figure 3–4, ROS topics are hosted on a central controlling compute. Each physical component publishes its sensor data to a designated topic. This data serves two primary purposes. For movable parts such as the gripper, cobot, and linear track, the data is used to update their status in the digital system at a constant frequency. Additionally, both this data and information from non-movable components like cameras and force-torque sensors feed into the central controlling algorithm, enabling real-time decision-making.

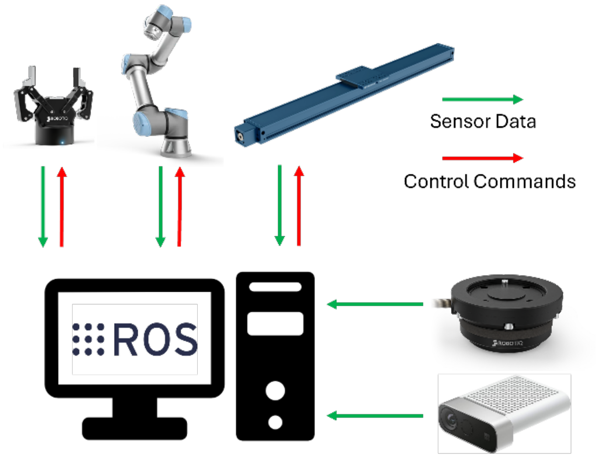

Figure 3–4: Communication system architecture

Once the controlling algorithm processes the incoming data and makes decisions, it publishes control commands to topics monitored by the movable parts. These components then execute the commands and provide immediate feedback, creating a responsive and dynamic system.

3.3. System Architecture

The design of modern human-robot collaboration (HRC) workstations has become a central focus in both research and industry, evolving from traditional segregated robot cells to shared environments where humans and robots work together in close proximity. Early reviews (Robla-Gomez et al., 2017) highlighted the move towards fenceless, flexible workstations, along with the associated safety challenges that arise when humans and robots share the same space.

More recently, the concept of “Symbiotic Human-Robot Collaboration” (SHRC) has gained prominence, placing greater emphasis on seamless and adaptive integration of human and robotic capabilities within a single workspace. Although collaborative robotics can enhance productivity and safety, many current HRC systems continue to face difficulties in coping with dynamic human behaviours and rapidly changing conditions (Barravecchia et al., 2025). These limitations have, in turn, restricted the full potential of collaborative workstations.

In response, researchers are increasingly identifying key features necessary for truly symbiotic HRC workstations. These include real-time adaptability to both the human and task context, intuitive and natural modes of interaction, and effective shared control between the human and the robot. New design methodologies are being proposed to ensure these features are embedded in practice.

The direction of recent literature is clear: next-generation HRC workstations must prioritise safety, flexibility, and a high degree of responsiveness to their human partners. By embracing these principles, the field is moving towards workspaces where humans and robots can cooperate efficiently, safely, and adaptively.

This project aims to develop a machine-learned control system for collaborative industrial robots, building upon the foundation of the developed HRC workstation. The overall architecture of the robot control system, which heavily incorporates machine-learned algorithms, consists of a loop execution of four essential sub-systems.

As illustrated in Figure 3–5, the process begins with machine-learned task analysis, where the system assesses the current environment and determines the next task. Once confirmed, the related objects and actions are identified. The machine-learned object pose estimation system then recognizes and locates the object to be manipulated within the 3D environment. This information is subsequently passed to the machine-learned trajectory planning system, which devises a collision-free path for the robot to reach its destination. The final step, robot trajectory execution, is handled by the robot's internal motor controller using traditional methods, as machine learning is not necessary for this stage. Upon completion of the trajectory, the current task concludes, and the robot returns to the task analysis step to begin the next cycle.

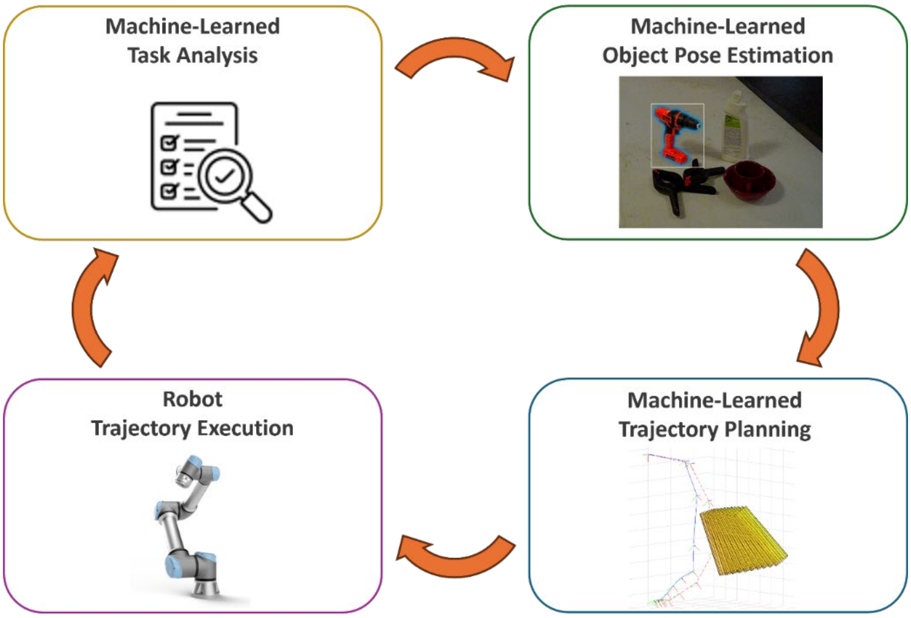

Figure 3–5: The machine-learned collaborative industrial robot control system architecture

The focus of this project's machine-learned system development is on object pose estimation and trajectory planning, which are detailed in Chapter 4 and Chapter 5, respectively. While task analysis methods are not extensively studied due to their job-specific nature and lack of a universal solution, two machine-learned task analysis methods based on state-of-the-art algorithms are developed to validate the overall workflow, as described in Chapter 6.

3.4. System Testing Application

To validate the constructed physical and digital system, making sure their functions can support the machine-learned control systems that will be developed, a simple application was implemented to simulate an industrial collaborative manufacturing task. This testing scenario focused on human-robot collaborative packaging and quality checking, mirroring real-world smart manufacturing environments where product specifications frequently change, and quality assessments often require human intervention due to their complexity or cost-prohibitive automation.

The application showcased a human operator sharing an HRC workbench with a cobot to complete product transferring, checking, packaging, and storage tasks. The workpiece in this scenario is a CNC-processed aluminium tube. The entire process, illustrated in Figure 3–6, comprises six stages, each represented by three scenes: a real-time status display of the digitalised system, a real-life record of human-cobot interactions, and the cobot's vision system output.

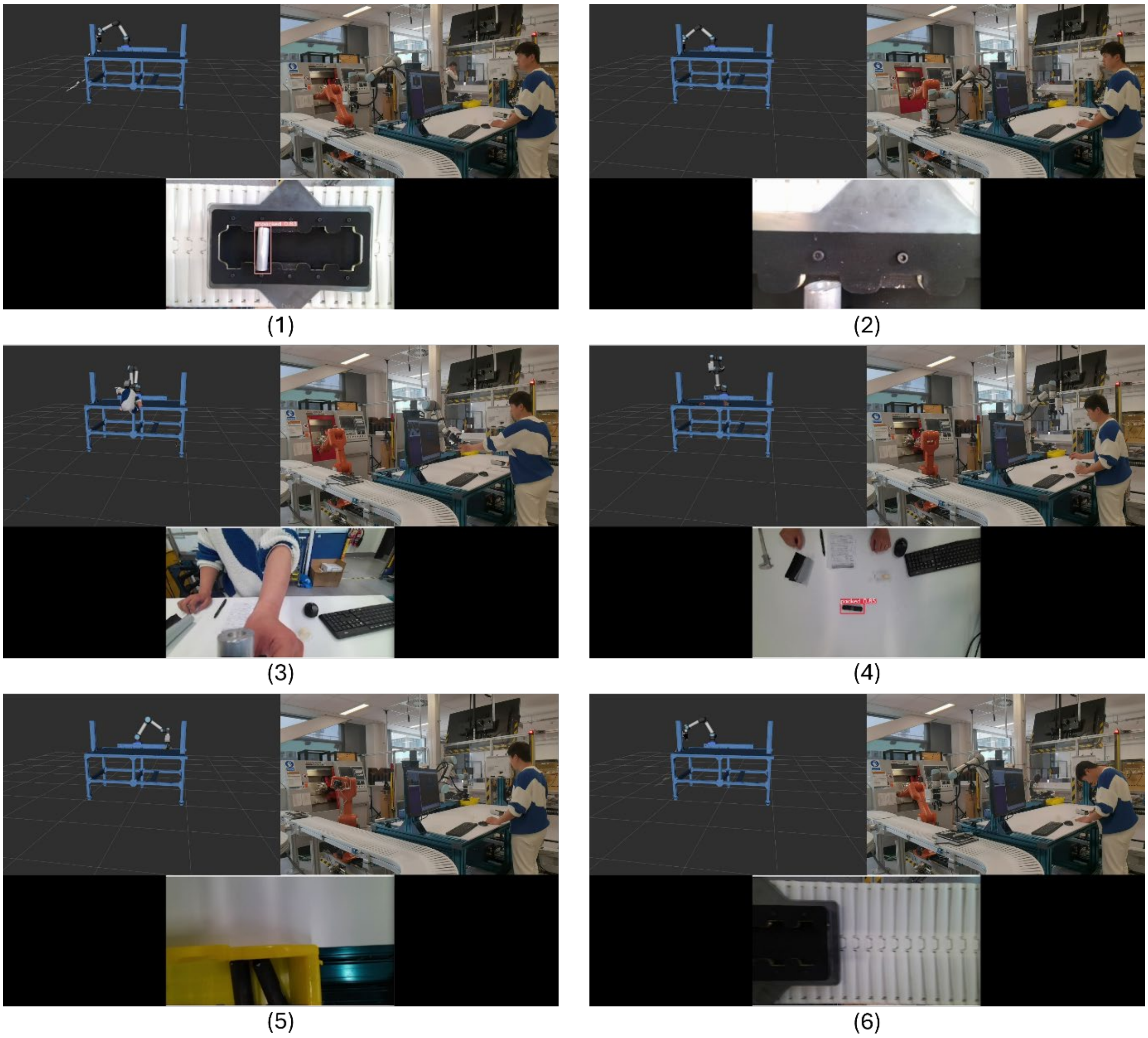

Figure 3–6: Process of the system testing application

The real-time status display, shown on a monitor during operation, provides data feedback from various system components, thereby enhancing human trust in the robot. The real-life record offers an intuitive illustration of human-cobot interaction. The vision system display powered by a YOLOv5 object detection algorithm forms the core of the cobot's smart decision-making system.

For accurate workpiece manipulation, the cobot calculates three key parameters from the camera capture: 2D center coordinates, planar rotation magnitude, and object-to-camera vertical distance. The 2D center coordinates is derived from the object detection bounding box, with the workpiece's symmetry ensuring its center aligns with the box's center. Using known camera intrinsics, the system then calculates the workpiece's 2D center coordinates on the holding plate.

The planar rotation of the object is also determined using the bounding box information. After detection, the workpiece is isolated from the background through segmentation and conceptualised as a rectangular block. Since the minimal rectangular bounding box's edges always align parallel to the image border, it can be used as a fixed reference frame. The rotation angle of the workpiece is then derived by measuring the angle between the segmented block and the bounding box. Finally, the vertical distance from the camera to the workpiece is measured using the depth sensor reading.

Figure 3–6 illustrates the six stages of the human-robot collaborative manufacturing process. Each stage is detailed below:

(1) Workpiece detection: The cobot positions itself above the conveyor belt, awaiting the transfer of the processed workpiece from the CNC processing station to the HRC workstation. Upon detection and stabilisation of the workpiece, the system calculates its 2D center coordinates, planar rotation magnitude, and vertical distance from the camera. Although the cobot's central position on the workbench initially places the conveyor belt out of reach, the addition of a linear track extends its working envelope, ensures task completion.

(2) Workpiece pick up: Following location calculation, the system communicates the target coordinates to the cobot via ROS topic. The cobot then plans and executes a trajectory to the target. Upon reaching the destination, it signals the gripper to close and secure the workpiece.

(3) Workpiece handover: After the gripper closes, it signals the cobot and linear track to move in tandem to the handover position. Once in position, the system begins analysing force/torque sensor readings. It first records the force and torque generated by the workpiece's weight, then awaits changes in all directions. When a human operator touches the workpiece, triggering sensor reading changes, the gripper opens to transfer the workpiece.

(4) Workpiece checking and packaging: The human operator takes over, inspecting the workpiece's appearance and dimensions before packaging it in black packing paper. During this process, the cobot assumes an observation position, elevating its camera to encompass the entire workbench within its FOV. This provides space for the human to work while monitoring the packaged workpiece. Once the packaged workpiece is placed on the workbench, the cobot recognises it and proceeds with localisation and picking, similar to the process described in step (1) and (2).

(5) Workpiece storage: The cobot transports the packaged workpiece to the storage area. It positions the workpiece slightly above the yellow storage box before releasing the gripper, allowing the package to drop into place.

(6) Return to beginning: The cobot and linear track return to their initial observation position above the conveyor belt, ready to receive the next incoming workpiece. This marks the completion of one full cycle of the process.

This comprehensive application demonstrates the integration of advanced vision systems, collaborative robotics, and human expertise in a smart manufacturing environment, highlighting the potential and effectiveness of the constructed digital and physical HRC workstation. For contributing to the society, providing ideas to other researchers and demonstrating the project to more audience, all the code, files and building procedures of this system is open-sourced, available on GitHub https://github.com/WanqingXia/HRC_robot.

3.5. Chapter Summary

This study presents the development and implementation of a comprehensive HRC workstation, integrating essential components to enable advanced sensing and acting capabilities. The HRC workstation incorporates a range of key components, each carefully selected and integrated to optimise performance. Sensory systems, including RGB-D cameras and force sensors, provide comprehensive environmental awareness. Actuators, including 6DoF cobot, linear track and parallel gripper, are used for precision and adaptability. High-performance processing units enable real-time data analysis and decision-making. Each component's role and integration within the system is detailly reported, offering a clear understanding of the workstation's architecture and functionality. This thorough documentation serves as a valuable resource for future developments and adaptations of the system.

Another important aspect of the work involved the digitalisation of the physical workstation. This is achieved by creating a highly accurate digital counterpart and establishing a robust communication framework between the physical and digital systems through ROS (Robot Operating System) topics. This bidirectional data flow enables real-time monitoring, simulation, and optimisation of the workstation's performance.

The collaborative industrial robot control system developed in this project represents a significant advancement in human-robot collaboration for manufacturing. Its key features of adaptability and generalisability, powered by cutting-edge ML algorithms, showcase the potential for future manufacturing scenarios that demand mass personalisation. By integrating advanced object pose estimation, trajectory planning, and task analysis methods, the system demonstrates remarkable flexibility in handling diverse tasks and environments. This adaptability is essential for modern manufacturing processes that require frequent changes in production lines and product specifications.

To demonstrate the practical applications and capabilities of the developed HRC workstation, a test case scenario focusing on human-robot collaborative quality checking and product packaging was conducted. This application showcases real-time coordination between human workers and robotic systems, adaptive task allocation based on real-time sensory input, enhanced quality control through the combination of human expertise and robotic precision, and improved efficiency in packaging processes.

The success of this test case highlights the potential for widespread adoption of such collaborative robot control systems in real-world industrial environments. It demonstrates how collaborative robotics can enhance productivity, ensure consistent quality, and improve workplace ergonomics.

Through the testing application, significant advantages of utilising the system were observed, encompassing four key areas: Enhanced Perception, Synergistic Collaboration, Safe Innovation, and Continuous Improvement. These areas represent the core capabilities enabled by the integration of physical and digital systems.

Enhanced Perception allows for a comprehensive real-time visualisation of the system's status. This capability transcends the limitations of physical observation, providing human operators with access to rich, multidimensional data streams. Operators can now perceive intricate details of robotic operations, environmental conditions, and process parameters that would otherwise remain invisible. This heightened awareness facilitates more informed decision-making and enables proactive system management.

Synergistic Collaboration fosters a new level of human-robot interaction, building a foundation of trust and efficiency. The digital system provides humans with a preview of the robot's intended actions, demystifying its operation and enhancing operator confidence. Simultaneously, the robot gains the ability to sense and interpret human actions, leveraging the digital system to make intelligent decisions and predictions. This bidirectional flow of information creates a harmonious working environment where human intuition and robotic precision complement each other.

Safe Innovation represents a paradigm shift in system development and testing. The digital system serves as a sandbox for implementing and rigorously testing new algorithms and strategies. This approach eliminates the need to interrupt physical operations, significantly reducing downtime and potential safety risks. Engineers and developers can push the boundaries of innovation, exploring novel approaches and optimisations in a risk-free virtual environment before deploying them to the physical system.

Continuous Improvement embodies the system's capacity for self-evolution. The digital component acts as a sophisticated data collection and analysis hub, gathering operational data from the physical system. This wealth of information undergoes thorough analysis, uncovering insights into system performance, efficiency bottlenecks, and potential areas for enhancement. The digital system thus becomes a catalyst for ongoing refinement, driving iterative improvements that keep the entire system at the cutting edge of performance and capability.

Chapter 4
Novel Object 6D Pose Estimation

6D pose estimation is key components in several applications including robot manipulation. Accurate 6D pose estimation of novel objects is essential for enhancing the adaptability and effectiveness of collaborative robots. This chapter introduces a novel multi-stage pipeline for 6D pose estimation of novel objects. The proposed method addresses current research gaps in existing approaches, which are limited by their object isolation strategies, accuracy, generalisation ability, and long execution time.

To thoroughly explore this new approach, the chapter is structured into four main sections. This chapter begin with an overview of 6D pose estimation, discussing why it is critical for this project, highlighting the challenges faced by existing methods and the need for improvement. This is followed by a detailed description of the proposed multi-stage pipeline, explaining its architecture and the rationale behind each stage. The third section presents the experimental setup and results, providing a comprehensive evaluation of the pipeline's performance in terms of accuracy, generalisation, and computational efficiency. At the end, this chapter concludes with a summary that synthesises the research findings and discusses the implications of this work.

4.1. Overview

Object 6D pose estimation is a one of the key tasks in CV and robotics, involving the determination of an object's position and orientation in three-dimensional space. This capability is essential for various applications like AR and autonomous driving, but particularly important in robot manipulation, where precise understanding of an object's pose enables robots to interact with their environment effectively. In manufacturing, warehouse automation, and even household robotics, accurate 6D pose estimation allows robots to grasp, move, and manipulate objects with precision.

Traditionally, 6D pose estimation methods have relied on learning-based approaches, training specialised neural networks on specific object datasets. While effective for known objects, these approaches face significant limitations when encountering novel items. For example, PoseCNN (Xiang et al., 2018) proposed a CNN-based pipeline for pose estimation and published the YCB-Video dataset. Later methods like FFB6D (Y. He et al., 2021), trained and tested on the YCB-Video dataset, achieved a score of 96.6 points. These approaches, while often groundbreaking and demonstrating excellent accuracy on trained datasets, lack generalisability to new objects.

This limited generalisability creates two critical problems: the need for extensive time-consuming training and the demand for large datasets. The time-consuming nature of training neural networks is unavoidable. When new objects become available, retraining the entire network is the only option, significantly increasing the time and computational resources required. Creating comprehensive 6D pose estimation datasets for each new item poses significant barriers to scalability and real-world application. Unlike image classification, detection, or segmentation tasks, the ground truth values for 6D pose cannot be evaluated by the human eye alone. It often requires expert knowledge and extensive measurements or specifically designed object positioning to obtain accurate ground truth data (Michel et al., 2018). This limitation becomes increasingly apparent as the demand for flexible and adaptive robotic systems grows in smart manufacturing.

In recent years, the field of ML has witnessed a paradigm shift towards more generalised models capable of handling a wider range of tasks and scenarios. This trend has extended to the domain of 6D pose estimation, where researchers are now focusing on developing algorithms that can estimate poses for novel objects without extensive retraining or dataset creation.

This shift towards novel object 6D pose estimation represents a significant advancement in the field. Recent algorithms, such as MegaPose (Labbé et al., 2022) and ZeroPose (J. Chen et al., 2023), exemplify this new approach. These methods often utilise CAD models to generate object templates—rendered images of the object from different viewpoints—and then recover the 6D pose with the help of these object templates. In this process, neural networks are still used but no longer embed object-specific features. Instead, they embed generalised techniques such as aligning the object in the image with the templates. Novel object pose estimation also faces significant challenges. Generalising to new objects requires adaptability to a large variability in shape, texture, and lighting conditions, as well as the ability to handle severe occlusions that can be encountered in real-world applications.

As research in novel object 6D pose estimation continues to advance, it holds the potential to revolutionise robotic manipulation across numerous applications, from flexible manufacturing to assistive technologies. The ongoing development of these generalised approaches not only addresses current limitations but also paves the way for more versatile and intelligent robotic systems capable of understanding and interacting with their environment in increasingly sophisticated ways.

Existing novel object 6D pose estimation methods can be simplified into two main steps: object detection and pose alignment. Since the CAD models are typically available, the first step is to detect and recognise the required object in the image. After the object is identified, its pose can be calculated by either matching it to the CAD model or to the generated templates. The performance of these methods often hinges on the object detection step. CNOS (CAD-based Novel Object Segmentation) (Nguyen et al., 2023) is a widely used method to detect, classify, and segment objects from images. It encodes templates of all objects in database into matrices, calculating the similarity between these encoded templates and mask proposals (generated by SAM) to identify objects. For each detection, all proposed masks and encoded templates need to be compared in pairs. When there is a large number of objects and proposals, both the detection speed and accuracy can be significantly affected.

In this work, a novel pipeline is proposed, aiming to address the challenges in novel object 6D pose estimation. The proposed three-stage method encompasses object detection and segmentation, mask selection, and pose estimation. Compared to existing methods, the proposed approach features better generalisability when facing a large number of objects in the database and demonstrates improved accuracy.

4.2. Proposed Method

In this work, a three-stage method is proposed, the proposed method employs four pre-trained neural networks, combined seamlessly to finish the pipeline from a query image and an object name to its 6D pose in the image. As shown in Figure 4–1, the three stages of the proposed method are detection + segmentation, mask selection and 6D pose estimation. Each stage is introduced in detail in the following sections.

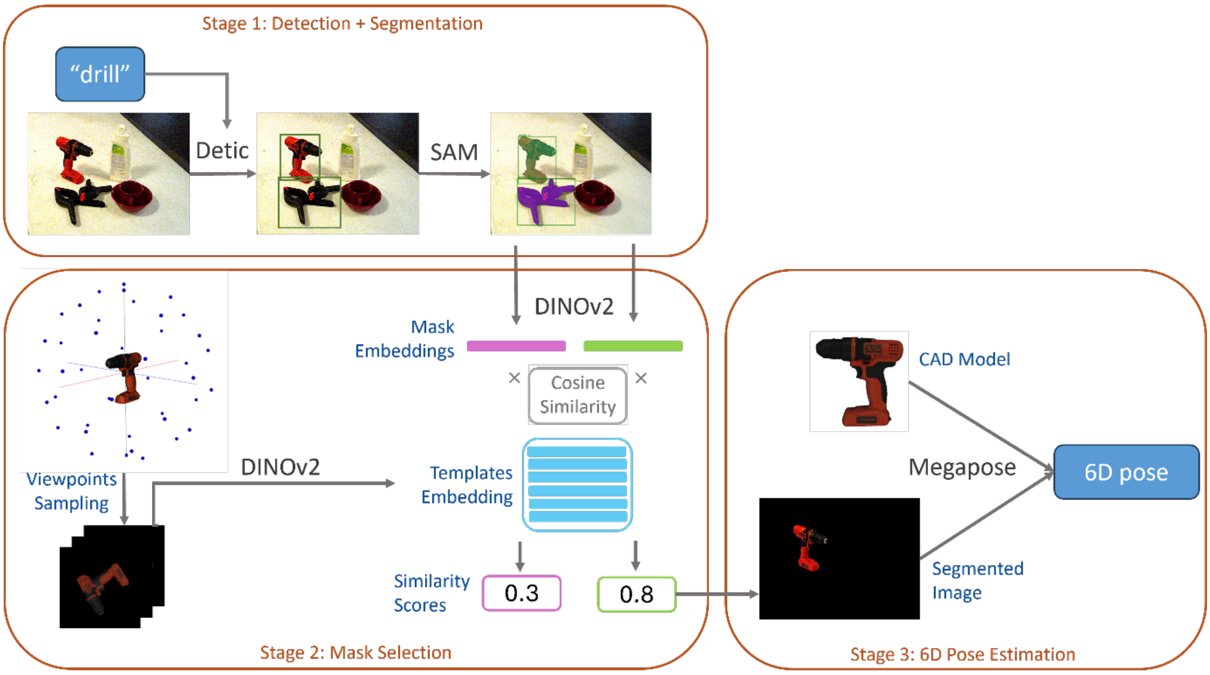

Figure 4–1: The proposed three-staged system architecture

4.2.1. Detection and Segmentation

The proposed pipeline begins with the first stage: identifying the object requiring pose estimation within the given image. This task is accomplished through the combined use of two powerful tools: Detic and SAM. Detic is a pre-trained object detector that can detect any class of object when provided with descriptive class names. To leverage this capability, each object in the database is assigned a descriptive name. When presented with an input image and the name of the target object, Detic predicts and generates bounding boxes around instances of that object within the image.

Following Detic, SAM takes over to perform segmentation. Using the bounding boxes provided by Detic, SAM isolates the object from its background, effectively separating it from the rest of the image. This two-step process is necessary because SAM, while highly effective at segmentation, cannot perform object-level segmentation on its own. As illustrated in Figure

4–2, without the guiding bounding boxes from Detic, SAM tends to segment objects at a more refined level than is suitable for the subsequent steps in the pipeline.

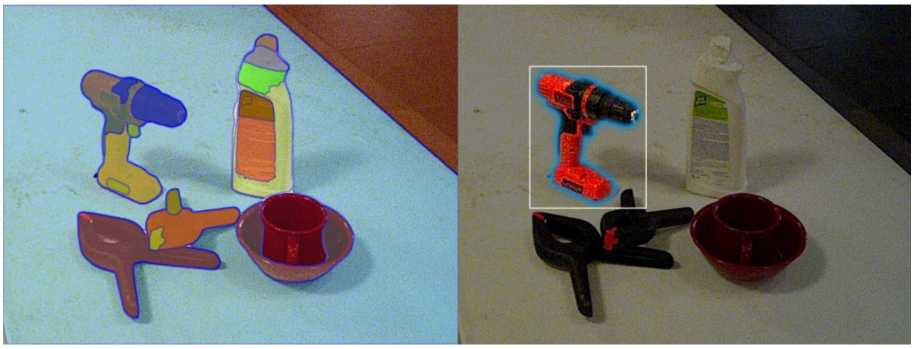

Figure 4–2: Comparison of SAM's output without (left) and with (right) bounding box

4.2.2. Mask Selection

While powerful in terms of generalisation, Detic can sometimes lead to misidentifications when dealing with visually similar objects. For instance, as shown in Figure 4–1, both a drill and clamps were identified as "drill" by Detic. To address this issue and enhance the accuracy of object identification, a mask selection stage has been introduced. This stage draws inspiration from CNOS but improves upon it in terms of mask selection efficiency.

The mask selection process begins with a database of N_o objects, each associated with a CAD model. For each object, V viewpoints are sampled uniformly on an enclosing sphere centred at the object's CAD model. From each viewpoint, a 224x224 template image is rendered and encoded using DINOv2, a pre-trained vision transformer capable of encoding images into a vector of C tokens. In this implementation, V is set to 48 and C to 1024.

For each segmentation result generated in the first stage, the isolated object is rescaled to 224x224 and similarly encoded by DINOv2, producing a vector of tokens C_s. This token vector is then used to calculate the cosine similarity with the encoded templates of the target object. The similarity scores against all templates are aggregated into a single number by taking their mean, representing the overall score for the segmented object. This calculation can be formulated as follows:

$$\text{Score} = \frac{1}{48}\sum_{i=1}^{48}\frac{Cs \cdot C_i}{|Cs||C_i|}$$

This approach offers significant advantages over CNOS. By comparing the segmented object only against templates of the target object, rather than against all objects in the database, the proposed method reduces computational complexity by a factor of N_o, where N_o is the number of objects in the database. Furthermore, if only one object segmentation is generated in the first stage, this selection process can be skipped entirely, further improving efficiency.

4.2.3. Pose Estimation

The final stage of the pipeline involves 6D pose estimation, which is conducted using the identified object segmentation. To enhance estimation accuracy, the segmented object is placed against a fully black background, effectively eliminating any potential interference from the original image context.

MegaPose is then employed to calculate the 6D pose using this isolated object image and the corresponding CAD model. As illustrated in Figure 4–3, MegaPose generates templates T^M by rendering the CAD model from M viewpoints. In this work, M = 576. The coarse network then classifies which rendered template best matches the input image. Given an initial pose estimate T^K, the refiner renders the objects at the estimated pose $T_{CO,1}$ (blue axes) along with 3 additional viewpoints $T_{CO,2}$ $T_{CO,3}$ $T_{CO,4}$ (green axes) defined such that the camera z-axis intersects the anchor point O. The refiner network consumes the concatenation of the observed and rendered images and predicts an updated pose estimate Tk+1. The refinement can be executed multiple times for better result.

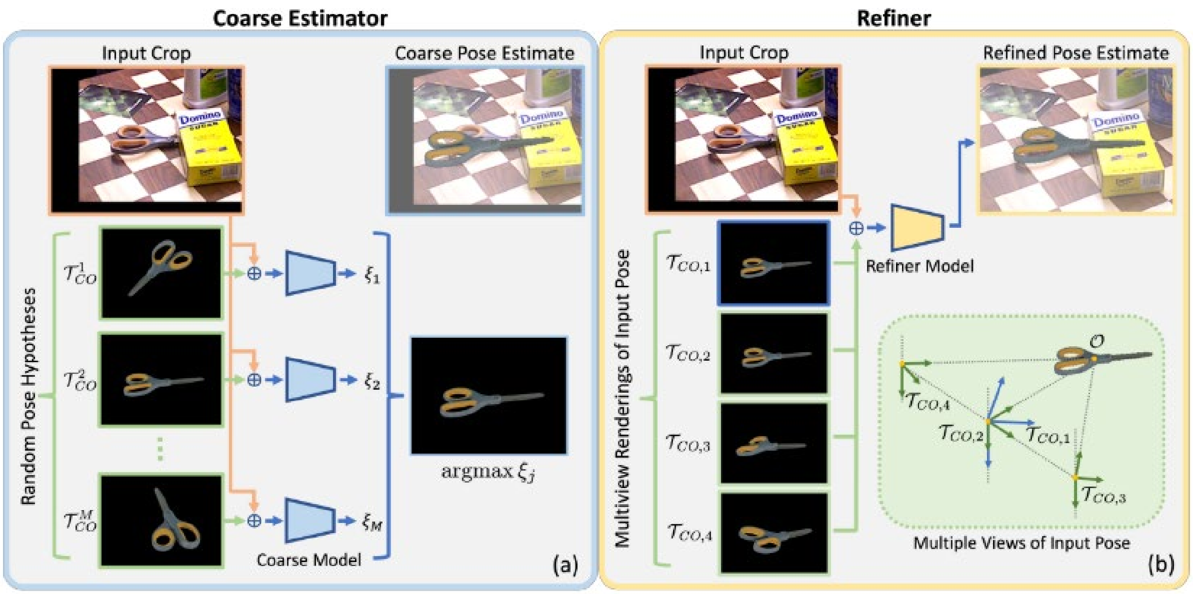

Figure 4–3: Processing pipeline of MegaPose (Labbé et al., 2022)

In this implementation, the coarse estimation step of MegaPose has been modified for improved efficiency. Rather than generating templates on-the-fly, M viewpoints and corresponding templates are pre-generated offline for each object in the database. During the coarse estimation process, these pre-generated templates are loaded from disk, scaled, and transformed as needed before being fed into MegaPose's coarse estimator. This optimisation significantly reduces processing time and computational costs by eliminating the need for real-time template generation.

4.3. Experiment

This section presents the experimental evaluation of the proposed method using publicly available datasets widely employed in 6D pose estimation research. Results benchmarking this approach against other methods are presented. Additionally, an ablation study demonstrates the significance of each stage in the proposed pipeline, highlighting their individual contributions to overall performance.

4.3.1. Testing Dataset and Metrics

The YCB-video dataset (Xiang et al., 2018) is a widely recognised benchmark in the field of robotic manipulation and CV. As illustrated in Figure 4–4, it features a diverse collection of 21 objects from the Yale-CMU-Berkeley (YCB) object set, captured in various poses and configurations. This dataset was specifically chosen for this work due to several key factors that align with the research objectives and practical considerations.

Figure 4–4: Objects in the YCB-Video dataset (Xiang et al., 2018)

One of the primary reasons for selecting the YCB-video dataset is the accessibility of its objects. Many of the items featured in the dataset are readily available and can be found in the LISMS lab, making it convenient for real-world testing and validation of our algorithms. This accessibility allows for seamless transition between simulated environments and physical implementations, enhancing the practicality and applicability of this research.

Furthermore, the YCB-video dataset offers a comprehensive range of object categories, encompassing both industrial and household items. This broad coverage provides a realistic representation of the diverse objects that robotic systems may encounter in various settings. By including such a wide array of objects, the dataset enables us to develop and evaluate algorithms that can generalise well across different object types and environments.

Of particular interest to this work is the inclusion of the "drill" object in the YCB-video dataset. This specific item plays a key role in the case study, making the dataset especially relevant to the research goals. The presence of the drill allows this work to focus on a practical, real-world object that has significant implications for industrial applications.

In addition to its content, the YCB-video dataset is often evaluated using the BOP (Benchmark for 6D Object Pose Estimation) challenge metric (Michel et al., 2018). This metric, known as Average Recall (AR), provides a comprehensive assessment of pose estimation performance. The AR is calculated as the average of three components: AR_{VSD} (Average Recall for Visible Surface Discrepancy), AR_{MSSD} (Average Recall for Maximum Symmetry-Aware Surface Distance), and AR_{MSPD} (Average Recall for Maximum Symmetry-Aware Projection Distance). This combined metric offers a balanced evaluation of pose estimation accuracy across different aspects, making it a robust measure for comparing and validating our algorithms against existing state-of-the-art methods.

4.3.2. Experiment Result

The experiment results of the proposed method on the YCB-Video dataset, along with benchmarks against other methods, are presented in Table 4-1. The comparison includes various approaches such as CosyPose (Labbé et al., 2020), SurfEmb (Haugaard & Buch, 2022), OSOP (Shugurov et al., 2022), Zephyr (Okorn et al., 2021), MegaPose (Labbé et al., 2022), and GigaPose (Nguyen et al., 2024). Some of these methods utilise CNOS (Nguyen et al., 2023) for initial object detection. It's worth noting that certain techniques require additional modules for pose refinement, such as BFGS (Broyden-Fletcher-Goldfarb-Shanno), ICP (Besl & McKay, 1992), and GenFlow (Moon et al., 2024).

While methods like CosyPose and SurfEmb were primarily designed for seen object pose estimation and are not applicable to novel objects, they provide valuable insight into the performance of state-of-the-art algorithms in this domain. For pose refinement methods, the table specifies whether RGB-D input is required, as depth data offers additional dimensional information that can significantly enhance the performance of pose refinement algorithms.

Table 4-1: Experiment result of the proposed method benchmarking with other methods

Pose Initialisation		**Pose Refinement**			**Performance**	
Method	Novel Objects	Method	Novel Objects	RGB-D Input	YCB-V	Time
CosyPose	×	CosyPose			57.4	-
SurfEmb	×	BFGS			64.7	-
SurfEmb	×	BFGS+ICP	✓	✓	80.6	-
OSOP	✓	Multi-Hyp.	✓		33.2	-
OSOP	✓	MH+ICP	✓	✓	57.2	-
(PPF, Sift) + Zephyr	✓	-	✓	✓	51.6	-
CNOS + MegaPose	✓	-	✓		28.1	15.5 s
CNOS + MegaPose	✓	MegaPose	✓	✓	62.1	21.9 s
CNOS + MegaPose	✓	GenFlow	✓	✓	63.3	20.8 s
CNOS + GigaPose	✓	-	✓		27.8	0.4 s
CNOS + GigaPose	✓	MegaPose	✓	✓	66.1	7.7 s
CNOS + GigaPose	✓	GenFlow	✓	✓	65.2	10.6 s
Proposed Method	✓	MegaPose	✓	✓	**75.8**	**5.08 s**

This experiment compares the performance of the proposed method against SOTA novel object pose estimation algorithms, primarily MegaPose and GigaPose. The results reveal interesting insights into the effectiveness and efficiency of these approaches.

Without pose refinement, both MegaPose and GigaPose demonstrate poor performance. However, when pose refinement is applied, MegaPose achieves a 63.3 AR score, but at the cost of significant processing time, exceeding 20 seconds per estimation. GigaPose slightly outperforms MegaPose in terms of AR score while substantially reducing processing time.

The proposed method demonstrates a significant improvement over these existing approaches. It achieves a 75.8 AR score, higher than the current best of 66.1, while reducing processing time to 5.08 seconds. This represents a considerable advancement in both accuracy and efficiency. However, it is worth noting that despite this improvement, the processing time still precludes real-time application of the algorithm in its current form.

It is important to acknowledge that while the proposed method achieves a high AR score, it still falls short of the performance demonstrated by seen object pose estimation methods like SurfEmb. This gap is expected, as novel object pose estimation methods are still evolving in their attempt to match or surpass the performance of methods trained on known objects. This underscores the need for continued development in the field of novel object pose estimation to further narrow this performance gap.

4.3.3. Ablation Study

This ablation study aims to evaluate the contribution and significance of various system components within the proposed method using the YCB-Video dataset. As presented in Table 4-2, the study focuses on two key metrics: the AR score, which measures accuracy, and time consumption, which assesses efficiency. These metrics are calculated for different combinations of system components, providing insight into their individual and collective impact on the overall performance.

Table 4-2: Experiment result for different combinations of system components

Method	AR Score	Time
Detic + MegaPose*	64.3	5.00 s
Detic + SAM + MegaPose*	64.2	5.05 s
Detic + DINOv2 + MegaPose*	70.5	5.01 s
Detic + SAM + DINOv2 + MegaPose*	**75.8**	5.08 s

Table 4-2 presents a comprehensive analysis of the proposed method's components, with Megapose* referring to the modified, more efficient version of the original MegaPose algorithm developed in this work. The most basic viable pose estimation approach, combining Detic for 2D object detection and MegaPose* for pose recovery, forms the cornerstone of the proposed method. However, this basic configuration achieves only a 64.3 AR score.

Interestingly, the addition of SAM to this basic setup slightly decreases the AR score by 0.1 and increases processing time by 0.05 seconds. This suggests that merely isolating objects from the background without correct identification does not enhance accuracy. In contrast, incorporating DINOv2 instead of SAM boosts the score significantly by 6.2 points, underscoring the importance of the mask selection process in filtering out objects that mis-detected by Detic.

The full pipeline, integrating all four components, achieves the highest AR score of 75.8, a further 5.3-point increase compared to the configuration without SAM. This demonstrates that background removal does indeed improve pose estimation accuracy, but only when the segmented objects are correctly identified.

From an efficiency standpoint, the basic model (Detic and MegaPose* only) requires 5 seconds of processing time, while the full system takes 5.08 seconds. This marginal increase in processing time despite the addition of two components (SAM and DINOv2) indicates that Detic and MegaPose* are significantly more computationally expensive. Therefore, to achieve real-time operation, future optimisation efforts should prioritise these two algorithms.

4.4. Chapter Summary

This chapter introduces an innovative three-stage approach for estimating the 6D pose of novel objects, harnessing the capabilities of several large pre-trained vision models. The proposed method demonstrates remarkable versatility, as it can localise any unseen object within an image, given only the camera intrinsics and the object's CAD model. Without additional training, this approach surpasses current SOTA algorithms for novel object 6D pose estimation on the YCB-Video dataset, exhibiting superior accuracy and efficiency.

Many recent works on 6D pose estimation utilise multi-stage pipelines. This is largely because most 6D pose estimation algorithms are designed to estimate pose alone, requiring object detection or segmentation to be performed as a separate step prior to pose estimation. For example, MegaPose employs a recursive render-and-mask selection process to segment the target object. While effective, this additional stage contributes significantly to MegaPose's computational overhead, making it rather time-consuming. In contrast, my proposed method eliminates the need for this segmentation stage, resulting in a more efficient and faster pipeline.

Similarly, GigaPose relies on CNOS as an object detector. CNOS works by rendering multiple views of an object and comparing the similarity between the rendered images and the target to locate the object. This reliance on multiple renderings introduces unnecessary complexity and increases inference time. I observed this drawback across many existing approaches, which is why I opted to design a streamlined pipeline that does not depend on CNOS or similar view-based methods.

Another example is ZeroPose, which incorporates a customised SAM pipeline for target object segmentation. Again, this multi-stage approach introduces additional computational cost and potential error propagation between stages.

By designing a unified pipeline that avoids separate detection or segmentation modules, my approach reduces inference time and mitigates the complexity associated with multi-stage frameworks. This not only improves efficiency but also simplifies deployment and maintenance in real-world applications.

An in-depth ablation study was conducted to analyse the significance and contribution of each system component. By systematically removing individual elements from the complete system, the study revealed that each component plays a unique and vital role. The results clearly indicate that the removal of any single component leads to a substantial decline in overall system performance.

Despite its impressive performance, the proposed pipeline faces limitations in terms of pose estimation efficiency. The full pipeline requires 5.08 seconds to process, which falls short of real-time operation requirements, thus constraining its applicability in real-world industrial scenarios. Through the ablation study, it was determined that the Detic and MegaPose* processes consume the majority of processing time. Enhancing or replacing these two methods could potentially yield significant improvements in processing efficiency.

The significance of this work extends beyond the development of a single method or pipeline for novel object pose estimation. It opens up new avenues for more effective utilisation of large pre-trained vision models. As vision models continue to grow in size and complexity, combining them in sophisticated ways can lead to substantial time savings in model training while achieving remarkably good results.

For those interested in further verification, development, or integration of this work into their own projects, the source code and demonstration images are publicly available on GitHub at https://github.com/WanqingXia/DetAnyPose. This accessibility promotes transparency and encourages collaboration within the research community.

Chapter 5
Proactive Trajectory Planning

Robot trajectory planning for obstacle avoidance has long been a critical research topic. In human-robot collaborative production environments, dynamic obstacle avoidance takes on paramount importance due to its direct impact on worker health and safety. This chapter introduces a novel robot trajectory planning method for proactive free-moving obstacle avoidance in 3D environments. The proposed approach addresses current research gaps in existing methods, which often lack the ability to predict human motions and plan ahead, thereby limiting the efficiency of collaborative robots.

To comprehensively explore this new approach, the chapter is structured into five main sections. It begins with an overview of robot trajectory planning, discussing the advantages of proactive obstacle avoidance compared to reactive methods and its significance in collaborative robot control. This is followed by a problem definition, where the challenge is formally described as a Markov Decision Process (MDP) suitable for DRL applications. The third section provides a detailed description of the proposed method and reward function. The fourth section presents the experimental setup and results, evaluating the method's performance in terms of success rate and time efficiency. The chapter concludes with a summary that synthesises the research findings and discusses the implications of this work.

5.1. Overview

Robotic manipulators are increasingly becoming integral components of the manufacturing industry worldwide, enhancing production efficiency and enabling the mass production of products at more affordable prices. Research indicates that integrating the adaptability and flexibility of humans with the precision and endurance of robots can result in a level of productivity that surpasses what could be achieved by either human or robotic systems alone (Matheson et al., 2019). To actualise cost-competitive manufacturing through cobot system, it is imperative to remove the physical barriers that currently separate humans and robots in future smart manufacturing environments. This will facilitate closer interactions between humans and robots (Robla-Gomez et al., 2017). However, it is important that the safety of human operators is not compromised in the pursuit of higher productivity. This necessitates advancements in reliable 3D perception, high-level planning, and sophisticated control within manufacturing systems (Y. Lee et al., 2021). Specifically, for cobots, it is vital to have a dependable environment perception system capable of detecting and locating the human body among other potential obstacles in the workspace. Additionally, these manipulators must possess the decision-making capabilities required to complete tasks in constrained environments. Chapter 3 showcase a human-robot collaborative assembly scenario set up in LISMS. The UR5e robot is controlled by a desktop computer equipped with a high-performance graphics card, necessary for running sophisticated AI control algorithms.

Currently, collision-free cobot control can be facilitated using traditional trajectory planning algorithms supplemented by a set of sensors. While a robot's state is readily accessible, sensors predominantly function to track human movement. The robot modifies its behaviour in response to human actions based on its state, which is an approach referred to as reactive trajectory planning (Haddadin et al., 2010). Due to its inherently responsive nature, the robot is frequently required to decelerate or, in some instances, halt to mitigate potential impacts when in proximity to humans. Such systems, although simple to implement, ensure safety in human-robot interactions. Nonetheless, the introduction of slowed movements and increased idle times for the robot can significantly diminish manufacturing efficiency. Accordingly, there is a compelling need for a trajectory planning algorithm that preserves the safety of human-robot collaboration without sacrificing operational efficiency.

To address this, a shift from reactive to proactive trajectory planning is advocated. Proactive systems, taking into account the dynamic nature of human movement throughout the manufacturing process, can forecast an operator's future actions and delineate a collision-free path, thereby obviating the need for the robot to decelerate or stop (Lasota & Shah, 2015). Traditional algorithms, which typically map a fixed global path from start to finish, lack the necessary adaptability to account for deviations in the human operator's predicted actions. Recently, RL has gained traction as a robust approach in trajectory planning. By enabling robots to learn tasks through trial and error, it minimises the need for algorithmic redesign or fine-tuning across different environments (Wu et al., 2019). Furthermore, RL algorithms operate by assessing the current state of the system to generate subsequent actions. This negates the need for extensive prediction of human actions and allows for local trajectory adjustments, aligning well with the proactive planning requirements of human-robot collaborative manufacturing

Recent advancements in robotics have witnessed the increasing incorporation of DRL into the trajectory planning of robotic arms, signifying a paradigm shift in both the complexity of tasks that robots can perform and the efficiency with which they execute them. Designing an appropriate reward function is one of the main challenges in DRL (Hadfield-Menell et al., 2017). The reward function serves as a compass guiding the robotic agent through the intricate landscape of possible actions towards the desired behaviour. A carefully crafted reward function can remarkably accelerate the learning process, enabling robots to carry out tasks with high precision. Another critical aspect of leveraging DRL in robotic trajectory planning is the challenge of the simulation-to-real (sim-to-real) transfer (Ibarz et al., 2021). This involves bridging the gap between the model's performance in a controlled simulation and its application in a real-world scenario, which is a leap that often presents significant hurdles due to the unpredictable nature of real-world dynamics. Efforts in this domain have concentrated on developing robust algorithms capable of generalising the training received in simulation to operate seamlessly when deployed, thus mitigating the risks and costs associated with direct real-world training.

In this section, a virtual training environment that involves robotic arm and free-moving obstacles is constructed, addressing the challenge of proactive trajectory planning through the application of the DRL algorithm. A novel reward structure that effectively guides the robot to achieve its target without collisions in an environment populated with dynamic obstacle. Additionally, a methodology that facilitates the transition from simulation to real-world application, mitigating the sim-to-real transfer problem is introduced. This approach enables the safe training of robot manipulators in simulated settings, safeguarding against potential damage to both the robot and human operators resulting from unforeseen actions.

5.2. Problem Definition

DRL and MDP are closely intertwined in the field of AI and decision-making. DRL is a powerful approach for solving complex decision-making problems, while MDP provide a mathematical framework for modelling these problems. In essence, DRL algorithms are designed to solve MDPs, particularly those that are large and complex. In this section, we define the problem of collision-free trajectory planning within the framework of MDP and explain its relationship with DRL methodologies.

MDP is a mathematical framework for modelling decision-making in a partially controllable environment. MDP is used to find policies that specify the best action to take in a given state to maximise the long-term reward. It is the fundamental concept in RL and is widely used for solving various problems in fields such as robotics, economics, manufacturing, and AI (Sutton & Barto, 2018). A MDP can be defined by a tuple $< \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma >$, which consists of a state set $\mathcal{S}$, an action set $\mathcal{A}$, a reward function $\mathcal{R}(s_t, a_t)$, a state transition function $\mathcal{P}(\, s_{t+1} \mid s_t, a_t\,)$, and the discount factor γ. An agent's behaviour is defined by a policy π, which maps states to a probability distribution over the actions $\pi: \mathcal{S} \rightarrow \mathcal{P}(\mathcal{A})$. The return from a state is defined as the sum of discounted future rewards:

$$\mathcal{R}_t = \sum_{i=t}^{T} \gamma^{(i-t)} r(s_i, a_i) \tag{5-1}$$

The goal of RL is to learn an optimal policy that instructs the agent on the best action to take in each state to maximise its long-term reward. The objective function J is often used to denote the performance of a policy in RL. It refers to the expected cumulative reward that an agent would receive starting from a state s and following a policy π, is given as:

$$J(\pi) = E_{\pi}\left[\sum_{i=t}^{T} \gamma^{(i-t)} R_i \,\middle|\, S_t = s\right] \tag{5-2}$$

In this work, the goal is to train the UR5e manipulator and let it learn to reach a goal pose in the 3D space autonomously by planning and executing a collision-free trajectory. As illustrated in Figure 5–1(a), end-effector's pose is represented by $\mathcal{P}_e = [p_e, \phi_e]^T$, where p_e being the three-dimensional translation and ϕ_e being the Euler angle rotation in the Cartesian space related to the world frame. The joint angles $\Theta = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$ denotes the angle of each robot joint in radians. Thus, pose of the end-effector $\mathcal{P}_e$ at any given time t can be calculated by using forward kinematics: $\mathcal{P}_t = Forward\_Kinematics(\Theta)$. The action of cobot can be determined to be success when its end-effector's pose aligned with the goal pose. Similar with the end-effector pose $\mathcal{P}_e$, the goal pose can be defined as $\mathcal{P}_g = [p_g, \phi_g]^T$, related to the world frame, shown in Figure 5–1(c). To move the manipulator to achieve the desired pose, the rotation of each robot joint needs to be controlled. Thus, our action space in MDP is defined as the change in angular position of each joint $\mathcal{A} = \Delta\Theta = (\Delta\theta_1, \Delta\theta_2, \Delta\theta_3, \Delta\theta_4, \Delta\theta_5, \Delta\theta_6)$. The joint angles at next time step $t+1$ can be found by $\Theta_{t+1} = \Theta_t + \Delta\Theta$.

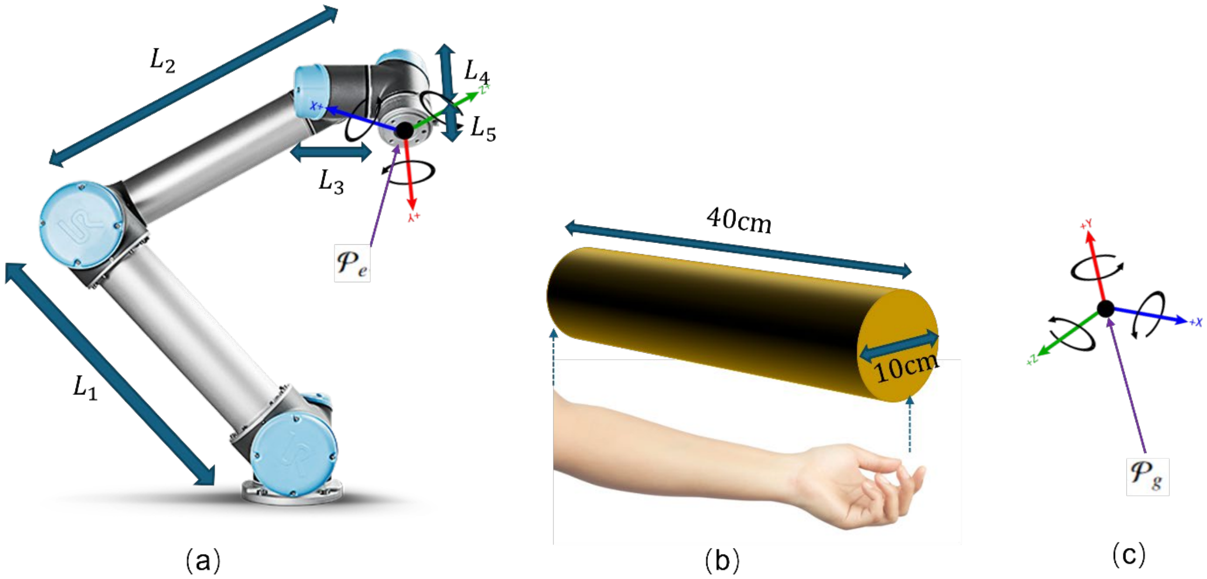

Figure 5–1:Illustration of the parameters for cobot, obstacle and goal

To ensure dynamic obstacle avoidance without collision, it is imperative to monitor and observe the obstacle's state continuously. In this research, an important assumption is that all objects within the workbench area remain static except for the human operator's arm, which is the sole dynamic obstacle. This assumption is made based on the design of the HRC workbench as explained in Chapter 3. This allows for a simplification in our model, where the operator's hand and forearm are represented as a cylindrical dynamic obstacle. The dimensions of this cylinder—10 cm in diameter and 40 cm in length—mirror the approximate size of an adult's arm, shown in Figure 5–1(b). For locating this obstacle within the space, its 6D pose is defined as $\mathcal{P}_o = [p_o, \phi_o]^T$, where p_o and ϕ_o are the relative translation and rotation of the obstacle's center of mass related to the world frame. Proactive trajectory planning requires not only the current position but also a prediction of the obstacle's future state. Therefore, the model incorporates both the position and the velocity of the obstacle, the latter denoted as $\mathcal{V}_o = [v_o, \omega_o]^T$, represents the linear and angular velocity of the obstacle's center of mass related to the origin.

After establishing the dynamic obstacle's position and velocity, the robot still lacks sufficient information for collision-free trajectory planning. This shortfall arises because the robot cannot sense other surrounding obstacles in the environment, such as the workbench. This will cause the model to do more trails in the environment to "guess" where the obstacles are, significantly reduce the learning efficiency and accuracy. To address this, the distances from each robot manipulator link to the surrounding static objects are provided as known parameters. This additional information equips the robot's learning model with the capability to avoid collisions with both static and dynamic obstacles, as well as with its own links. As illustrated in Figure 5–1(a), the first link of the robot (the shoulder link) is stationary. Therefore, it is unnecessary to account for its distance to obstacles, thus the considerations are confined to the remaining five links. The distances for these links are represented as $\mathcal{D} = (\mathrm{d}_1, \mathrm{d}_2, \mathrm{d}_3, \mathrm{d}_4, \mathrm{d}_5)$, represents the smallest Euclidean distance between any point on the obstacle and link L1-L5, respectively.

In summation, the model accounts for a state vector composed of 35 variables, which includes the end-effector pose, robot joint angles, goal pose, obstacle pose, obstacle velocity, and the distances of links to potential collision objects, denoted as $\mathcal{S} =< \mathcal{P}_e, \Theta, \mathcal{P}_g, \mathcal{P}_o, \mathcal{V}_o, \mathcal{D} >$. The significance of each symbol within this state vector is delineated in Table 5-1. The proactive nature of the model is further evidenced by its consideration of state space variables, denoted as $\mathcal{P}_o, \mathcal{V}_o, and\ \mathcal{D}$. These variables are contingent upon the present state of obstacles. However, given the dynamic nature of the environment, the model must anticipate future obstacle states and formulate trajectory plans accordingly.

Table 5-1: Definition of the state variables

SYMBOLS	DESCRIPTION
$\boldsymbol{\mathcal{P}_e}$	$x_e, y_e, z_e, \alpha_e, \beta_e, \gamma_e$
$\boldsymbol{\Theta}$	$\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$
$\boldsymbol{\mathcal{P}_g}$	$x_g, y_g, z_g, \alpha_g, \beta_g, \gamma_g$
$\boldsymbol{\mathcal{P}_o}$	$x_o, y_o, z_o, \alpha_o, \beta_o, \gamma_o$
$\boldsymbol{\mathcal{V}_o}$	$v_x, v_y, v_z, \omega_\alpha, \omega_\beta, \omega_\gamma$
$\boldsymbol{\mathcal{D}}$	d_1, d_2, d_3, d_4, d_5

To this end, the states $\mathcal{S}$ and actions $\mathcal{A}$ in MDP have been defined, where γ is a constant value defined empirically and $\mathcal{P}$ in DRL is always learned by the deep neural network. The last and most important part of the MDP model is the reward function $\mathcal{R}$, which acts as the solution to solve the defined problem, is presented in the next section.

5.3. Proposed Method

To solve the formulated MDP, a DRL algorithm is employed. This section explains the selected DRL algorithm and justifies its suitability for the defined problem. A reward function is then designed based on the chosen algorithm. Finally, a method developed using the digitalised HRC workstation is introduced to address the sim-to-real transfer challenge.

5.3.1. Soft-actor Critic

The SAC algorithm is a state-of-the-art DRL algorithm that optimises a stochastic policy in an off-policy way, while taking into account the entropy of the policy to encourage exploration (Haarnoja et al., 2018). SAC consist of two main components: an actor learns to map states to actions, and a critic learns to evaluate the selected actions by estimating the Q-value, which represents the expected return after taking an action in a given state and thereafter following a certain policy.

Figure 5–2 shows details of the SAC algorithm. SAC introduces the concept of "soft" policy iteration, where the policy is not only evaluated and improved based on the expected return but also on the entropy of the policy. Entropy is a measure from information theory that quantifies the amount of randomness in the policy. A higher entropy means more exploration, as the policy is more random and thus tries out a wider range of actions. This is particularly useful in environments with many local optima to prevent premature convergence to suboptimal policies. To address the overestimation bias of Q-values in actor-critic methods, SAC uses two Q-networks and takes the minimum of their estimated Q-values to reduce overoptimistic value estimates. In addition to the twin Q-networks, SAC also includes a separate value network that estimates the value of a state. The value function is trained to minimise the mean-squared error between its predictions and the target values, which are computed using the minimum Q-value from the twin Q-networks adjusted by the entropy term.

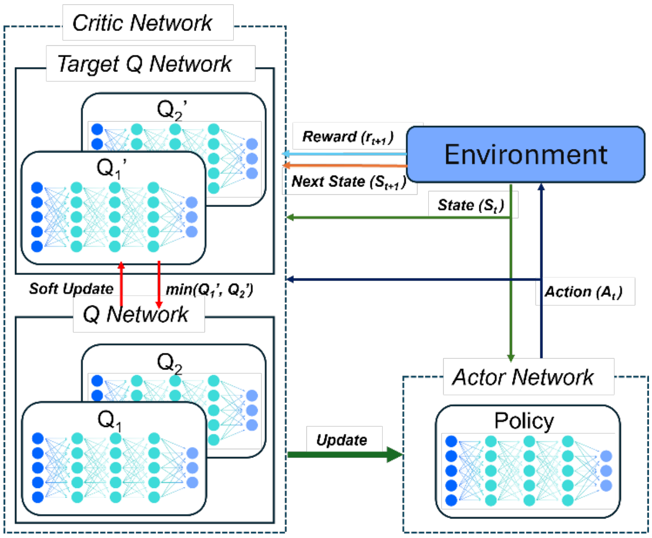

Figure 5–2: The SAC algorithm's structure

Given the distinctive advantages over other DRL algorithms, SAC is selected to train the cobot control model. The first advantage is its robust stability. SAC models are tempered by an entropy regularisation term throughout the training process, promoting extensive exploration of the state space. This results in a stable and robust model--an essential quality for ensuring safety in human-robot collaborative manufacturing. The second advantage is sample efficiency. As an off-policy algorithm, SAC capitalises on historical data collected from previous environmental interactions. This attribute is particularly beneficial for our application, as the exploration space for a 6-DOF robot is expansive, and efficiently navigating this space is crucial for effective learning. Lastly, SAC demonstrates exceptional performance within environments characterised by continuous action spaces. This aligns perfectly with the specific setting in this work, which involves a six-joint robot navigating around dynamic obstacles. The synergy between SAC's capabilities and the environment's demands suggests that the SAC algorithm is ideally suited to the requirements of this study.

5.3.2. Reward Function Design

The cornerstone of this study lies in the innovative design of the reward function. While prior research has explored the training of robotic manipulators using RL and DRL, this work is the first to tackle the challenge of avoiding dynamically moving obstacles in a 3D space solely through DRL, without relying on supplementary algorithms such as the potential field method. The distinctive reward design is instrumental in directing the model to accomplish its objectives using a vanilla DRL algorithm. This not only simplifies the learning process but also facilitates the transferability of our method to different robotic platforms and environments.

As mentioned earlier, the goal is to navigate the robot's end-effector from an initial pose $\mathcal{P}_e$ to a goal pose $\mathcal{P}_g$, controlling the joint angles Θ without any collision with free-moving obstacles. To this end, the reward function is designed to be composed of five components:

Collision Reward r_c: Penalises the robot if the end-effector collides with anything, ensuring safe manoeuvring. The algorithm determines Whether a collision occurred based on the distances between robot links and other objects $\mathcal{D}$. If any of these values is equal to or smaller than zero, it indicates the manipulator collides, the training episode immediately stops and penalises the model with a -500 reward. The equation for collision reward is given in Equation 5-3.

$$r_c = \begin{cases} -500 & \textit{if collision} \\ 0 & \textit{otherwise} \end{cases} \tag{5-3}$$

Success Reward r_s: Rewards the robot for successfully reaching the goal pose, reinforcing the completion of the task. If the manipulator successfully reaches the goal, the training episode also immediately stops and rewards the model with a 200 reward, given in Equation 5-4.

$$r_s = \begin{cases} 200 & \textit{if success} \\ 0 & \textit{otherwise} \end{cases} \tag{5-4}$$

Translational Reward r_t: Provides incremental feedback based on the proximity of the end-effector's current pose $\mathcal{P}_e$ to the goal pose $\mathcal{P}_g$, guiding the trajectory. The translational reward is calculated by the Euclidean distance between two points with a scale of -70, given in Equation 5-5 and 5-6.

$$D_t = \sqrt{(x_e - x_g)^2 + (y_e - y_g)^2 + (z_e - z_g)^2} \tag{5-5}$$

$$r_t = \text{-}70 D_t \tag{5-6}$$

Rotational Reward r_r: Encourages the correct orientation of the end-effector, aligning it with the desired goal orientation. Since the end-effector and goal rotation in the model states are in Euler angle format, it is first converted to quaternions: $(\alpha_e, \beta_e, \gamma_e) \rightarrow (i_e, j_e, k_e, w_e)$, $(\alpha_g, \beta_g, \gamma_g) \rightarrow (i_g, j_g, k_g, w_g)$. Then the rotational reward is calculated by the smallest distance between two quaternions with a scale of -30, given in Equation 5-7 and 5-8.

$$D_r = 2 \times \arccos\left(|i_e i_g + j_e j_g + k_e k_g + w_e w_g|\right) \tag{5-7}$$

$$r_r = \text{-}30 D_r \tag{5-8}$$

Distance Reward r_d: Penalises moving towards the obstacle, ensuring safe distance while moving. For every moveable link on the robot, it is penalised for moving closer to the obstacle and rewarded for moving away from the obstacle when the link distance is smaller than 0.2 meters. If the link distance is bigger than 0.2 meters, it is considered safe, and the reward is 0.

$$r_d = \sum_{n=1}^{5} \begin{cases} W_i \times \Delta\text{distance}_i & \text{if } distance_i < 0.2 \\ 0 & otherwise \end{cases} \tag{5-9}$$

The weight of each link in the distance reward is different in this work, determined by the total link distance reward weight and each link's physical weight. From link 1 to link 5, their physical weights are approximately 8, 2.4, 1.2, 1.2, 0.2 Kilograms. The total link distance reward weight is set to 50, and the detailed reward weight for each link is given in Equation 5-10. It needs to be noticed that the link weights are set to encourage the algorithm to optimise for energy efficiency by preferring movements that involve lighter links over heavier ones, whenever possible. They do not perfectly reflect the true dynamics of the robotic arm.

$$W_i = \frac{W_i{}^p}{\sum W_i{}^p} \times 50 = [30.77,9.23,4.62,4.62,0.77] \tag{5-10}$$

In the proposed DRL model, the reward weights are tuned for translational and rotational distance through a systematic trial-and-error process. Determining the optimal balance between translational and rotational rewards was challenging, as it influences whether the model prioritises achieving goal translation or rotations. Initially, these weights are set to a total of -100. This baseline allows adjusting the weights for collision and success in subsequent trials. Then the weight proportion is tested with nine different scenarios, varying translational rewards from -10 to -90, while keeping other parameters constant. After training each scenario for one million episodes, the resulting reward curves is illustrated in Figure 5–3, where each curve correlates with a distinct weight setting. For example, the line labelled "T: -20, O: -80" represents the scenario with a translational weight of -20 and a rotational weight of -80. The testing result indicated that a translational reward weight of -70 and a rotational weight of -30 yielded the most balanced and effective learning outcomes.

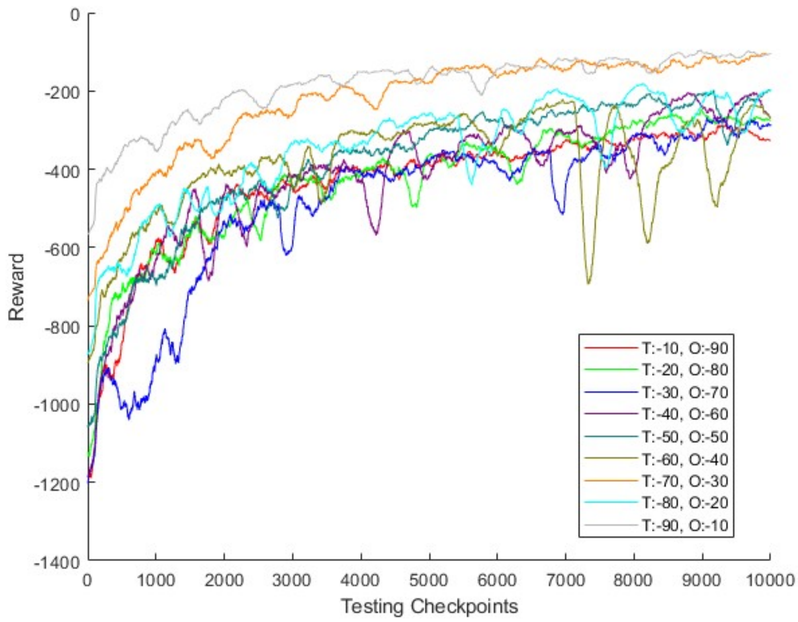

Figure 5–3: Reward curves for different translational and rotational weight

Setting the weights for collision and success was less complex. Based on the total weight of -100 for translational and rotational distances, -500 for collision and +200 for success were selected after simple preliminary testing. The final hyperparameter, the total link distance weight, affects the manipulator's behaviour regarding obstacles. A weight that is too high results in excessive avoidance behaviour, preventing the manipulator from reaching its goal pose. Conversely, a weight that is too low increases the risk of collision. After further experiment, it is determined that a weight of 50 struck the right balance. To provide a clear understanding of the reward calculation process, it is formulated in the accompanying pseudocode Algorithm 1.

Algorithm 1: Computing Reward

```
for each step in episode do
    if step is 0 then
        previous distance ← 0
        reward ← 0
    end if
    if collision then
        return collision weight
    end if
    if success then
        return success weight
    end if
    link distance ← get current link distance
    Δlink distance ← link distance − previous distance
    previous distance ← link distance
    translational distance ← calc_tran_dist (current pose, goal pose)
    rotational distance ← calc_rot_dist (current pose, goal pose)
    reward ← reward + translational weight · translational distance
    reward ← reward + rotational weight · rotational distance
    for each Δdistance_i in Δlink distance do
        if Δdistance_i < 0.2 then
            reward ← reward + link distance weight_i · Δdistance_i
        end if
    end for
end for
return reward
```

5.3.3. Sim-to-real Transfer

A pivotal challenge in deploying DRL algorithms is the sim-to-real transfer, also known as domain adaptation. This entails the successful migration of a policy trained in a simulated environment to the real world. While simulation offers a safe, controllable, and cost-effective training ground, discrepancies between the simulation and real-world scenarios—collectively known as the "reality gap"—-can severely hinder the performance and reliability of a robot when executing tasks such as obstacle avoidance. The reality gap encompasses differences in dynamics, perception, and environmental interactions between the simulated and physical worlds.

When transitioning from a theoretical model to practical applications, there are two critical assumptions. The first is that the position and velocity of the obstacle can be measured with high accuracy by the sensors. The second assumption specifies that the acceleration of the obstacle remains relatively consistent between consecutive measurements; significant deviations would render the prediction of the obstacle's future state unreliable. These assumptions are essential for the practical deployment of the trained model, ensuring that the robot can effectively forecast and navigate around the moving obstacle presented by the operator's arm.

In this study, through system identification and the implementation of precision position commanding, a dual approach is proposed to minimise the reality gap. In the system identification process, a detailed model of the actual environment is used, as introduced in Chapter 3. It needs to be noticed that the workbench is simplified to a rectangular block to reduce the computational complexity for distance calculation and collision checking in this environment.

In addition to system identification, precision position commanding is used to exert direct control over the robot's joint positions, instead of using joint velocity or torque control. Given that simulations can only approximate physical properties such as friction, mass, and inertia, torque or speed control could result in imprecise outcomes. By contrast, the proposed model exerts direct control over the angular displacement of each joint. Coupled with the insights gained from system identification, this approach ensures that the robot's state post-execution aligns precisely with real-world behaviour. Figure 5–4 illustrates the training and deployment framework, highlighting the strategies we employ to minimise the reality gap.

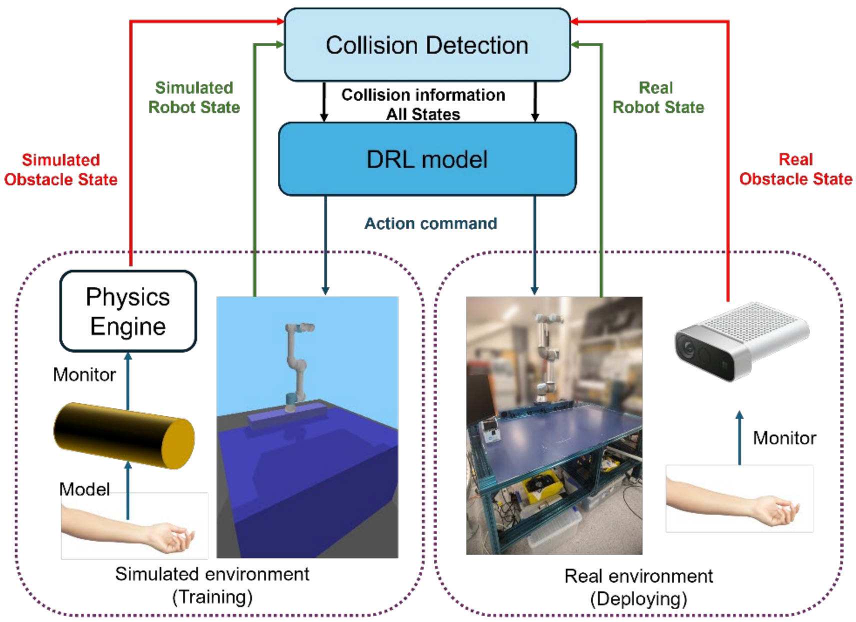

Figure 5–4: Proposed framework for training and deploying the DRL model

5.4. Experiment

In this section, the training and testing experiment setup in a simulated environment are introduced, as well as the experiment procedures. Subsequent analysis of the training results and test data set will follow, proofing the performance of the proposed method.

5.4.1. Training Environment Setup

During the model training, PyBullet was key in creating the simulated environment. Renowned for its effectiveness in simulations, robotics, and machine learning, PyBullet enabled a training environment that surpassed the limitations of real-world settings. This approach not only sped up the training process but also prevented potential harm to both the robot and humans from unexpected movements during training.

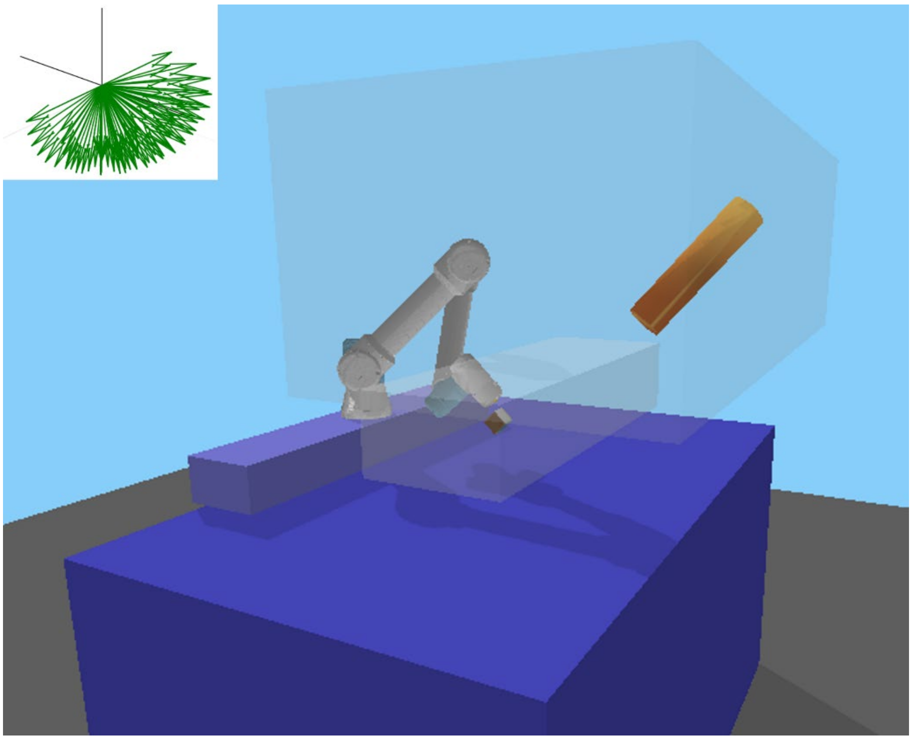

Figure 5–5: Goal and obstacle sampling range in simulated environment

Figure 5–5 visualises the training environment setup, including the cobot, linear track and workbench. There are other items in the figure visualising conceptual objects, such as the multicoloured cube, serving as a visual guide for the goal pose. The smaller, transparent cuboid below represents the target position sampling area. Arrows at the top left corner indicate the range for sampling target rotations, with the robot's end-effector designed to avoid pointing upwards or backward, aligning with practical constraints in real-world robot grasping. When the robot is reaching for the goal pose, the action is considered successful when the translational distance is within 3cm, and the rotational distance is within 5 degrees. The yellow cylinder in the simulation represents the moving obstacle, symbolising a human co-worker's hand and forearm; its darker end represents the hand, and the lighter end, the elbow. The larger transparent cuboid above shows the moving obstacle's sampling range.

During training, each episode begins with a randomly selected goal translation and rotation. The starting and ending poses for the obstacle are also randomly chosen. Throughout the episode, the obstacle moves and rotates at a constant speed from its starting to end pose over two seconds, then remains stationary. Training examples are generated through direct interaction between the robot and the environment. At each step of this interaction, the robot executes actions, consequently inducing changes in the environment's state. The volume of our training data hinges on two primary factors: the number of episodes and the number of steps per episode. A significantly large number for the training episodes to encompass all possible poses of the human arm during interaction with the manipulator in the large operational 3D space. This extensive coverage is critical for training the model to handle a diverse array of real-world scenarios. Thereby fostering robustness, adaptability and generalisability to a broad spectrum of human-arm and robotic-manipulator collaboration scenarios. Additional hyperparameters fine-tuned to optimise training results are detailed in Table 4-2.

Table 4-2: List of training hyperparameters

Hyperparameter	Value
Episodes	6,000,000
Steps per Episode	100
Buffer Size	10,000,000
Learning Rate	0.0001
γ	0.95
Batch Size	256

5.4.2. Testing Environment Setup

Even though we have validation episodes during the training, the size of it is limited to 100, and the goal translation and orientation are totally random. It cannot represent the real performance of the trained model. In response, a test set needs to be developed to evaluate the trained model. For constructing the test set, the data need to be consistent and cover enough scenarios to ensure the model is tested thoroughly. As shown in Figure 5–5, the goal pose is sampled within a cuboid area with 1.0m, 0.3m, and 0.2m in length, width, and height, respectively. The goal positions are uniformly sampled every 0.05m within the goal area, and for each of these goal positions, five random orientations are sampled. This methodology resulted in a total of 3675 points ($21 \times 7 \times 5 \times 5 = 3675$) within the test set. Similar with the training environment, the obstacle in testing environment moves randomly and reset for each testing case. The sampled testing goal position, orientation and the obstacle moving trajectory are recorded, ensures every trained model is tested on the same dataset.

While metrics like minimum distance to obstacles and episode completion time do provide meaningful insights on a per-episode level, their value diminishes when averaged over a substantial volume of testing episodes. This is because averaging these metrics can obscure important nuances and variations in performance that are essential for a comprehensive evaluation of safety and effectiveness. For instance, outliers in minimum distance or completion time can significantly affect their average values, potentially leading to misleading interpretations of the algorithm's overall performance. Thus, during model testing, the success rate over 3675 cases is used as the metric to determine the performance of a model.

5.4.3. Experiment Result

During the training process, it is necessary to periodically validate the model and save it if it surpasses previous performance levels. The periodic validation results also indicate the trend of the training process. As outlined in Table 4-2 the training encompasses 6 million episodes. The model is validated for every 1,000 training episodes using a 100-episode validation set, resulting in 6,000 data points. The model is saved when it achieves a higher success rate than before. In Figure 5–6, the mean reward, success rate, mean episode length and reward standard deviation is plotted for these data points.

The most important plot among all four plots in Figure 5–6 is (a), the mean reward curve. This is because the reward is calculated by several factors, hence the reward curve shows the comprehensive performance of a model. It can be told from Figure 5–6 (a) that initially the mean reward hovers around -800, indicating frequent collisions by the robot with either the obstacle or itself. This leads to a consistent success rate of zero (shown in Figure 5–6 (b)) a very short mean episode length (shown in Figure 5–6 (c)). However, after approximately 300 validation runs, there is a notable shift: the reward drops, and the mean episode length jumps to around 80 steps. This suggests the robot has started to learn how to avoid the moving obstacle, though it has not yet mastered reaching its goal. As training progresses, the mean reward gradually improves, eventually stabilising at about -80. Correspondingly, the mean episode length decreases and levels out at around 10 steps. The success rate also shows significant improvement, rising from zero and stabilising above 95%. This progression indicates that the robot is not only learning to achieve its target but is also becoming more efficient in doing so, minimising the time taken to complete tasks. Figure 5–6 (d) illustrates the standard deviation of rewards across validation runs. The trend initially rises from near zero to a peak, then decreases to approximately 200. There is significant variation in rewards between runs, which can be attributed to two factors: the model's success rate in achieving the goal and the inherent variability of each run. Given that the validation points' and obstacles' pose are randomised, some variation is expected. However, if the model could achieve a 100% success rate under all conditions, the reward standard deviation curve would likely be smoother.

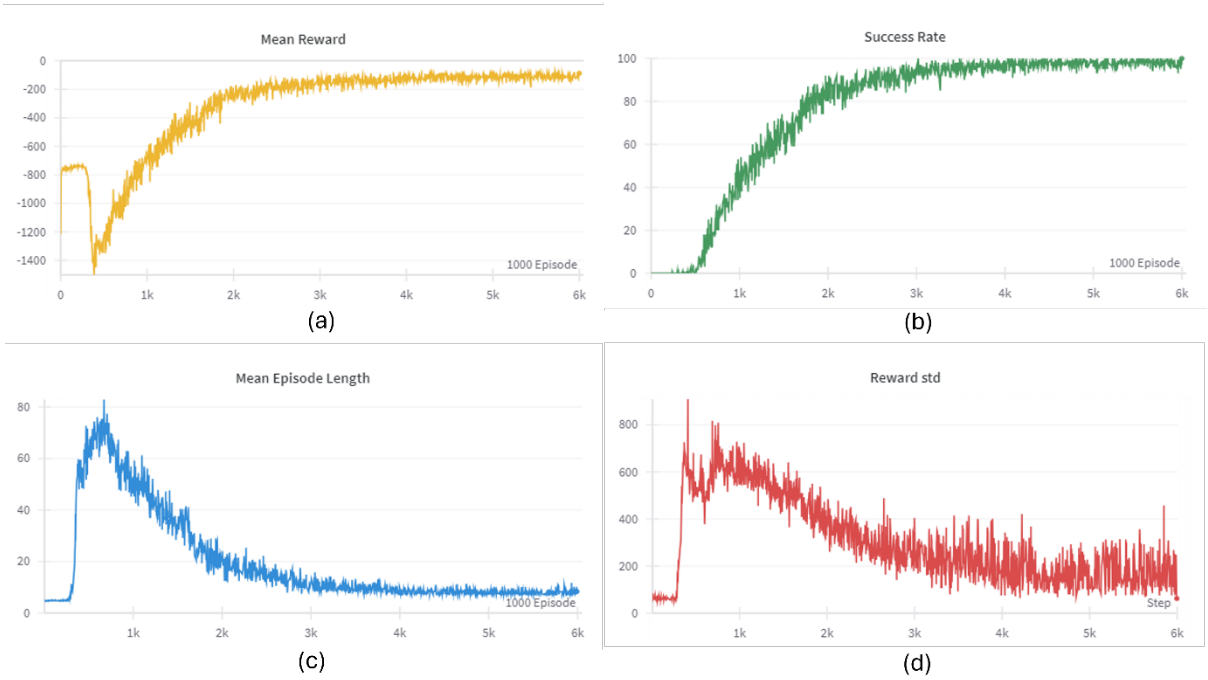

Figure 5–6: Mean reward, success rate, mean episode length and reward std obtained during validation

During the training-validation process, the model's capability can be assessed through the reward, success rate, and episode length curves. These metrics stabilise at certain levels, indicating that the model's performance has plateaued, and further training episodes are unlikely to yield significant improvements. Consequently, the best-performing model is saved upon completion of the training process. To provide an intuitive measure of the model's effectiveness, it is then evaluated using the test set. In this study, the trained model achieved an 96.5% success rate on the test dataset, demonstrating its robust performance and validates the success rate achieved during training.

5.4.4. Discussion

Although the model has been tested using a generated test dataset, it cannot conclusively assert that it has effectively learned obstacle avoidance. This uncertainty arises because the model achieved a 96.5% success rate, yet the obstacles' starting and ending poses were randomly generated. Consequently, it is plausible that the model's success was due to chance rather than genuine learning. To address this, an additional model was trained in the same environment but without obstacles (red dash lines in Figure 5–7) and subsequently tested it on the same dataset. The end-effector's trajectories from both models are compared in the same test scenario, alongside the obstacle's trajectory, as illustrated in Figure 5–7. This comparison distinctly demonstrates that the model trained with dynamic obstacle avoidance consistently maintains a safe distance from a moving obstacle, whereas the model lacking this training collided with the obstacle.

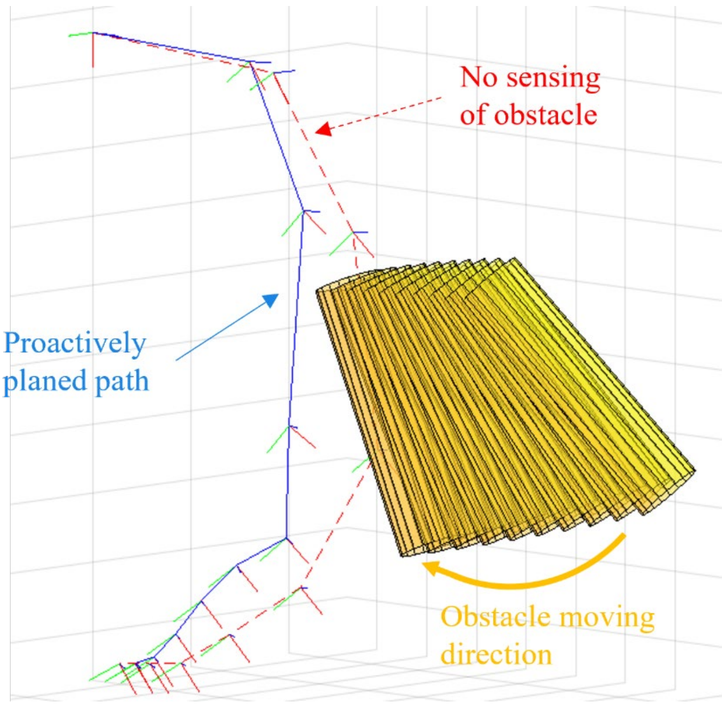

Figure 5–7: Robot end-effector's trajectory obtained from two models on the same testing case

5.5. Chapter Summary

This chapter presents a significant advancement in the field of robot trajectory planning and execution, offering a novel DRL-based approach for dynamic obstacle avoidance. The use of the SAC algorithm, combined with a well-crafted reward function, has proven effective in enabling a robotic manipulator to navigate around moving obstacles safely and efficiently, achieving success navigation in 96.5% of all cases. Additionally, a framework of transferring the trained model seamlessly to the physical HRC workstation is proposed based on the environment design and assumptions. The framework aims to reduce the training time and eliminate risks by train the robot control model on the digitalised HRC workstation.

Reward function design varies significantly across prior works in this field. The most common and basic reward structures typically include success, collision, and translational components. In scenarios that require obstacle avoidance, a distance-based reward is often incorporated. My work introduces two key innovations to this area.

A major advancement in my approach is the introduction of a rotational reward. Previous works have largely focused on guiding the TCP to a target position, with little regard for the final orientation of the robot. However, the robot's end pose is critical, particularly when the task requires the robot to interact with objects at specific angles or poses, as is often the case in HRC. To address this, I developed a rotational reward that explicitly encourages correct orientation upon reaching the target. While this adds complexity due to the increased dimensionality of the state space, it is essential for real-world applicability and more nuanced manipulation tasks.

The second key difference lies in the formulation of the distance reward. While some previous methods include a distance reward, it is typically based solely on the distance between the robot TCP and nearby obstacles. In contrast, I designed the distance reward to consider the minimum distance between all robot links and the obstacles. This naturally accounts for the contribution of each link to the overall reward, promoting collision awareness across the entire robot rather than just the TCP. As a result, the system achieves improved safety and energy efficiency, as it encourages the robot to avoid collisions in a holistic manner.

In summary, by incorporating a rotational reward and a more comprehensive distance reward, my approach addresses critical shortcomings in prior works and leads to more reliable, efficient, and versatile robot behaviour.

This work offers several directions for future completion and extension. While the trained model achieves a high success rate, it falls short of 100% accuracy. Significant effort is required to investigate the remaining 3.5% of failure cases, determining whether these instances are solvable but beyond the current model's capabilities, or if they represent inherently unsolvable scenarios.

The model's performance in a physical environment remains untested, primarily due to safety concerns. In real-world applications, human skeleton motion data would be captured by RGB-D cameras and processed through separate vision algorithms. This introduces additional uncertainties to the system due to potential inaccuracies in data collection and processing, which could further increase the likelihood of failure. Although the robot is equipped with reactive stop mechanisms that triggers when collisions are sensed, this alone cannot guarantee human safety from potential impacts. Consequently, conducting experiments under these circumstances presents significant ethical challenges.

For anyone who is interested to further verify and develop this work, or use this work as a part of their own project, the source code of this project is publicly accessible on GitHub: https://github.com/WanqingXia/UR-gym.

Chapter 6
Case Studies

The previously developed robot vision and trajectory planning system has proven effective in datasets. However, to be utilised in real-life collaborative manufacturing environments, it needed further validation. This chapter describes how the developed systems are integrated into the overall system structure. Different task analysis algorithms are also implemented, completing the machine-learned collaborative robot control system.

The chapter explores three case studies. The first case study incorporates GPT4o as the task analysis algorithm and completed the detail implementation of the overall system architecture, demonstrates the system's capability in robot assisted picking for household items. The second showcases its generalisability to fully novel objects. These two case studies share the workflow but focus on different aspects of capabilities of the develop system. The last case study uses a human action analysation module as the task analysis method, showcasing a proactive and symbiotic cobot control system. These case studies also provide insightful ideas for future applications of the developed system.

6.1. Case Study 1: Robot Assisted Picking for Household Items

In Chapter 3, this project's high-level system structure is presented, featuring a loop execution of four key components: task analysis, object pose estimation, trajectory planning, and trajectory execution. With the novel machine-learned algorithms developed for object pose estimation (Chapter 4) and robot trajectory planning (Chapter 5), the system execution pipeline can now be described in greater detail.

This detailed structure, as depicted in Figure 6–1, incorporates the object pose estimation workflow (stages 2, 3, and 4) while introducing a task specific machine-learned task analysis algorithm: human intention prediction. The intention prediction module, implemented as Stage 1, leverages the capabilities of GPT4o, a multimodal LLM developed by OpenAI. This advanced LLM enables the system to not only understand human commands but also predict the underlying intentions and hidden meanings within the operator's spoken sentences.

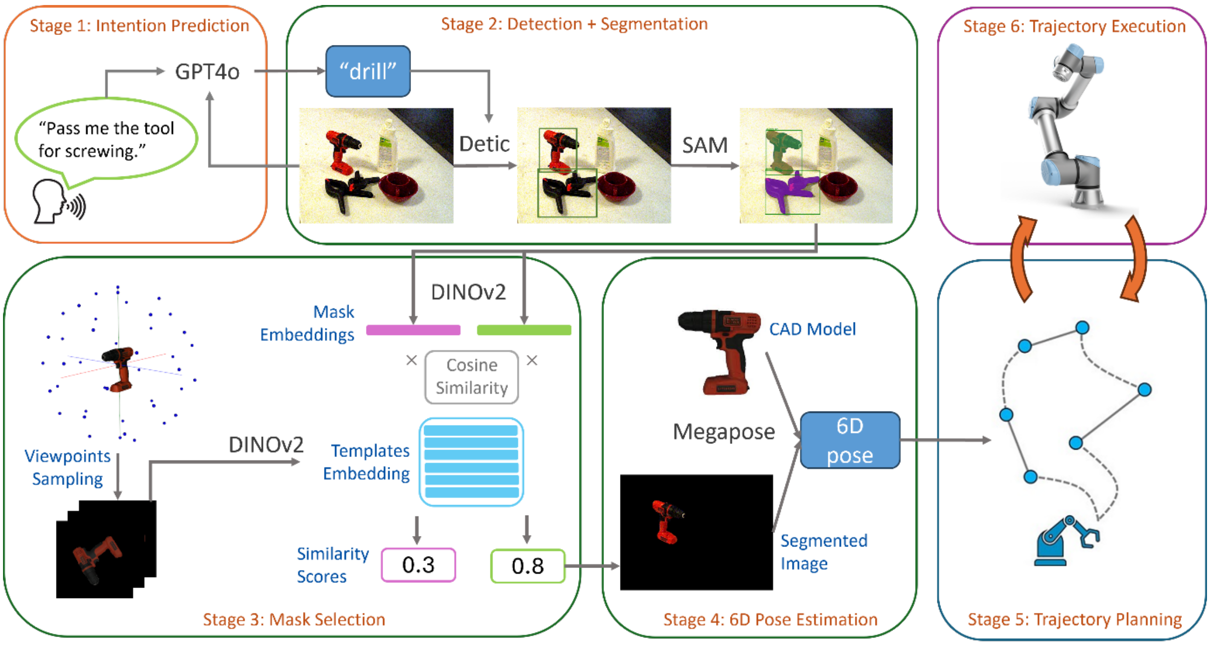

Figure 6–1: The detailed task execution workflow of the collaborative robot control system

The integration of GPT4o brings significant advantages over traditional command understanding techniques. However, as a closed-source commercial model, it cannot be deployed locally and must be accessed through OpenAI's API services. As introduced in Chapter 3, the system utilises the Azure Kinect camera that mounted on the UR5e robot, to capture both audio and visual input. When an operator issues a command, such as "Pass me the tool for screwing" during a task, the camera's integrated microphone array picks up the audio, which is then converted to text. Simultaneously, the camera captures an image of the workbench and the objects present.

Both the converted text and the captured image are sent to the GPT4o API for processing. The LLM analyses this multimodal input to determine which object the operator requires and whether it is available on the workbench. If the requested item is present, its name is returned and passed to the second stage, initiating the object pose estimation process. This sophisticated approach allows for more intuitive and context-aware human-robot interaction, enabling the robot to respond effectively to natural language requests while considering the visual environment.

Another detail worth noting is the relationship between the Stage 5: Trajectory Planning and Stage 6: Trajectory Execution. In the high-level system structure, their relationship is simplified as one-off execution, where the trajectory planning algorithm plans a global solution and executed by the robot. However, the machine-learned trajectory planning algorithm developed in this work performs online planning. The system keeps monitoring on the surrounding environment and plan trajectory incrementally. Thus, the process will keep alternating between trajectory planning and execution until the destination is successfully achieved.

To validate and present the developed collaborative industrial robot control system, this case study is designed with a robot assisted picking scenario. While this project focuses on industrial applications of cobot, the system is first tested on household items that available in the LISMS lab to validate the usability of the proposed method in real-life.

In the initial phase of the process, the cobot wait in a fixed pose, its end-effector angled at 45 degrees towards the workbench, as shown in Figure 6–2(a). This positioning ensures that both the human operator's hands and the objects on the workbench fall within the camera's FOV. From this standby position, the cobot awaits voice instructions to initiate its next action. In the meantime, the camera's position in the world frame is determined, as it is mounted on the robot's end-effector joint.

Once a voice command is received and processed, GPT4o identifies the required object. Then the object poses estimation algorithm springs into action and estimates the object pose in camera coordinate system. Transforming its pose to world coordinate frame is necessary to guide the gripper to reach the target pose. While this process traditionally involves a series of intricate matrix calculations and measurements, the digitalisation of the entire workbench, as described in Chapter 3, significantly simplifies and enhances the accuracy of these computations.

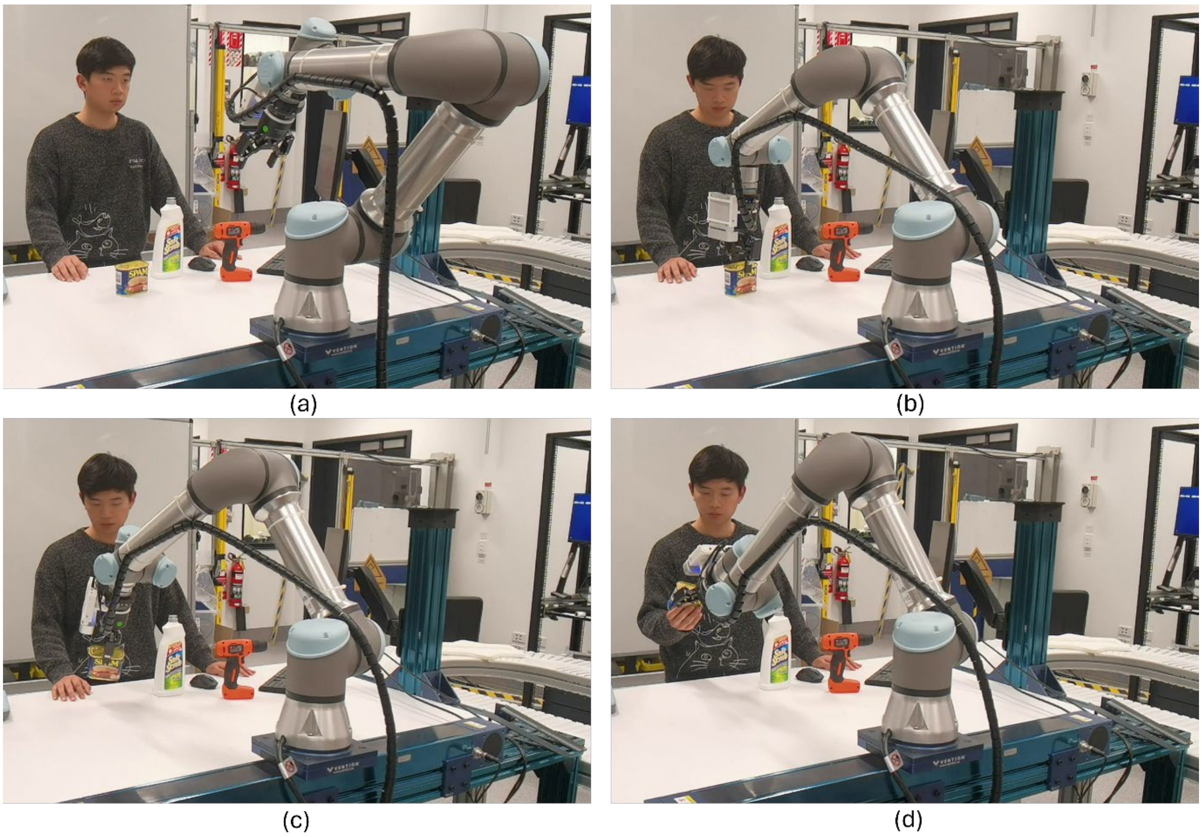

Figure 6–2: Robot waiting position (a) and execution sequence (b-d)

With the object 6D pose, the cobot trajectory planning and execution system can navigates towards the target object, grasps it, and proceeds to hand it over to the human co-worker, as shown in Figure 6–2(b-d). The cobot releases its grip only when it detects the force of the human taking the object. Following this transfer, the cobot returns to its initial waiting position, ready for the next instruction.

It's worth noting that while this picking process shares similarities with the test application described in Chapter 3, it represents an advancement in efficiency. The current process utilises two machine-learned algorithms developed in this project. This streamlined approach enhances the overall performance and responsiveness of the cobot system.

This case study demonstrates the sophisticated capabilities of the proposed system using two common household items: a spam can and a bleach bottle. To test GPT4o's decision-making abilities based on both text and visual input, the human commands were designed to avoid explicitly naming the objects. For instance, when requesting the spam can, the human worker stated, "I am hungry, give me something to eat." Similarly, to request the bleach bottle, "I need to do some cleaning job." In both scenarios, GPT4o successfully interpreted the implicit meaning of these commands and correctly identified the appropriate object from those available on the workbench.

The system's pose estimation capabilities are illustrated in Figure 6–3, which shows the results for both the spam can and the bleach bottle. For clarity, these results are visualised by rendering each object's CAD model in the estimated pose. While the rendered models do not align perfectly with the objects' original positions, the accuracy achieved is sufficient to guide the robot's gripping action effectively. This level of precision underscores the system's potential for practical applications.

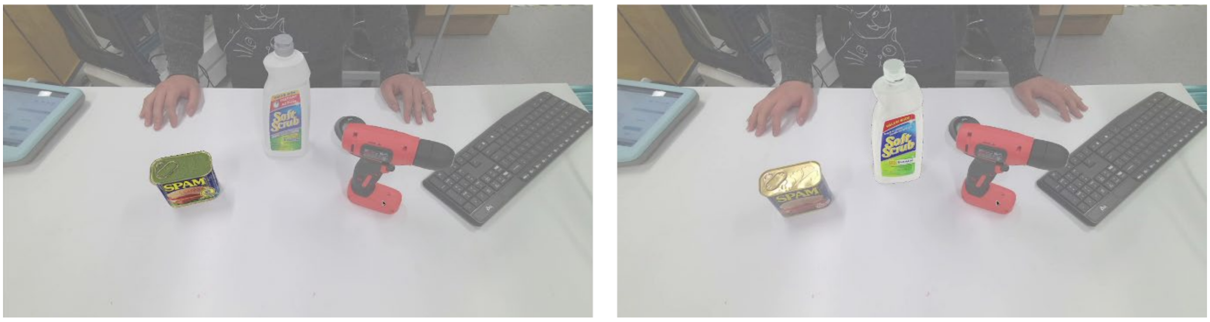

Figure 6–3: 6D pose estimation result of spam can (left) and bleach bottle (right)

By successfully handling these everyday items, the case study not only validates the effectiveness of the proposed system but also highlights its potential applications in assistive robotics. The ability to understand context-dependent requests and accurately manipulate common household objects demonstrates the system's versatility and its potential to enhance HRC in domestic and industrial settings alike.

6.2. Case Study 2: System adaption to New Objects

This case study further evaluates the adaptability and robustness of the proposed novel object pose estimation system, addressing concerns about its performance with truly novel objects. The study acknowledges that while the system has been tested on objects from the YCB-Video dataset, the novelty of these objects is questionable due to potential exposure in other training datasets.

To rigorously test the system's capability with genuinely novel objects, a custom-designed item was created: a yellow worm gear mounted on a black holder with a handle. This object was specifically designed and 3D printed to ensure it had never appeared in any existing dataset.

During the inference phase, the object's CAD model was imported into the project database for rendering template images.

The experimental setup for this case study mirrors the previous one, as illustrated in Figure 6–4. Notably, the power drill is intentionally placed in front of the worm gear, introducing an additional layer of complexity. This arrangement not only tests the system's generalisability to novel objects but also its robustness in handling occlusions, presenting a significantly more challenging scenario than dealing with unobstructed novel objects alone.

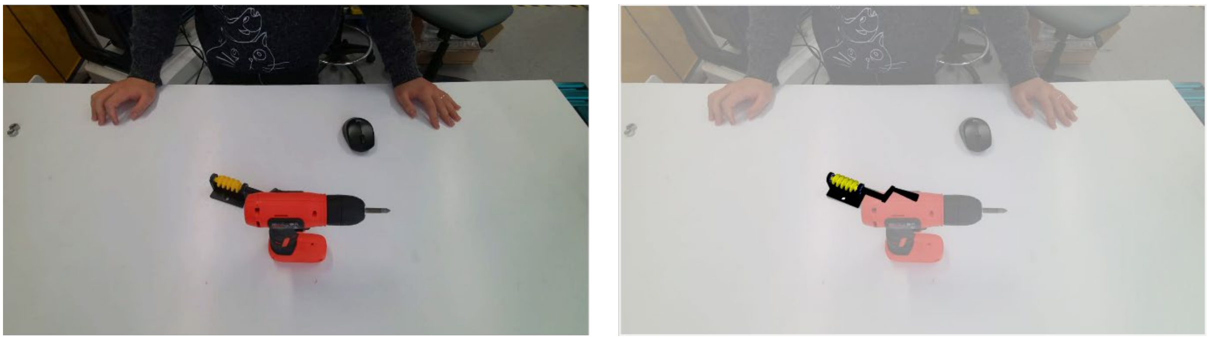

Figure 6–4: The captured scene (left) and estimation result (right)

As evidenced in Figure 6–4, the proposed system successfully estimated the pose of the custom-designed object, even under partial occlusion. This achievement demonstrates the system's ability to adapt to entirely new objects and its resilience in managing complex, real-world situations where target objects may be partially obscured.

This case study provides compelling evidence of the system's generalisability and adaptability. By successfully handling a truly novel, custom-designed object under challenging conditions, the system proves its potential for deployment in diverse industrial settings where it may encounter unfamiliar objects and complex spatial arrangements. The results underscore the robustness of the proposed approach in bridging the gap between laboratory testing and real-world application in human-robot collaborative environments.

6.3. Case Study 3: Human-Robot Collaborative Assembly

This case study presents an advanced human-robot collaborative assembly scenario, showcasing a proactive approach to robot assistance. The key difference lies in how the robot determines which object to pick up. As illustrated in Figure 6–5, the intention prediction module from Figure 6–1 is replaced by a task prediction module. This modification allows the system to operate without relying on human voice commands, representing a significant advancement from traditional to symbiotic collaborative robot control system.

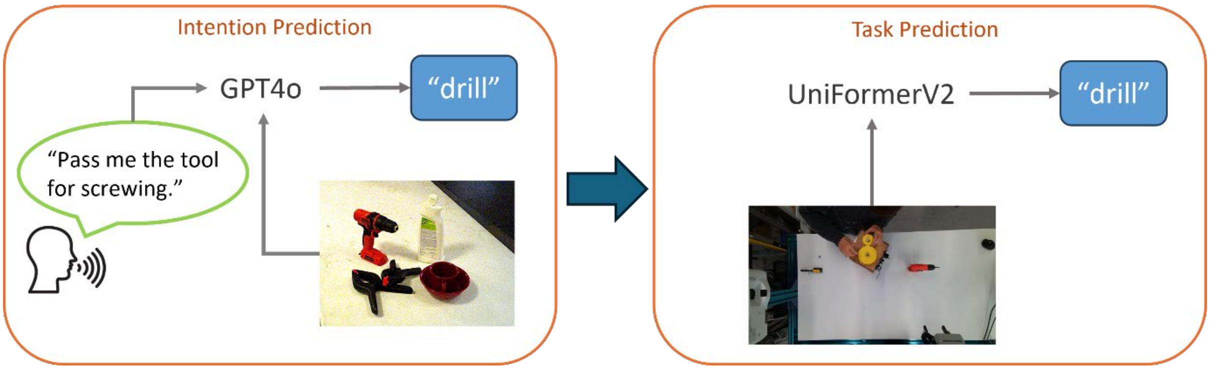

Figure 6–5: Replacing the intention prediction with task prediction module

At the core of this system is the UniFormerV2 (K. Li et al., 2022) model, a SOTA algorithm for spatiotemporal representation learning, which captures and understands patterns in data that vary across both space and time. Trained on the HA-ViD (H. Zheng et al., 2023) dataset, which features representative industrial human-robot collaborative assembly scenarios, the model analyses the assembly sequence in real-time. This capability enables the robot to predict the next assembly action and identify the required tools, enhancing the efficiency of the collaborative process.

To overcome the limitations of a single camera mounted on the robot, such as restricted FOV and potential occlusions, an additional camera is employed and mounted on the ceiling. As shown in Figure 6–6, this setup provides comprehensive coverage of the workbench area, ensuring reliable data input for the task analysis and prediction algorithm.

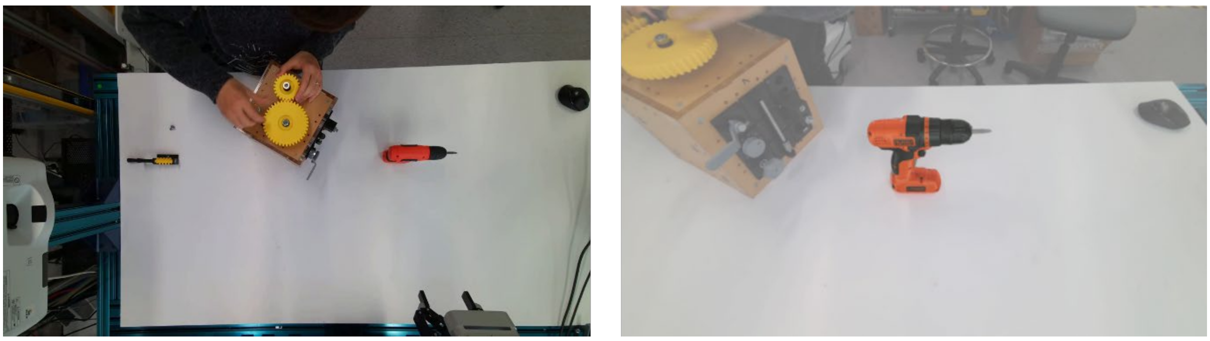

Figure 6–6: Top view for task prediction (left) and front view pose estimation result (right)

The case study focuses on a specific assembly task involving the installation of two round gears and a worm gear onto an assembly box, along with smaller components like washers, bolts, screws, and nuts. The assembly sequence begins with the round gears, followed by the worm gear installation, which requires a power drill for tightening screws.

As the worker progresses through the assembly, the UniFormerV2 model continuously analyses the task state. When it detects that the worker is about to install the worm gear, it anticipates the next step of tightening the screw with the predefined assembly sequence. Recognising that this step requires a power drill, the system activates the pose estimation algorithm to locate the tool. The robot then retrieves and hands over the power drill to the worker.

This case study highlights the adaptability of the proposed robot control structure outlined in Figure 6–1. By replacing the intention prediction stage with a task prediction module, the system demonstrates its flexibility in accommodating various input methods while maintaining its core functionality of accurate object pose estimation and collaborative assistance.

6.4. Discussion

These three case studies collectively demonstrate the versatility, robustness, and potential of the proposed robot control system in various HRC scenarios. Each study focuses on distinct aspects of the system's capabilities:

Case Study 1 showcases the integration of GPT4o for human intention prediction and scene understanding, highlighting its compatibility with the novel object pose estimation algorithm. This study emphasises the system's ability to interpret complex human intentions and translate them simple required object names.

Case Study 2 delves into the vision system's adaptability and generalisability, specifically testing its performance with certified novel objects. This study also demonstrates the system's robustness against occlusions, a common challenge in real-world environments. The successful handling of a custom-designed, 3D-printed object underscores the system's ability to work with truly unfamiliar items.

Case Study 3 represents a significant advancement towards symbiotic and proactive HRC. By replacing the human intention prediction module with a task prediction module, this study showcases the system's ability to operate autonomously without relying on explicit human commands. This proactive approach enables the robot to make decisions and provide assistance during collaboration, marking a shift towards more intuitive and efficient human-robot interaction.

These case studies not only validate the current capabilities of the proposed system but also provide valuable insights into its potential applications and future developments. As mentioned, the flexibility of the system architecture, particularly the final stage shown in Figure 6–1: A five-stage robot control structure for HRC, allows for further expansion. The robot's role could be extended beyond simple assistance to include more complex sequences of actions, enabling it to collaborate on tasks that a single human worker cannot complete alone.

This adaptability opens up possibilities for deploying the system in a wide range of industrial scenarios, from assembly lines to logistics operations, where the robot can seamlessly integrate with human workflows, anticipate needs, and provide timely support. The demonstrated capabilities in handling novel objects, understanding complex intentions, and operating proactively position this system as a promising solution for advancing human-robot collaboration across various industries.

For those interested in further verification, development, or integration of this work into their own projects, the source code and demonstration images are publicly available on GitHub at https://github.com/WanqingXia/HRC_DetAnyPose. This accessibility promotes transparency and encourages collaboration within the research community. However, it needs to notice that the original code does not guarantee fully replication of this project as some of the information are not correct. For example, the GPT4o API is a paid function, thus the API access token is not available in the source code.

Chapter 7 Conclusions and Future Work

This project has designed and implemented a collaborative industrial robot system leveraging machine learning technologies. Aimed at developing an adaptive collaborative robot control system to address the challenges posed by smart manufacturing, four objectives were established in Chapter 1. These objectives include (1) developing a digital model of the HRC workstation; (2) creating a novel objects 6D pose estimation pipeline; (3) designing a robot trajectory planning algorithm for proactive collision avoidance, and (4) implementing case studies to validate the developed systems. Previous chapters have reviewed the research gaps underlying these objectives and comprehensively reported on the implementation details for each.

This chapter begins with a recap of the research project, summarising and concluding the findings for each objective. It then outlines the societal contributions of this research. Based on the author's findings and challenges encountered during the research, the limitations of this work are presented. Finally, the author provides recommendations for future work, informed by the identified limitations and personal insights.

7.1. Recap of Research

This research aims to develop a collaborative industrial robot system tailored for mass personalised production. Machine-learned algorithms are leveraged to meet the demanding requirements of adaptability, generalisability, safety, and working efficiency in futuristic manufacturing environments.

The study begins by examining the growing trend of mass personalisation in the manufacturing industry, highlighting the necessity of integrating cobots and ML technologies to address this shift. With this foundation, the research establishes its main goal and breaks it down into four primary objectives, each with corresponding sub-objectives to measure progress and success.

A comprehensive literature review in Chapter 2 identifies key research gaps that have previously hindered the widespread deployment of cobots in mass personalised manufacturing settings. Chapter 3 then details the development of a HRC workstation, which serves as the foundational platform for this research, and presents the overall system architecture.

Chapter 4 and Chapter 5 delve into two critical aspects of the system: object pose estimation and trajectory planning, respectively. These components form the core of the cobot control system and are essential for achieving the project's objectives. Finally, Chapter 6 addresses the last objective by combining all the algorithms developed throughout this work with SOTA task analysis methods. This chapter validates the application of the cobot control system through relevant case studies, demonstrating its effectiveness in real-world scenarios.

7.2. Contributions

The research presented in this project offers significant contributions to the field of collaborative robot control systems, advancing the boundaries of HRC through innovative systems and algorithms. These advancements can be summarised in four key areas.

Firstly, the project establishes a solid foundation with the development of an HRC workstation. This workstation, encompassing both physical and digital components, serves as a model for creating shared human-robot workspaces. The system demonstrates notable advantages in enhanced perception, synergistic collaboration, safe innovation, and continuous improvement.

Secondly, the research introduces a novel object 6D pose estimation pipeline. This innovative approach leverages SOTA computer vision algorithms to detect and localise objects in 3D space using only simple CAD models as input, without requiring additional training or fine-tuning.

The method achieves an 75.8 AR score on the YCB-Video dataset, outperforming other leading novel object 6D pose estimation algorithms. While the processing time of 5.08 seconds per object does not meet real-time inferencing requirements, it still offers a faster solution compared to existing methods, providing a computationally efficient approach to visually guided robot grasping.

The third major contribution is a proactive trajectory planning solution designed for efficient robot operation in stochastic environments. This method utilises simulation-based RL that can be directly deployed on real robots, reducing training time while ensuring the safety of both humans and robots. The trained model achieves a 96.5% success rate for collision-free trajectory planning in environments with randomly moving human arms as obstacles. By incorporating predictive capabilities, this proactive planning approach significantly enhances robot task execution efficiency.

Finally, the project culminates in the development of a comprehensive collaborative industrial robot system. This system integrates all the aforementioned contributions along with customised task analysis methods, enabling symbiotic and proactive collaboration between humans and robots. The resulting system addresses key requirements for mass personalised manufacturing, including adaptability, scalability, generalizability, and improved working efficiency. Three case studies demonstrate the power and potential of this futuristic collaborative system, positioning it as a benchmark for future developments in the mass personalised manufacturing industry.

7.3. Limitations

The developed cobot control system and its sub-systems, while functional, still face several limitations due to time and resource constraints. These limitations, identified but not addressed in this work, primarily concern operation safety, scalability and generalisability, and real-time application capabilities.

The operation safety of the system is a significant concern, particularly regarding the developed trajectory planning algorithm. Although the trained RL model boasts a success rate of 96.5%, the remaining 3.5% of failed cases indicate that the planning trajectories are not yet sufficiently safe for all scenarios. Moreover, the model's effectiveness may further decline when confronted with multiple moving obstacles, as it was primarily trained to handle only one.

Scalability and generalisability are common challenges in ML algorithms, often designed for specific conditions and trained on designated datasets. In this work, the RL model is currently limited to functioning with the UR5e robot. While the reward functions designed can be applied to other robots, retraining would be necessary. The object pose estimation algorithm, though proposed to address generalisation issues and work on any object without retraining, still struggles in certain scenarios. For instance, it may fail to distinguish between visually similar objects of different sizes, as it relies on appearance rather than dimensional awareness.

The real-time application of the system presents another significant limitation. As shown in Table 4-2, the proposed algorithm typically requires around 5 seconds to estimate an object's pose. This duration falls far short of the 0.2-0.3 second requirement for real-time applications, which is often a minimum standard for vision algorithms in production environments. Further investigation revealed that the most time-consuming component is Detic, a large visual language model trained with transformers. While Detic offers high generalisability, its substantial model size inevitably leads to longer processing times, accounting for approximately 4.5 seconds of the total estimation time.

7.4. Recommended Future Work

Based on the research findings, identified limitations, and the author's personal insights, several recommendations for future work have been proposed to further develop this project. These improvements, which could not be addressed in the current work due to resource and time constraints, may bring the project closer to practical application in the manufacturing industry.

The first and foremost recommendation focuses on enhancing the safety level of the system, particularly by increasing the success rate of the RL model-controlled trajectory planning system to 100%. Given that the RL model is an unexplainable black box, direct modification of model parameters is not feasible. However, the model's training in a virtual environment with a physics engine allows for visualization of robot movements and the surrounding environment. This enables case-by-case analysis of failed scenarios to identify factors hindering the RL model's success rate. Additionally, more complex conditions need to be considered, such as multiple human body parts and objects moving in shared workspaces. For instance, a scenario where a human worker uses both hands, with one hand holding a tool, presents a challenge that the current model, trained on the assumption of a single human arm without tools, cannot address. Handling these complex conditions may necessitate significant changes in both the reward function and the RL algorithm.

To improve the generalisability and reduce the inference time of the object pose estimation system, replacing the Detic object detector is recommended. While Detic's ability to detect instances in images using human language prompts is powerful and generalises well, it is not ideal for industrial applications. The requirement to assign descriptive names to objects poses two main challenges in industrial settings: the need to use everyday language for naming, which precludes the use of specific model names or codes, and the difficulty in distinguishing similar-looking industrial parts with subtle differences. The recommendation is to develop a new zero-shot detector with expandable output channels, capable of learning to detect new object instances without updating the entire network or affecting the detection accuracy of other objects. Although such a detector is not currently available and would be challenging to develop, it is expected to significantly reduce inference time by limiting object categories and eliminating the need for text prompt inputs.

These proposed improvements aim to address the critical aspects of safety, generalisability, and real-time performance. By focusing on these areas, the project could potentially overcome its current limitations and move closer to practical implementation in manufacturing environments.

7.5. Concluding Remarks

This research project has made substantial progress in advancing collaborative robotics for mass personalised manufacturing. By developing an integrated system that combines innovative algorithms for object pose estimation, trajectory planning, and human-robot collaboration, the project has pushed the boundaries of what is possible in adaptive and efficient manufacturing environments.

The achievements in this work represent a significant step forward in creating symbiotic relationships between humans and robots in industrial settings. The novel approaches developed here not only improve upon existing methods but also open new avenues for future research and development in the field.

While challenges remain, particularly in areas of operational safety, generalisation and real-time processing, the foundation laid by this project provides a clear direction for future advancements. The insights gained and the systems developed serve as a valuable benchmark for the industry, pointing the way towards more flexible, safe, and efficient manufacturing processes.

As the manufacturing landscape continues to evolve towards mass personalisation, the work presented here stands as a testament to the potential of collaborative robotics. It offers a glimpse into a future where humans and robots work seamlessly together, adapting to changing demands with agility and precision. This research not only contributes to the academic field but also has the potential to revolutionize industrial practices, bringing us closer to the realisation of truly adaptive and responsive manufacturing systems.

Appendix I

```xml
<?xml version="1.0"?>
<robot name="workbench_assemble" xmlns:xacro="https://ros.org/wiki/xacro">

    <xacro:arg name="robot" default="workbench_assemble" />
    <xacro:property name="robot" value="$(arg robot)" />
    <xacro:property name="prefix" value="" />
    <!-- workbench-->
    <xacro:include filename="$(find vention_conveyor_description)/urdf/workbench_table.urdf.xacro" />

    <link name="world" />

    <!--joint that connects world and workbench-->
    <joint name="world_joint" type="fixed">
        <parent link="world" />
        <child link="workbench_base" />
        <origin xyz="0.0 0.0 0.0" rpy="0.0 0.0 0.0" />
    </joint>
    <xacro:workbench_table prefix="" />

    <!-- UR5e -->
    <xacro:include filename="$(find ur_description)/urdf/inc/ur5e_macro.xacro" />
    <xacro:arg name="transmission_hw_interface" default="hardware_interface/PositionJointInterface" />
    <xacro:arg name="joint_limit_params" default="$(find ur_description)/config/ur5e/joint_limits.yaml"/>
    <xacro:arg name="kinematics_params" default="$(find ur_description)/config/ur5e/default_kinematics.yaml"/>
    <xacro:ur5e_robot prefix="${prefix}"
    transmission_hw_interface="$(arg transmission_hw_interface)"
    joint_limits_parameters_file="$(arg joint_limit_params)"
    kinematics_parameters_file="$(arg kinematics_params)" />

    <!--joint that connects workbench and ur5e-->
    <joint name="attachment_joint" type="fixed">
        <parent link="workbench_mount" />
        <child link="${prefix}base_link" />
        <origin xyz="0.0 0.0 0.00" rpy="0 0.0 3.14159" />
    </joint>

    <!-- ############## FT SENSOR ############### -->
    <!-- gripper coupling (ISO 9409-1-50-4-M / Robotiq AGC-CPL-062-002_20171219) -->
    <xacro:property name="coupling_radius" value="0.0375" />
    <xacro:property name="coupling_outer_length" value="0.014" />
    <xacro:property name="coupling_inner_length" value="0.008" />
    <xacro:property name="coupling_mass" value="0.100" />
    <xacro:include filename="$(find robotiq_ft_sensor)/urdf/robotiq_ft300.urdf.xacro" />
    <xacro:robotiq_ft300 parent="${prefix}tool0" prefix="${prefix}">
        <origin xyz="0 0 ${-coupling_inner_length / 2}" rpy="0 0 ${-pi/2}" />
    </xacro:robotiq_ft300>
```

```xml
    <!-- ############### CAMERA MOUNT ################ -->
    <xacro:include filename="$(find azure_kinect_ros_driver)/urdf/camera_mount.urdf.xacro" />
    <xacro:azure_camera_mount prefix="${prefix}" parent="${prefix}robotiq_ft_frame_id" >
        <origin xyz="0.05 0.0375 0.04" rpy="${pi/2} ${pi} 0"/>
    </xacro:azure_camera_mount>

    <!-- ############### CAMERA ################ -->
    <xacro:include filename="$(find azure_kinect_ros_driver)/urdf/azure_kinect.urdf.xacro" />
    <xacro:azure_kinect prefix="${prefix}" parent="${prefix}camera_link" >
        <origin xyz="0.05 -0.04 0.106" rpy="0 0 ${-pi/2}"/>
    </xacro:azure_kinect>

    <!-- ############### GRIPPER ################ -->
    <xacro:include filename="$(find robotiq_85_description)/urdf/robotiq_85_gripper.urdf.xacro" />
    <joint name="${prefix}robotiq_coupler_joint" type="fixed">
        <parent link="${prefix}tool_link"/>
        <child link="${prefix}gripper_coupler_link"/>
        <origin xyz="0.05 0.025 0.04" rpy="${-pi/2} 0 0" />
    </joint>

    <link name="${prefix}gripper_coupler_link">
        <visual>
            <geometry>
                <cylinder radius="0.04" length="0.017"/>
            </geometry>
            <material name="black">
                <color rgba="1 1 1 1.0"/>
            </material>
        </visual>
        <collision>
            <geometry>
                <cylinder radius="0.04" length="0.017"/>
            </geometry>
        </collision>
    </link>

    <xacro:robotiq_85_gripper prefix="${prefix}" parent="${prefix}gripper_coupler_link" >
        <origin xyz="0 0 -0.008" rpy="0 ${pi/2} ${-pi/2}" />
    </xacro:robotiq_85_gripper>

</robot>
```

Appendix II

Journal publication - Xia, W., Zheng, H., Xu, W., & Xu, X. (2025). Large vision-language models enabled novel objects 6D pose estimation for human-robot collaboration. Robotics and Computer-Integrated Manufacturing, 95, 103030. https://doi.org/10.1016/j.rcim.2025.103030

Robotics and Computer–Integrated Manufacturing 95 (2025) 103030

Contents lists available at ScienceDirect

Robotics and Computer-Integrated Manufacturing

journal homepage: www.elsevier.com/locate/rcim

Large vision-language models enabled novel objects 6D pose estimation for human-robot collaboration

Wanqing Xia *, Hao Zheng, Weiliang Xu, Xun Xu

Department of Mechanical and Mechatronics Engineering, The University of Auckland, Auckland, New Zealand

ARTICLE INFO

Keywords:
Machine vision
Vision-language models
6d pose estimation
Human-robot collaboration
Machine learning

ABSTRACT

Six-Degree-of-Freedom (6D) pose estimation is essential for robotic manipulation tasks, especially in human-robot collaboration environments. Recently, 6D pose estimation has been extended from seen objects to novel objects due to the frequent encounters with unfamiliar items in real-life scenarios. This paper presents a three-stage pipeline for 6D pose estimation of previously unseen objects, leveraging the capabilities of large vision-language models. Our approach consists of vision-language model-based object detection and segmentation, mask selection with pose hypothesis generated from CAD models, and refinement and scoring of pose candidates. We evaluate our method on the YCB-Video dataset, achieving a state-of-the-art Average Recall (AR) score of 75.8 with RGB-D images, demonstrating its effectiveness in accurately estimating 6D poses for a diverse range of objects. The effectiveness of each operation stage is investigated in the ablation study. To validate the practical applicability of our approach, we conduct case studies on a real-world robotic platform, focusing on object pick-up tasks by integrating our 6D pose estimation pipeline with human intention prediction and task analysis algorithms. Results show that the proposed method can effectively handle novel objects in our test environments, as demonstrated through the YCB dataset evaluation and case studies. Our work contributes to the field of human-robot collaboration by introducing a flexible, generalizable approach to 6D pose estimation, enabling robots to adapt to new objects without requiring extensive retraining—a vital capability for advancing human-robot collaboration in dynamic environments. More information can be found in the project GitHub page: https://github.com/WanqingXia/HRC_DetAnyPose.

1. Introduction

Human-robot collaboration (HRC) refers to the synergistic collaboration between human workers and robots [1,2]. This collaboration aims to combine the strengths of both humans (flexibility, problem-solving, and adaptability) and robots (precision, strength, and endurance) to enhance overall productivity and efficiency [3,4]. This collaboration often involves advanced sensors, artificial intelligence (AI), and adaptive control systems that enable robots to work safely alongside humans, sharing the same workspace and tasks [5]. Central to this collaboration is the accurate 6D pose estimation of objects, a key task in computer vision (CV) and robotics that determines an object's position and orientation in three-dimensional space. In the context of HRC, precise 6D pose estimation empowers robots to interact effectively with their environment, grasping and manipulating objects with high accuracy [6,7], thereby enhancing the overall efficiency and capabilities of HRC.

Traditionally, 6D pose estimation methods have relied on learning-based approaches, training specialized neural networks on specific object datasets [8–12]. While effective for known objects, these approaches face significant limitations when encountering novel items. This limited generalizability creates two critical problems: the need for extensive time-consuming training and the demand for large datasets. The time-consuming nature of training neural networks is unavoidable. The manufacturing industry is heading towards mass personalisation, where new parts are introduced to the manufacturing system frequently. When new objects become available, retraining the entire network is the only option, significantly increasing the time and computational resources required. Creating comprehensive 6D pose estimation datasets for each new item poses significant barriers to scalability and real-world application. Unlike image classification, detection, or segmentation tasks, the ground truth values for 6D pose cannot be evaluated by the human eye alone. It often requires expert knowledge and extensive measurements or specifically designed object positioning to obtain

* Corresponding author.
E-mail address: wanqing.xia@auckland.ac.nz (W. Xia).

https://doi.org/10.1016/j.rcim.2025.103030
Received 23 November 2024; Received in revised form 21 February 2025; Accepted 10 April 2025
Available online 26 April 2025

Appendix III

Conference publication - Xia, W., Lu, Y., Xu, W., & Xu, X. (2024). Deep reinforcement learning based proactive dynamic obstacle avoidance for safe human-robot collaboration. Manufacturing Letters, 41, 1246-1256.

ELSEVIER

Available online at www.sciencedirect.com

Manufacturing Letters

Manufacturing Letters 41 (2024) 1246–1256

52nd SME North American Manufacturing Research Conference (NAMRC 52, 2024)

Deep reinforcement learning based proactive dynamic obstacle avoidance for safe human-robot collaboration

Wanqing Xia[a,*], Yuqian Lu[a], Weiliang Xu[a], Xun Xu[a]

[a] *The University of Auckland, 20 Symond Street, Auckland 1010, New Zealand*

Abstract

Ensuring the health and safety of human operators is paramount in manufacturing, particularly in human-robot collaborative environments. In this paper, we present a deep reinforcement learning-based trajectory planning method for a robotic manipulator designed to avoid collisions with human body parts in real-time while achieving its goal. We modelled the human arm as a freely moving cylinder in 3D space and formulated the dynamic obstacle avoidance problem as a Markov decision process. The algorithm was tested in a simulated environment that closely mimics our laboratory environment, with the goal of training a deep reinforcement learning model for autonomous task completion. A composite reward function was developed to balance the effects of different environmental variables, and the soft-actor critic algorithm was employed. The trained model demonstrated a 93% success rate in avoiding dynamic obstacles while achieving its goals when tested on a generated data set.

Keywords: *Human-robot collaboration; Dynamic obstacle avoidance; Deep reinforcement learning; Reward engineering*

1. Introduction

Robotic manipulators are increasingly becoming integral components of the manufacturing industry worldwide, enhancing production efficiency and enabling the mass production of products at more affordable prices. Research indicates that integrating the adaptability and flexibility of humans with the precision and endurance of robots can result in a level of productivity that surpasses what could be achieved by either human or robotic systems alone [19]. To actualise cost-competitive manufacturing through human-robot collaboration, it is imperative to remove the physical barriers that currently separate humans and robots in future smart manufacturing environments. This will facilitate closer interactions between humans and robots [24]. However, it is crucial that the safety of human operators is not compromised in the pursuit of higher productivity. This necessitates advancements in reliable 3D perception, high-level planning, and sophisticated control within manufacturing systems [14]. Specifically, for human-robot collaborative manipulators, it is vital to have a dependable environment perception system capable of detecting and locating the human body among other potential obstacles in the workspace. Additionally, these manipulators must possess the decision-making capabilities required to complete tasks in constrained environments. In Fig 1, we showcase a human-robot collaborative assembly scenario set up in the Laboratory for Industry 4.0 Smart Manufacturing Systems (LISMS) at the University of Auckland. The UR5e robot featured in this scenario is outfitted with an RGB-depth (RGB-D) camera for environmental perception, a force and torque sensor, and a parallel gripper to facilitate task completion and handover to the human operator. Importantly, the UR5e robot is controlled by a desktop computer equipped with a high-performance graphics card, necessary for running sophisticated artificial intelligence control algorithms.

Corresponding author. Tel.: +64-0212993410
E-mail address: wxia612@aucklanduni.ac.nz (Wanqing Xia).

Appendix IV

Conference publication - Xia, W., Lu, Y., & Xu, X. (2023, November). Application and Evaluation of Soft-Actor Critic Reinforcement Learning in Constrained Trajectory Planning for 6DOF Robotic Manipulators. In 2023 29th International Conference on Mechatronics and Machine Vision in Practice (M2VIP) (pp. 1-6). IEEE.

2023 29th International Conference on Mechatronics and Machine Vision in Practice (M2VIP) | 979-8-3503-2562-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/M2VIP58386.2023.10413400

Application and Evaluation of Soft-Actor Critic Reinforcement Learning in Constrained Trajectory Planning for 6DOF Robotic Manipulators

Wanqing Xia
Mechanical and Mechatronics Engineering
The University of Auckland
Auckland, New Zealand
wxia612@aucklanduni.ac.nz

Yuqian Lu
Mechanical and Mechatronics Engineering
The University of Auckland
Auckland, New Zealand
yuqian.lu@auckland.ac.nz

Xun Xu
Mechanical and Mechatronics Engineering
The University of Auckland
Auckland, New Zealand
x.xu@auckland.ac.nz

***Abstract*—In the field of robotic manipulator operations, precise trajectory planning for the end-effector's position and orientation is crucial, especially in tasks such as grasping a bottle by its neck. This paper presents a novel approach utilizing Reinforcement Learning to address this issue. Specifically, we employed the Soft Actor-Critic and Hindsight Experience Replay algorithm to train a UR5e manipulator in a simulated environment, incorporating a unique design for state, action, and reward. Through comparative analysis with other reward function designs, we found that our trained Reinforcement Learning model generated a more efficient trajectory and achieved a significantly higher success rate. This study underscores the potential of our approach for enhancing the trajectory planning of robotic manipulator operations.**

***Index Terms*—Reinforcement Learning, Robot Control, Trajectory Planning**

I. INTRODUCTION

Robotic manipulators are presently utilized extensively across the globe within the manufacturing industry, bolstering high production efficiency. In comparison to traditional manufacturing settings, the future-oriented smart manufacturing system necessitates increased responsiveness, flexibility, adaptability, and cost-competitiveness [1]. However, the execution of intricate robotic tasks in smart manufacturing introduces challenges pertaining to reliable 3D perception, high-level planning, and sophisticated control [2]. The implementation of a task and navigation towards the stipulated pose requires planning a brief, collision-free trajectory for the robotic manipulator. The advent of neural networks has spurred numerous research initiatives aimed at resolving the trajectory planning problem via reinforcement learning (RL).

In recent years, various deep reinforcement learning (DRL) algorithms have been proposed for solving excessive amount of robotic tasks, such as bin picking [3], door opening [4], and object stacking [5]. These algorithms typically employ vision sensing and deep neural networks to enable the model to learn high-level, multi-step tasks through a trial-and-error approach. Nonetheless, this approach may precipitate a lack of flexibility as the task parameters are embedded within the neural network, and the model is limited to solving a specific problem. Moreover, the necessity of repeatedly learning the same basic actions can be time-consuming and inefficient. An alternative solution may involve decomposing tasks into smaller, more modular units to enhance the efficiency and flexibility of learning in robotics. For instance, a grasping task can be segmented into an action sequence involving 1 recognizing the object, 2 analyzing the most stable grasping point and pose, 3 moving the Tool Centre Point (TCP) of the robot to the desired pose, and finally, 4 closing the gripper. Employing rapid and reliable trajectory planning algorithms for action 3 allows the DRL model to more expeditiously learn actions 1 and 2. This lets us to decompose complex policies into a sequence of primitive actions that can be independently learned and reused, thus potentially increasing the efficiency and flexibility of the learning process for robotic tasks.

The primary focus of this paper is the application of reinforcement learning to solve the trajectory planning problem for a robotic manipulator, with particular emphasis on the fundamental reaching action constrained by the TCP position and orientation. Although RL has been previously deployed in trajectory planning, most approaches solely consider reaching the desired position devoid of collisions, neglecting TCP orientation. Nonetheless, it is imperative to constrain both the target location and the spatial orientation in robotic manipulation for a comprehensive spectrum of tasks. Our objective is to formulate an RL model capable of planning a trajectory and controlling the manipulator to closely achieve the strictly constrained goal TCP position and orientation for

References

Aaltonen, I., Salmi, T., & Marstio, I. (2018). Refining levels of collaboration to support the design and evaluation of human-robot interaction in the manufacturing industry. *Procedia CIRP*, *72*(March), 93–98. https://doi.org/10.1016/j.procir.2018.03.214

Abbas, A. N., & Beleznai, C. (2024). TalkWithMachines: Enhancing Human-Robot Interaction Through Large/Vision Language Models. *2024 Eighth IEEE International Conference on Robotic Computing (IRC)*, 253–258.

Ahn, M., Brohan, A., Brown, N., Chebotar, Y., Cortes, O., David, B., Finn, C., Fu, C., Gopalakrishnan, K., & Hausman, K. (2022). Do as i can, not as i say: Grounding language in robotic affordances. *ArXiv Preprint ArXiv:2204.01691*.

Ajoudani, A., Zanchettin, A. M., Ivaldi, S., Albu-Schäffer, A., Kosuge, K., & Khatib, O. (2018). Progress and prospects of the human–robot collaboration. *Autonomous Robots*, *42*(5), 957–975. https://doi.org/10.1007/s10514-017-9677-2

Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., & Reynolds, M. (2022). Flamingo: a visual language model for few-shot learning. *Advances in Neural Information Processing Systems*, *35*, 23716–23736.

Alexander Obaigbena, Oluwaseun Augustine Lottu, Ejike David Ugwuanyi, Boma Sonimitiem Jacks, Enoch Oluwademilade Sodiya, Obinna Donald Daraojimba, & Oluwaseun Augustine Lottu. (2024). AI and human-robot interaction: A review of recent advances and challenges. *GSC Advanced Research and Reviews*, *18*(2), 321–330. https://doi.org/10.30574/gscarr.2024.18.2.0070

Badillo, S., Banfai, B., Birzele, F., Davydov, I. I., Hutchinson, L., Kam-Thong, T., Siebourg-Polster, J., Steiert, B., & Zhang, J. D. (2020). An Introduction to Machine Learning. *Clinical Pharmacology and Therapeutics*, *107*(4), 871–885. https://doi.org/10.1002/cpt.1796

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., & Henighan, T. (2022). Training a helpful and harmless assistant with reinforcement learning from human feedback. *ArXiv Preprint ArXiv:2204.05862*.

Barravecchia, F., Bartolomei, M., Mastrogiacomo, L., & Franceschini, F. (2025). Designing

Symbiotic Human–Robot Collaboration in assembly tasks. *Production Engineering*, 1–33.

Bauer, W., Bender, M., & Martin, B. (2016). Lightweight robots in manual assembly – best to start simply. *Frauenhofer-Institut Für Arbeitswirtschaft Und Organisation IAO, Stuttgart, 1*.

Bay, H., Tuytelaars, T., & Van Gool, L. (2006). Surf: Speeded up robust features. *Computer Vision–ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7-13, 2006. Proceedings, Part I 9*, 404–417.

Besl, P. J., & McKay, N. D. (1992). Method for registration of 3-D shapes. *Sensor Fusion IV: Control Paradigms and Data Structures*, *1611*, 586–606.

Bisk, Y., Zellers, R., Gao, J., & Choi, Y. (2020). Piqa: Reasoning about physical commonsense in natural language. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(05), 7432–7439.

Bohlin, R., & Kavraki, L. E. (2000). Path planning using Lazy PRM. *Proceedings-IEEE International Conference on Robotics and Automation*, *1*(April), 521–528. https://doi.org/10.1109/ROBOT.2000.844107

Boschetti, G., Faccio, M., & Granata, I. (2023). Human-Centered Design for Productivity and Safety in Collaborative Robots Cells: A New Methodological Approach. *Electronics (Switzerland)*, *12*(1), 1–16. https://doi.org/10.3390/electronics12010167

Brock, O. (2002). Elastic Strips : A Framework for Motion Generation in Human Environments. *The International Journal of Robotics Research*, *21*(12), 1031–1052. http://ijr.sagepub.com/content/21/12/1031.short

Bukschat, Y., & Vetter, M. (2020). EfficientPose: An efficient, accurate and scalable end-to-end 6D multi object pose estimation approach. *ArXiv Preprint*. http://arxiv.org/abs/2011.04307

Cao, Z., Simon, T., Wei, S.-E., & Yaser, S. (2017). Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-End Object Detection with Transformers. *Lecture Notes in Computer Science*

(Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), *12346 LNCS*, 213–229.

Chand, S., & Lu, Y. (2023). Dual task scheduling strategy for personalized multi-objective optimization of cycle time and fatigue in human-robot collaboration. *Manufacturing Letters*, *35*, 88–95. https://doi.org/10.1016/j.mfglet.2023.08.064

Chen, J., Sun, M., Bao, T., Zhao, R., Wu, L., & He, Z. (2023). ZeroPose: CAD-Model-based Zero-Shot Pose Estimation. *ArXiv Preprint*. http://arxiv.org/abs/2305.17934

Chen, L.-C., Papandreou, G., Schroff, F., & Adam, H. (2017). Rethinking Atrous Convolution for Semantic Image Segmentation. *ArXiv Preprint*. http://arxiv.org/abs/1706.05587

Chen, L. C., Papandreou, G., Kokkinos, I., Murphy, K., & Yuille, A. L. (2018). DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *40*(4), 834–848. https://doi.org/10.1109/TPAMI.2017.2699184

Chen, L. C., Zhu, Y., Papandreou, G., Schroff, F., & Adam, H. (2018). Encoder-decoder with atrous separable convolution for semantic image segmentation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *11211 LNCS*, 833–851. https://doi.org/10.1007/978-3-030-01234-2_49

Chen, L., Jiang, Z., Cheng, L., Knoll, A. C., Zhou, M., & Lu, B. (2022). Deep Reinforcement Learning Based Trajectory Planning Under Uncertain Constraints. *Frontiers in Neurorobotics*, *16*(May), 1–10. https://doi.org/10.3389/fnbot.2022.883562

Chen, P., Pei, J., Lu, W., & Li, M. (2022). A deep reinforcement learning based method for real-time path planning and dynamic obstacle avoidance. *Neurocomputing*, *497*, 64–75. https://doi.org/10.1016/j.neucom.2022.05.006

Chen, Y., Ge, Y., Li, Y., Ge, Y., Ding, M., Shan, Y., & Liu, X. (2024). Moto: Latent motion token as the bridging language for robot manipulation. *ArXiv Preprint ArXiv:2412.04445*.

Chiacchio, F., Petropoulos, G., & Pichler, D. (2018). The Impact of Industrial Robots on EU Employment and Wages: A Local Labour Market Approach. *McKinsey Global Institute*, *10*(March), 1–28.

Dijkstra, E. W. (2022). A note on two problems in connexion with graphs. In *Edsger Wybe*

Dijkstra: his life, work, and legacy (pp. 287–290).

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). an Image Is Worth 16X16 Words: Transformers for Image Recognition At Scale. *ICLR 2021 - 9th International Conference on Learning Representations*.

Driess, D., Xia, F., Sajjadi, M. S. M., Lynch, C., Chowdhery, A., Wahid, A., Tompson, J., Vuong, Q., Yu, T., & Huang, W. (2023). *Palm-e: An embodied multimodal language model*.

Duan, K., Bai, S., Xie, L., Qi, H., Huang, Q., & Tian, Q. (2019). CenterNet: Keypoint triplets for object detection. *Proceedings of the IEEE International Conference on Computer Vision*, *2019-Octob*, 6568–6577. https://doi.org/10.1109/ICCV.2019.00667

El-Shamouty, M., Wu, X., Yang, S., Albus, M., & Huber, M. F. (2020). Towards Safe Human-Robot Collaboration Using Deep Reinforcement Learning. *Proceedings - IEEE International Conference on Robotics and Automation*, 4899–4905. https://doi.org/10.1109/ICRA40945.2020.9196924

Elbanhawi, M., & Simic, M. (2014). Sampling-based robot motion planning: A review. *IEEE Access*, *2*, 56–77. https://doi.org/10.1109/ACCESS.2014.2302442

Ferguson, D., Likhachev, M., & Stentz, A. (2005). A guide to heuristic-based path planning. *Proceedings of the International Workshop on Planning under Uncertainty for Autonomous Systems, International Conference on Automated Planning and Scheduling (ICAPS)*, 1–10. http://cs.cmu.edu/afs/cs.cmu.edu/Web/People/maxim/files/hsplanguide_icaps05ws.pdf

Fischler, M. A., & Bolles, R. C. (1981). Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, *24*(6), 381–395.

Flacco, F., Kröger, T., De Luca, A., & Khatib, O. (2012). A depth space approach to human-robot collision avoidance. *Proceedings - IEEE International Conference on Robotics and Automation*, 338–345. https://doi.org/10.1109/ICRA.2012.6225245

Fox, D., Burgard, W., & Thrun, S. (1997). The dynamic window approach to collision avoidance. *IEEE Robotics and Automation Magazine*, *4*(1), 23–33.

https://doi.org/10.1109/100.580977

Fujimoto, S., Van Hoof, H., & Meger, D. (2018). Addressing Function Approximation Error in Actor-Critic Methods. *35th International Conference on Machine Learning, ICML 2018*, *4*, 2587–2601.

Galin, R., & Meshcheryakov, R. (2019). Automation and robotics in the context of Industry 4.0: The shift to collaborative robots. *IOP Conference Series: Materials Science and Engineering*, *537*(3), 0–5. https://doi.org/10.1088/1757-899X/537/3/032073

Garcia-Garcia, A., Orts-Escolano, S., Oprea, S., Villena-Martinez, V., & Garcia-Rodriguez, J. (2017). A Review on Deep Learning Techniques Applied to Semantic Segmentation. *ArXiv Preprint*, 1–23. http://arxiv.org/abs/1704.06857

Ge, S. S., & Cui, Y. J. (2002). Dynamic motion planning for mobile robots using potential field method. *Autonomous Robots*, *13*(3), 207–222. https://doi.org/10.1023/A:1020564024509

Georgiou, T., Liu, Y., Chen, W., & Lew, M. (2020). A survey of traditional and deep learning-based feature descriptors for high dimensional data in computer vision. *International Journal of Multimedia Information Retrieval*, *9*(3), 135–170. https://doi.org/10.1007/s13735-019-00183-w

Girshick, R. (2015). Fast R-CNN. *Proceedings of the IEEE International Conference on Computer Vision*, *2015 Inter*, 1440–1448. https://doi.org/10.1109/ICCV.2015.169

Girshick, R., Donahue, J., Darrell, T., & Malik, J. (2014). Rich feature hierarchies for accurate object detection and semantic segmentation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 580–587. https://doi.org/10.1109/CVPR.2014.81

Gu, S., Lillicrap, T., Sutskever, I., & Levine, S. (2016). Continuous Deep Q-Learning with Model-based Acceleration. *International Conference on Machine Learning*, 2829–2838.

Haarnoja, T., Zhou, A., Abbeel, P., & Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *35th International Conference on Machine Learning, ICML 2018*, *5*, 2976–2989.

Haddadin, S., Urbanek, H., Parusel, S., Burschka, D., Roßmann, J., Albu-Schäffer, A., & Hirzinger, G. (2010). Real-time reactive motion generation based on variable attractor dynamics and shaped velocities. *IEEE/RSJ 2010 International Conference on Intelligent*

Robots and Systems, IROS 2010 - Conference Proceedings, 3109–3116. https://doi.org/10.1109/IROS.2010.5650246

Hadfield-Menell, D., Milli, S., Abbeel, P., Russell, S., & Dragan, A. (2017). Inverse Reward Design. *Proceedings of the 31st International Conference on Neural Information Processing Systems*, *Nips*. https://doi.org/10.48550/arXiv.1711.02827

Han, X., Chen, S., Fu, Z., Feng, Z., Fan, L., An, D., Wang, C., Guo, L., Meng, W., & Zhang, X. (2025). Multimodal fusion and vision-language models: A survey for robot vision. *ArXiv Preprint ArXiv:2504.02477*.

Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, *42*(1–3), 335–346.

Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *Systems Science and Cybernetics*, *4*(2), 100–107.

Haugaard, R. L., & Buch, A. G. (2022). SurfEmb: Dense and Continuous Correspondence Distributions for Object Pose Estimation with Learnt Surface Embeddings. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *2022-June*, 6739–6748. https://doi.org/10.1109/CVPR52688.2022.00663

He, K., Gkioxari, G., Dollar, P., & Girshick, R. (2017). Mask R-CNN. *2017 IEEE International Conference on Computer Vision (ICCV)*, 2980–2988. https://doi.org/10.1109/ICCV.2017.322

He, Y., Huang, H., Fan, H., Chen, Q., & Sun, J. (2021). FFB6D: A Full Flow Bidirectional Fusion Network for 6D Pose Estimation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 3002–3012. https://doi.org/10.1109/CVPR46437.2021.00302

Huang, W., Abbeel, P., Pathak, D., & Mordatch, I. (2022). Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. *International Conference on Machine Learning*, 9118–9147.

Huang, W., Xia, F., Xiao, T., Chan, H., Liang, J., Florence, P., Zeng, A., Tompson, J., Mordatch, I., & Chebotar, Y. (2022). Inner monologue: Embodied reasoning through planning with language models. *ArXiv Preprint ArXiv:2207.05608*.

Humayun, M. (2021). Industrial Revolution 5.0 and the Role of Cutting Edge Technologies.

International Journal of Advanced Computer Science and Applications, *12*(12), 605–615. https://doi.org/10.14569/IJACSA.2021.0121276

Hurst, A., Lerer, A., Goucher, A. P., Perelman, A., Ramesh, A., Clark, A., Ostrow, A. J., Welihinda, A., Hayes, A., & Radford, A. (2024). Gpt-4o system card. *ArXiv Preprint ArXiv:2410.21276*.

Ibarz, J., Tan, J., Finn, C., Kalakrishnan, M., Pastor, P., & Levine, S. (2021). How to train your robot with deep reinforcement learning: lessons we have learned. *International Journal of Robotics Research*, *40*(4–5), 698–721. https://doi.org/10.1177/0278364920987859

Ibrahim, H., Alsafi, H., & Babar, M. Z. (2025). *LLMBot: Multi-Agent Robotic Systems for Adaptive Task Execution*.

Iqbal, T., & Riek, L. D. (2017). Coordination dynamics in multihuman multirobot teams. *IEEE Robotics and Automation Letters*, *2*(3), 1712–1717. https://doi.org/10.1109/LRA.2017.2673864

ISO, I. S. O. (2011a). ISO 10218-1: 2011 robots and robotic devices—safety requirements for industrial robots—part 1: Robots. *International Organization for Standardization*.

ISO, I. S. O. (2011b). ISO 10218-2: 2011: Robots and robotic devices–Safety requirements for industrial robots–Part 2: Robot systems and integration. *Geneva, Switzerland: International Organization for Standardization*, *3*.

ISO, I. S. O. (2016). ISO/TS 15066: 2016: Robots and robotic devices–collaborative robots. *International Organization for Standardization: Geneva, Switzerland*.

Jaillet, L., & Siméon, T. (2004). A PRM-based motion planner for dynamically changing environments. *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, *2*, 1606–1611. https://doi.org/10.1109/iros.2004.1389625

Jeong, H., Lee, H., Kim, C., & Shin, S. (2024). A survey of robot intelligence with large language models. *Applied Sciences*, *14*(19), 8868.

Kadir, B. A., Broberg, O., & Conceição, C. S. da. (2019). Current research and future perspectives on human factors and ergonomics in Industry 4.0. *Computers and Industrial Engineering*, *137*(August), 106004. https://doi.org/10.1016/j.cie.2019.106004

Kalakrishnan, M., Chitta, S., Theodorou, E., Pastor, P., & Schaal, S. (2011). STOMP:

Stochastic trajectory optimization for motion planning. *Proceedings - IEEE International Conference on Robotics and Automation*, 4569–4574. https://doi.org/10.1109/ICRA.2011.5980280

Karaman, S., & Frazzoli, E. (2011). Incremental sampling-based algorithms for optimal motion planning. *Robotics: Science and Systems*, *6*, 267–274. https://doi.org/10.15607/rss.2010.vi.034

Kavraki, L. E., Švestka, P., Latombe, J. C., & Overmars, M. H. (1996). Probabilistic roadmaps for path planning in high-dimensional configuration spaces. *IEEE Transactions on Robotics and Automation*, *12*(4), 566–580. https://doi.org/10.1109/70.508439

Khatib, O. (1985). Real-time obstacle avoidance for manipulators and mobile robots. *Proceedings - IEEE International Conference on Robotics and Automation*, 500–505. https://doi.org/10.1109/ROBOT.1985.1087247

Kim, C. Y., Lee, C. P., & Mutlu, B. (2024). Understanding large-language model (llm)-powered human-robot interaction. *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 371–380.

Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A. C., Lo, W.-Y., Dollár, P., & Girshick, R. (2023). Segment Anything. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 3992–4003. https://doi.org/10.1109/ICCV51070.2023.00371

Kober, J., Bagnell, J. A., & Peters, J. (2013). Reinforcement learning in robotics: A survey. *International Journal of Robotics Research*, *32*(11), 1238–1274. https://doi.org/10.1177/0278364913495721

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). Large language models are zero-shot reasoners. *Advances in Neural Information Processing Systems*, *35*, 22199–22213.

Konda, V. R., & Tsitsiklis, J. N. (2003). On actor-critic algorithms. *Advances in Neural Information Processing Systems*, *42*(4), 1143–1166.

Koren, Y. (2010). *The Global Manufacturing Revolution Reconfigurable Manufacturing Systems and System Configuration Analysis Responsive Business Models for Global Manufacturing Enterprises Enterprise Globalization Strategies Where Are*

Manufacturing Enterprises Headed? http://adrge.engin.umich.edu/wp-content/uploads/sites/50/2013/08/12pgbook.pdf

Krüger, J., Lien, T. K., & Verl, A. (2009). Cooperation of human and machines in assembly lines. *CIRP Annals - Manufacturing Technology*, *58*(2), 628–646. https://doi.org/10.1016/j.cirp.2009.09.009

Kuffner, J. J., & La Valle, S. M. (2000). RRT-connect: an efficient approach to single-query path planning. *Proceedings - IEEE International Conference on Robotics and Automation*, *2*(April), 995–1001. https://doi.org/10.1109/robot.2000.844730

Kumar, A. (2007). From mass customization to mass personalization: A strategic transformation. *International Journal of Flexible Manufacturing Systems*, *19*(4), 533–547. https://doi.org/10.1007/s10696-008-9048-6

Labbé, Y., Carpentier, J., Aubry, M., & Sivic, J. (2020). CosyPose: Consistent Multi-view Multi-object 6D Pose Estimation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *12362 LNCS*, 574–591. https://doi.org/10.1007/978-3-030-58520-4_34

Labbé, Y., Manuelli, L., Mousavian, A., Tyree, S., Birchfield, S., Tremblay, J., Carpentier, J., Aubry, M., Fox, D., & Sivic, J. (2022). MegaPose: 6D Pose Estimation of Novel Objects via Render & Compare. *Proceedings of the 6th Conference on Robot Learning*, *205*(CoRL), 715–725.

Lasota, P. A., Fong, T., & Shah, J. A. (2017). A Survey of Methods for Safe Human-Robot Interaction. *A Survey of Methods for Safe Human-Robot Interaction*. https://doi.org/10.1561/9781680832792

Lasota, P. A., & Shah, J. A. (2015). Analyzing the effects of human-aware motion planning on close-proximity human-robot collaboration. *Human Factors*, *57*(1), 21–33. https://doi.org/10.1177/0018720814565188

LaValle, S. (1998). Rapidly-exploring random trees: A new tool for path planning. *Research Report 9811*.

Law, H., & Deng, J. (2018). CornerNet: Detecting Objects as Paired Keypoints. *European Conference on Computer Vision(ECCV)*, 765–781.

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2323. https://doi.org/10.1109/5.726791

Lee, J., Azamfar, M., Singh, J., & Siahpour, S. (2020). Integration of digital twin and deep learning in cyber-physical systems: Towards smart manufacturing. *IET Collaborative Intelligent Manufacturing*, *2*(1), 34–36. https://doi.org/10.1049/iet-cim.2020.0009

Lee, J., Bagheri, B., & Kao, H. A. (2015). A Cyber-Physical Systems architecture for Industry 4.0-based manufacturing systems. *Manufacturing Letters*, *3*, 18–23. https://doi.org/10.1016/j.mfglet.2014.12.001

Lee, Y., Hu, E. S., & Lim, J. J. (2021). IKEA Furniture Assembly Environment for Long-Horizon Complex Manipulation Tasks. *Proceedings - IEEE International Conference on Robotics and Automation*, *2021-May*(Icra), 6343–6349. https://doi.org/10.1109/ICRA48506.2021.9560986

Li, G., Yuan, C., Kamarthi, S., Moghaddam, M., & Jin, X. (2021). Data science skills and domain knowledge requirements in the manufacturing industry: A gap analysis. *Journal of Manufacturing Systems*, *60*(April), 692–706. https://doi.org/10.1016/j.jmsy.2021.07.007

Li, H., Gong, D., & Yu, J. (2021). An obstacles avoidance method for serial manipulator based on reinforcement learning and Artificial Potential Field. *International Journal of Intelligent Robotics and Applications*, 186–202.

Li, K., Wang, Y., He, Y., Li, Y., Wang, Y., Wang, L., & Qiao, Y. (2022). UniFormerV2: Spatiotemporal Learning by Arming Image ViTs with Video UniFormer. *ArXiv Preprint*, 1–24. http://arxiv.org/abs/2211.09552

Li, P., An, Z., Abrar, S., & Zhou, L. (2025). Large Language Models for Multi-Robot Systems: A Survey. *ArXiv Preprint ArXiv:2502.03814*.

Li, S., Wang, R., Zheng, P., & Wang, L. (2021). Towards proactive human–robot collaboration: A foreseeable cognitive manufacturing paradigm. *Journal of Manufacturing Systems*, *60*(July), 547–552. https://doi.org/10.1016/j.jmsy.2021.07.017

Li, X., Wang, H., Yi, L., Guibas, L. J., Abbott, A. L., & Song, S. (2020). Category-Level Articulated Object Pose Estimation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 3703–3712.

https://doi.org/10.1109/CVPR42600.2020.00376

Li, Y., Wang, G., Ji, X., Xiang, Y., & Fox, D. (2018). DeepIM: Deep iterative matching for 6D pose estimation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *11210 LNCS*, 695–711. https://doi.org/10.1007/978-3-030-01231-1_42

Li, Zhigang, Wang, G., & Ji, X. (2019). CDPN: Coordinates-based disentangled pose network for real-time RGB-based 6-DoF object pose estimation. *Proceedings of the IEEE International Conference on Computer Vision*, *2019-Octob*, 7677–7686. https://doi.org/10.1109/ICCV.2019.00777

Li, Zhijun, Liu, J., Huang, Z., Peng, Y., Pu, H., & Ding, L. (2017). Adaptive Impedance Control of Human – Robot. *IEEE Transactions on Industrial Electronics*, *64*(10), 8013–8022. http://ieeexplore.ieee.org.

Liang, J., Huang, W., Xia, F., Xu, P., Hausman, K., Ichter, B., Florence, P., & Zeng, A. (2023). Code as policies: Language model programs for embodied control. *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 9493–9500.

Likhachev, M., Gordon, G., & Thrun, S. (2003). ARA *: Anytime A * with Provable Bounds on. *Proceedings of Advances in Neural Information Processing Systems (NIPS 2003)*, 767–774. https://papers.nips.cc/paper/2382-ara-anytime-a-with-provable-bounds-on-sub-optimality.pdf

Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., & Wierstra, D. (2016). Continuous control with deep reinforcement learning. *4th International Conference on Learning Representations, ICLR 2016 - Conference Track Proceedings*.

Lin, H., Liu, Z., Cheang, C., Fu, Y., Guo, G., & Xue, X. (2022). SAR-Net: Shape Alignment and Recovery Network for Category-level 6D Object Pose and Size Estimation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *2022-June*, 6697–6707. https://doi.org/10.1109/CVPR52688.2022.00659

Lin, J., Wei, Z., Li, Z., Xu, S., Jia, K., & Li, Y. (2021). DualPoseNet: Category-level 6D Object Pose and Size Estimation Using Dual Pose Network with Refined Learning of Pose Consistency. *Proceedings of the IEEE International Conference on Computer Vision*, 3540–3549. https://doi.org/10.1109/ICCV48922.2021.00354

Lin, T.-Y., Dollar, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature Pyramid Networks for Object Detection. *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 936–944. https://doi.org/10.1109/CVPR.2017.106

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). Focal Loss for Dense Object Detection. *2017 IEEE International Conference on Computer Vision (ICCV)*, 2999–3007. https://doi.org/10.1109/ICCV.2017.324

Lin, X., Wang, D., Zhou, G., Liu, C., & Chen, Q. (2024). TransPose: 6D object pose estimation with geometry-aware Transformer. *Neurocomputing*, *589*(March), 127652. https://doi.org/10.1016/j.neucom.2024.127652

Liu, B., Jiang, Y., Zhang, X., Liu, Q., Zhang, S., Biswas, J., & Stone, P. (2023). Llm+ p: Empowering large language models with optimal planning proficiency. *ArXiv Preprint ArXiv:2304.11477*.

Liu, C., Chen, L. C., Schroff, F., Adam, H., Hua, W., Yuille, A. L., & Fei-Fei, L. (2019). Auto-deeplab: Hierarchical neural architecture search for semantic image segmentation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *2019-June*, 82–92. https://doi.org/10.1109/CVPR.2019.00017

Liu, H., & Wang, L. (2017). Human motion prediction for human-robot collaboration. *Journal of Manufacturing Systems*, *44*, 287–294. https://doi.org/10.1016/j.jmsy.2017.04.009

Liu, Q., Liu, Z., Xiong, B., Xu, W., & Liu, Y. (2021). Deep reinforcement learning-based safe interaction for industrial human-robot collaboration using intrinsic reward function. *Advanced Engineering Informatics*, *49*(July), 101360. https://doi.org/10.1016/j.aei.2021.101360

Liu, Q., Liu, Z., Xu, W., Tang, Q., Zhou, Z., & Pham, D. T. (2019). Human-robot collaboration in disassembly for sustainable manufacturing. *International Journal of Production Research*, *57*(12), 4027–4044. https://doi.org/10.1080/00207543.2019.1578906

Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C. Y., & Berg, A. C. (2016). SSD: Single shot multibox detector. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, *9905 LNCS*, 21–37. https://doi.org/10.1007/978-3-319-46448-0_2

Liu, X., Zheng, L., Shuai, J., Zhang, R., & Li, Y. (2020). Data-driven and AR assisted intelligent collaborative assembly system for large-scale complex products. *Procedia CIRP*, *93*(March), 1049–1054. https://doi.org/10.1016/j.procir.2020.04.041

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 9992–10002. https://ieeexplore.ieee.org/document/9710580/

Long, J., Shelhamer, E., & Darrell, T. (2015). Fully convolutional networks for semantic segmentation. *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 3431–3440. https://doi.org/10.1109/CVPR.2015.7298965

Lorenzini, M., Lagomarsino, M., Fortini, L., Gholami, S., & Ajoudani, A. (2023). Ergonomic human-robot collaboration in industry: A review. *Frontiers in Robotics and AI*, *9*(January), 1–24. https://doi.org/10.3389/frobt.2022.813907

Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, *60*, 91–110.

Lu, Y., Xu, X., & Wang, L. (2020). Smart manufacturing process and system automation – A critical review of the standards and envisioned scenarios. In *Journal of Manufacturing Systems* (Vol. 56, Issue September). https://doi.org/10.1016/j.jmsy.2020.06.010

Ma, Y. J., Liang, W., Wang, G., Huang, D.-A., Bastani, O., Jayaraman, D., Zhu, Y., Fan, L., & Anandkumar, A. (2023). Eureka: Human-level reward design via coding large language models. *ArXiv Preprint ArXiv:2310.12931*.

Ma, Y., Song, Z., Zhuang, Y., Hao, J., & King, I. (2024). A survey on vision-language-action models for embodied ai. *ArXiv Preprint ArXiv:2405.14093*.

Malik, A. A., & Bilberg, A. (2018). Digital twins of human robot collaboration in a production setting. *Procedia Manufacturing*, *17*, 278–285. https://doi.org/10.1016/j.promfg.2018.10.047

Martinez, J., Black, M. J., & Romero, J. (2017). On human motion prediction using recurrent neural networks. *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*, *2017-Janua*, 4674–4683. https://doi.org/10.1109/CVPR.2017.497

Matheson, E., Minto, R., Zampieri, E. G. G., Faccio, M., & Rosati, G. (2019). Human-robot collaboration in manufacturing applications: A review. *Robotics*, *8*(4), 1–25. https://doi.org/10.3390/robotics8040100

Michel, F., Brachmann, E., Kehl, W., Buch, A. G., Kraft, D., Drost, B., Vidal, J., Ihrke, S., Zabulis, X., Sahin, C., Manhardt, F., Tombari, F., Kim, T., & Rother, C. (2018). BOP: Benchmark for 6D Object Pose Estimation. (arXiv:1808.08319v1 [cs.CV]). *Eccv*. http://arxiv.org/abs/1808.08319

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., & Riedmiller, M. (2013). Playing Atari with Deep Reinforcement Learning. *ArXiv Preprint*, 1–9. http://arxiv.org/abs/1312.5602

Mnih, V., Puigdomènech Badia, A., Mirza, M., Harley, T., P. Lillicrap, T., Silver, D., & Kavukcuoglu, K. (2016). Asynchronous Methods for Deep Reinforcement Learning. *International Conference on Machine Learning*, *48*, 1928–1937. http://arxiv.org/abs/1301.3781

Moon, S., Son, H., Hur, D., & Kim, S. (2024). GenFlow: Generalizable Recurrent Flow for 6D Pose Refinement of Novel Objects. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.*, 10039–10049.

Nguyen, V. N., Groueix, T., Ponimatkin, G., Lepetit, V., & Hodan, T. (2023). CNOS: A Strong Baseline for CAD-based Novel Object Segmentation. *Proceedings - 2023 IEEE/CVF International Conference on Computer Vision Workshops, ICCVW 2023*, 2126–2132. https://doi.org/10.1109/ICCVW60793.2023.00227

Nguyen, V. N., Groueix, T., Salzmann, M., & Lepetit, V. (2024). GigaPose: Fast and Robust Novel Object Pose Estimation via One Correspondence. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9903–9913. http://arxiv.org/abs/2311.14155

Nguyen, V. N., Hu, Y., Xiao, Y., Salzmann, M., & Lepetit, V. (2022). Templates for 3D Object Pose Estimation Revisited: Generalization to New Objects and Robustness to Occlusions. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 6761–6770. https://doi.org/10.1109/CVPR52688.2022.00665

Ogenyi, U. E., Liu, J., Yang, C., Ju, Z., & Liu, H. (2021). Physical Human-Robot Collaboration: Robotic Systems, Learning Methods, Collaborative Strategies, Sensors, and Actuators.

IEEE Transactions on Cybernetics, *51*(4), 1888–1901. https://doi.org/10.1109/TCYB.2019.2947532

Okorn, B., Gu, Q., Hebert, M., & Held, D. (2021). ZePHyR: Zero-shot Pose Hypothesis Rating. *Proceedings - IEEE International Conference on Robotics and Automation*, *2021-May*(Icra), 14141–14148. https://doi.org/10.1109/ICRA48506.2021.9560874

Oktay, O., Schlemper, J., Folgoc, L. Le, Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N. Y., Kainz, B., Glocker, B., & Rueckert, D. (2018). Attention U-Net: Learning Where to Look for the Pancreas. *ArXiv Preprint*, *Midl*. http://arxiv.org/abs/1804.03999

Park, C., Pan, J., & Manocha, D. (2012). ITOMP: Incremental trajectory optimization for real-time replanning in dynamic environments. *ICAPS 2012 - Proceedings of the 22nd International Conference on Automated Planning and Scheduling*, 207–215. https://doi.org/10.1609/icaps.v22i1.13513

Paszke, A., Chaurasia, A., Kim, S., & Culurciello, E. (2016). ENet: A Deep Neural Network Architecture for Real-Time Semantic Segmentation. *ArXiv Preprint*, 1–12. https://doi.org/10.48550/arXiv.1606.02147v1

Peng, S., Zhou, X., Liu, Y., Lin, H., Huang, Q., & Bao, H. (2019). PVNet: Pixel-Wise Voting Network for 6DoF Object Pose Estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(6), 3212–3223. https://doi.org/10.1109/TPAMI.2020.3047388

Plaat, A., Wong, A., Verberne, S., Broekens, J., van Stein, N., & Back, T. (2024). Reasoning with large language models, a survey. *ArXiv Preprint ArXiv:2407.11511*.

Quinlan, S., & Khatib, O. (1993). Elastic bands: Connecting path planning and control. *Proceedings - IEEE International Conference on Robotics and Automation*, *2*, 802–807. https://doi.org/10.1109/robot.1993.291936

R. Novais, L., Maqueira, J. M., & Bruque, S. (2019). Supply chain flexibility and mass personalization: a systematic literature review. *Journal of Business and Industrial Marketing*, *34*(8), 1791–1812. https://doi.org/10.1108/JBIM-03-2019-0105

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning Transferable Visual Models From Natural Language Supervision. *Proceedings of Machine Learning Research*, *139*, 8748–8763.

Ravichandar, H., Polydoros, A. S., Chernova, S., & Billard, A. (2020). Recent advances in robot learning from demonstration. *Annual Review of Control, Robotics, and Autonomous Systems*, *3*(1), 297–330.

Ravichandran, Z., Robey, A., Kumar, V., Pappas, G. J., & Hassani, H. (2025). Safety Guardrails for LLM-Enabled Robots. *ArXiv Preprint ArXiv:2503.07885*.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 779–788. https://doi.org/10.1109/CVPR.2016.91

Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *Advances in Neural Information Processing Systems*, *28*. https://doi.org/10.4324/9780080519340-12

Robla-Gomez, S., Becerra, V. M., Llata, J. R., Gonzalez-Sarabia, E., Torre-Ferrero, C., & Perez-Oria, J. (2017). Working Together: A Review on Safe Human-Robot Collaboration in Industrial Environments. *IEEE Access*, *5*, 26754–26773. https://doi.org/10.1109/ACCESS.2017.2773127

Rodriguez-Guerra, D., Sorrosal, G., Cabanes, I., & Calleja, C. (2021). Human-Robot Interaction Review: Challenges and Solutions for Modern Industrial Environments. *IEEE Access*, *9*, 108557–108578. https://doi.org/10.1109/ACCESS.2021.3099287

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In *IEEE Access* (Vol. 9, pp. 234–241). https://doi.org/10.1007/978-3-319-24574-4_28

Sangiovanni, B., Rendiniello, A., Incremona, G. P., Ferrara, A., & Piastra, M. (2018). Deep Reinforcement Learning for Collision Avoidance of Robotic Manipulators. *European Control Conference (ECC)*, 2063–2068.

Schulman, J., Ho, J., Lee, A., Awwal, I., Bradlow, H., & Abbeel, P. (2016). Finding Locally Optimal, Collision-Free Trajectories with Sequential Convex Optimization. *Robotics: Science and Systems*, *9*(1), 1–10. https://doi.org/10.15607/rss.2013.ix.031

Schulman, J., Levine, S., Moritz, P., Jordan, M., & Abbeel, P. (2015). Trust region policy optimization. *International Conference on Machine Learning*, 1889–1897.

https://doi.org/10.3917/rai.067.0031

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal Policy Optimization Algorithms. *ArXiv Preprint*, 1–12. http://arxiv.org/abs/1707.06347

Sherwani, F., Asad, M. M., & Ibrahim, B. S. K. K. (2020). Collaborative Robots and Industrial Revolution 4.0 (IR 4.0). *2020 International Conference on Emerging Trends in Smart Technologies, ICETST 2020*, *0*. https://doi.org/10.1109/ICETST49965.2020.9080724

Shotton, J., Johnson, M., & Cipolla, R. (2008). Semantic texton forests for image categorization and segmentation. *2008 IEEE Conference on Computer Vision and Pattern Recognition*, *7*(PARTS A, B, AND C), 1–8. https://doi.org/10.1109/CVPR.2008.4587503

Shridhar, M., Manuelli, L., & Fox, D. (2023). Perceiver-actor: A multi-task transformer for robotic manipulation. *Conference on Robot Learning*, 785–799.

Shugurov, I., Li, F., Busam, B., & Ilic, S. (2022). OSOP: A Multi-Stage One Shot Object Pose Estimation Framework. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *2022-June*, 6825–6834. https://doi.org/10.1109/CVPR52688.2022.00671

Smierzchalski, R., & Michalewicz, Z. (2005). Path planning in dynamic environments. In *Innovations in Robot Mobility and Control* (pp. 135–153). Springer.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *International Conference on Machine Learning*, *15*(3), 190. https://doi.org/10.48550/arXiv.1905.11946

Tao, F., Cheng, J., Qi, Q., Zhang, M., Zhang, H., & Sui, F. (2018). Digital twin-driven product design, manufacturing and service with big data. *International Journal of Advanced Manufacturing Technology*, *94*(9–12), 3563–3576. https://doi.org/10.1007/s00170-017-0233-1

Tao, F., Qi, Q., Liu, A., & Kusiak, A. (2018). Data-driven smart manufacturing. *Journal of Manufacturing Systems*, *48*, 157–169. https://doi.org/10.1016/j.jmsy.2018.01.006

Tian, M., Ang, M. H., & Lee, G. H. (2020). Shape Prior Deformation for Categorical 6D Object Pose and Size Estimation. *Lecture Notes in Computer Science (Including Subseries*

Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), *12366 LNCS*, 530–546. https://doi.org/10.1007/978-3-030-58589-1_32

Tuptuk, N., & Hailes, S. (2018). Security of smart manufacturing systems. *Journal of Manufacturing Systems*, *47*(May), 93–106. https://doi.org/10.1016/j.jmsy.2018.04.007

Varatharasan, V., Shin, H. S., Tsourdos, A., & Colosimo, N. (2019). Improving Learning Effectiveness for Object Detection and Classification in Cluttered Backgrounds. *2019 International Workshop on Research, Education and Development on Unmanned Aerial Systems, RED-UAS 2019*, 78–85. https://doi.org/10.1109/REDUAS47371.2019.8999695

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention Is All You Need. *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, *30*, 4752–4758. https://dl.acm.org/doi/10.1145/3583780.3615497

Vicentini, F. (2021). Collaborative Robotics: A Survey. *Journal of Mechanical Design, Transactions of the ASME*, *143*(4), 1–20. https://doi.org/10.1115/1.4046238

Villani, V., Pini, F., Leali, F., & Secchi, C. (2018). Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics*, *55*(March), 248–266. https://doi.org/10.1016/j.mechatronics.2018.02.009

Wang, B., Liu, Z., Li, Q., & Prorok, A. (2020). Mobile robot path planning in dynamic environments through globally guided reinforcement learning. *IEEE Robotics and Automation Letters*, *5*(4), 6932–6939. https://doi.org/10.1109/LRA.2020.3026638

Wang, C., Xu, D., Zhu, Y., Martin-Martin, R., Lu, C., Fei-Fei, L., & Savarese, S. (2019). DenseFusion: 6D object pose estimation by iterative dense fusion. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, *2019-June*, 3338–3347. https://doi.org/10.1109/CVPR.2019.00346

Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., & Anandkumar, A. (2023). Voyager: An open-ended embodied agent with large language models. *ArXiv Preprint ArXiv:2305.16291*.

Wang, H., Sridhar, S., Huang, J., Valentin, J., Song, S., & Guibas, L. J. (2019). Normalized object coordinate space for category-level 6D object pose and size estimation. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern*

Recognition, *2019-June*, 2637–2646. https://doi.org/10.1109/CVPR.2019.00275

Wang, Jiaqi, Shi, E., Hu, H., Ma, C., Liu, Y., Wang, X., Yao, Y., Liu, X., Ge, B., & Zhang, S. (2024). Large language models for robotics: Opportunities, challenges, and perspectives. *Journal of Automation and Intelligence*.

Wang, Jinjiang, Ma, Y., Zhang, L., Gao, R. X., & Wu, D. (2018). Deep learning for smart manufacturing: Methods and applications. *Journal of Manufacturing Systems*, *48*, 144–156. https://doi.org/10.1016/j.jmsy.2018.01.003

Wang, L., Gao, R., Váncza, J., Krüger, J., Wang, X. V., Makris, S., & Chryssolouris, G. (2019). Symbiotic human-robot collaborative assembly. *CIRP Annals*, *68*(2), 701–726. https://doi.org/10.1016/j.cirp.2019.05.002

Wang, Lihui. (2019). From Intelligence Science to Intelligent Manufacturing. *Engineering*, *5*(4), 615–618. https://doi.org/10.1016/j.eng.2019.04.011

Wang, Lihui, Liu, S., Liu, H., & Wang, X. V. (2020). Overview of human-robot collaboration in manufacturing. In *Proceedings of 5th International Conference on the Industry 4.0 Model for Advanced Manufacturing*. https://doi.org/10.1007/978-3-030-46212-3_2

Wang, Lirui, Ling, Y., Yuan, Z., Shridhar, M., Bao, C., Qin, Y., Wang, B., Xu, H., & Wang, X. (2023). Gensim: Generating robotic simulation tasks via large language models. *ArXiv Preprint ArXiv:2310.01361*.

Wang, Y., Mohamed, A., Le, D., Liu, C., Xiao, A., Mahadeokar, J., Huang, H., Tjandra, A., Zhang, X., Zhang, F., Fuegen, C., Zweig, G., & Seltzer, M. L. (2020). Transformer-Based Acoustic Modeling for Hybrid Speech Recognition. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, *2020-May*, 6874–6878. https://doi.org/10.1109/ICASSP40776.2020.9054345

Wang, Z., Cai, S., Chen, G., Liu, A., Ma, X. S., & Liang, Y. (2023). Describe, explain, plan and select: interactive planning with llms enables open-world multi-task agents. *Advances in Neural Information Processing Systems*, *36*, 34153–34189.

Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, *8*, 279–292.

Welfare, K. S., Hallowell, M. R., Shah, J. A., & Riek, L. D. (2019). Consider the Human Work Experience When Integrating Robotics in the Workplace. *ACM/IEEE International Conference on Human-Robot Interaction*, *2019-March*, 75–84.

https://doi.org/10.1109/HRI.2019.8673139

Willia, R. J. (1992). Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning. *Machine Learning*, *8*(3), 229–256. https://doi.org/10.1023/A:1022672621406

Wu, K., Yuan, S., Esfahani, M. A., & Wang, H. (2019). Depth-based obstacle avoidance through deep reinforcement learning. *ACM International Conference Proceeding Series*, *Part F1476*, 102–106. https://doi.org/10.1145/3314493.3314495

Wuest, T., Weimer, D., Irgens, C., & Thoben, K. D. (2016). Machine learning in manufacturing: Advantages, challenges, and applications. *Production and Manufacturing Research*, *4*(1), 23–45. https://doi.org/10.1080/21693277.2016.1192517

Xiang, Y., Schmidt, T., Narayanan, V., & Fox, D. (2018). PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes. *Robotics: Science and Systems*. https://doi.org/10.15607/RSS.2018.XIV.019

Xiao, H., & Wang, P. (2023). Llm a*: Human in the loop large language models enabled a* search for robotics. *ArXiv Preprint ArXiv:2312.01797*.

Xiao, X., Liu, J., Wang, Z., Zhou, Y., Qi, Y., Jiang, S., He, B., & Cheng, Q. (2025). Robot learning in the era of foundation models: A survey. *Neurocomputing*, 129963.

Xu, L. Da, Xu, E. L., & Li, L. (2018). Industry 4.0: State of the art and future trends. *International Journal of Production Research*, *56*(8), 2941–2962. https://doi.org/10.1080/00207543.2018.1444806

Xu, X. (2012). From cloud computing to cloud manufacturing. *Robotics and Computer-Integrated Manufacturing*, *28*(1), 75–86. https://doi.org/10.1016/j.rcim.2011.07.002

Yan, J., Meng, Y., Lu, L., & Li, L. (2017). Industrial Big Data in an Industry 4.0 Environment: Challenges, Schemes, and Applications for Predictive Maintenance. *IEEE Access*, *5*, 23484–23491. https://doi.org/10.1109/ACCESS.2017.2765544

Zakharov, S., Shugurov, I., & Ilic, S. (2019). DPOD: 6D pose object detector and refiner. *Proceedings of the IEEE International Conference on Computer Vision*, *2019-Octob*, 1941–1950. https://doi.org/10.1109/ICCV.2019.00203

Zhao, C., Yuan, S., Jiang, C., Cai, J., Yu, H., Wang, M. Y., & Chen, Q. (2023). Erra: An embodied representation and reasoning architecture for long-horizon language-conditioned manipulation tasks. *IEEE Robotics and Automation Letters*, *8*(6), 3230–3237.

Zhao, H., Qi, X., Shen, X., Shi, J., & Jia, J. (2018). ICNet for Real-Time Semantic Segmentation on High-Resolution Images. *European Conference on Computer Vision*, 418–434. https://github.com/hszhao/ICNet%0Ahttp://arxiv.org/abs/1704.08545

Zhao, H., Shi, J., Qi, X., Wang, X., & Jia, J. (2017). Pyramid Scene Parsing Network. *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, *37*(178), 6230–6239. https://doi.org/10.1109/CVPR.2017.660

Zheng, H., Lee, R., & Lu, Y. (2023). HA-ViD: A Human Assembly Video Dataset for Comprehensive Assembly Knowledge Understanding. *Advances in Neural Information Processing Systems*, *36*(NeurIPS), 1–13.

Zheng, P., wang, H., Sang, Z., Zhong, R. Y., Liu, Y., Liu, C., Mubarok, K., Yu, S., & Xu, X. (2018). Smart manufacturing systems for Industry 4.0: Conceptual framework, scenarios, and future perspectives. *Frontiers of Mechanical Engineering*, *13*(2), 137–150. https://doi.org/10.1007/s11465-018-0499-5

Zhong, R. Y., Xu, X., Klotz, E., & Newman, S. T. (2017). Intelligent Manufacturing in the Context of Industry 4.0: A Review. *Engineering*, *3*(5), 616–630. https://doi.org/10.1016/J.ENG.2017.05.015

Zhu, Y., Li, M., Yao, W., & Chen, C. (2022). A Review of 6D Object Pose Estimation. *IEEE Joint International Information Technology and Artificial Intelligence Conference (ITAIC)*, *2022-June*, 1647–1655. https://doi.org/10.1109/ITAIC54216.2022.9836663

Zindler, F., Lucchi, M., Wohlhart, L., Pichler, H., & Hofbaur, M. (2022). Towards Dynamic Obstacle Avoidance for Robot Manipulators with Deep Reinforcement Learning. In *Mechanisms and Machine Science: Vol. 120 MMS*. Springer International Publishing. https://doi.org/10.1007/978-3-031-04870-8_11

Zsifkovits, H., Kapeller, J., Reiter, H., Weichbold, C., & Woschank, M. (2020). The opportunities and challenges of sme manufacturing automation: Safety and ergonomics in human-robot collaboration. In *Industry 4.0 for SMEs: Challenges, Opportunities and Requirements*. https://doi.org/10.1007/978-3-030-25425-4_6

Zucker, M., Ratliff, N., Dragan, A. D., Pivtoraiko, M., Klingensmith, M., Dellin, C. M., Bagnell, J. A., & Srinivasa, S. S. (2013). CHOMP: Covariant Hamiltonian optimization for motion planning. *International Journal of Robotics Research*, *32*(9–10), 1164–1193. https://doi.org/10.1177/0278364913488805